

**PROJECT HERA: A LIVELIHOOD TRAINING AND
RESOURCE MANAGEMENT PLATFORM FOR ANGAT NUEVA
ECIJA NGO**



A CAPSTONE AND RESEARCH PROJECT

by:

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December 2025

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RESOURCE MANAGEMENT PLATFORM FOR ANGAT NUEVA
ECIJA NGO**

A Capstone and Research Project
Presented to the Faculty of the
College of Information and Communications
Technology
Cabanatuan City, Nueva Ecija

In Partial
Fulfillment of the
Requirements for the degree
Bachelor of Science in Information Technology
With Specialization in
Web Systems Technology

by:
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Enrile, Janella Eunice C.
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December 2025



APPROVAL SHEET

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JOINT UNDERTAKING

We, **Lyndon Mark S. Cabang, Janella Eunice C. Enrile, Kurt M. Alfonso, Crister John F. Beltran**, all of legal age, Filipino, and students of Nueva Ecija University of Science and Technology (NEUST), after having been duly sworn in accordance with the law, do hereby state and depose:

1. That the undersigned are the authors/researchers of an academic project entitled *Project Hera: A Livelihood Training and Resource Management Platform for Angat Nueva Ecija NGO*, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology at Nueva Ecija University of Science and Technology (NEUST). This project was completed with the guidance and contribution of our adviser, **Cris Norman P. Olipas, Ed.D.**, and our course teacher, **Rose Anne G. Cochanco, MSIT**, who are hereby recognized as co-authors of this work in acknowledgment of their valuable intellectual and professional support throughout the development of the project.
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IN WITNESS WHEREOF, we have hereunto set on our hands this ____ day of _____, 20____ in the City of Cabanatuan, Philippines.

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College of Information and Communications Technology



CERTIFICATION OF ENGLISH CRITIC

This is to certify that the capstone and research project entitled **Project Hera: A Livelihood Training and Resource Management Platform for Angat Nueva Ecija NGO**, prepared and submitted by **Lyndon Mark S. Cabang, Janella Eunice C. Enrile, Kurt M. Alfonso, Crister John F. Beltran**, in partial fulfillment of the requirements for the degree, Bachelor of Science in Information Technology with Specialization in Web Systems Technology has been grammatically and technically edited by the undersigned. The said manuscript is acceptable to the rules of grammar and composition.

This certification has been issued to the abovementioned student-researcher for reference purposes only.

Cris Norman P. Olipas, Ed.D.
Research Critic/Editor

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Dedication

This project is dedicated
to the Almighty God for His guidance,
to our parents for their support,
to our advisers for their guidance,
and to all who encouraged and helped us
throughout this journey.

- The Researchers

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Abstract

Project HERA is a web-based Livelihood Training and Resource Management Portal developed to address the operational challenges of Angat Nueva Ecija NGO, particularly its reliance on manual processes for training coordination, resource tracking, and donation management. Using a convergent parallel mixed-method design, the study collected survey responses and interview insights from NGO staff, IT experts, donors, youth volunteers, and women beneficiaries. Results showed significant issues in time efficiency, data security, record loss, and transparency.

The system was developed using the Agile Scrum methodology, allowing iterative refinement based on user feedback. Key features include user management, donation monitoring, automated reports, and online training access. The platform was evaluated using ISO/IEC 25010:2023 standards and the System Usability Scale (SUS), demonstrating high functionality, strong performance, and excellent usability.

Keywords: Livelihood training, NGO management system, women empowerment, resource management, Agile Scrum, ISO/IEC 25010, usability evaluation, mixed-method research.

Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Empowering women through sustainable livelihood programs plays a crucial role in promoting economic independence and social development (Al Hakim et al., 2022; Panda & Gope, 2023; Acharya, 2024; Rawat & Rawat, 2024). However, many women, especially those in marginalized communities, continue to face significant challenges in accessing resources, training, and opportunities needed for self-sufficiency (Kanyagui et al., 2023; Malima, 2024). This situation reveals a critical gap between skills development and long-term livelihood sustainability, which limits economic empowerment and reduces the overall well-being of women in areas such as Nueva Ecija.

Angat Nueva Ecija NGO, originally established in 1997 as the Nueva Ecija NGO Partnership and Development, emerges as a community-driven organization formed by civic leaders committed to supporting underprivileged sectors, particularly women and children. Over time, the NGO has collaborated with local governments, private organizations, and

educational institutions to offer skills training in various areas, including sewing, mushroom production, baking, and cosmetology. These programs aim to equip women with the tools to start small businesses and achieve financial independence. Despite these efforts, many beneficiaries struggle to establish sustainable livelihoods after training due to the absence of post-training support, limited access to equipment, and challenges in securing funding or employment.

Although the NGO provides essential tools, including sewing machines, to various women's groups—even within correctional facilities—its operations remain largely manual and inefficient. The organization relies on paper-based documentation and basic tools, such as spreadsheets, to track donations, schedule training, and manage resources. This reliance on face-to-face coordination hinders effective tracking of beneficiary progress, slows resource distribution, and complicates impact assessment. The absence of a centralized, automated system increases the risk of errors, delays, and administrative burdens. Manual reporting consumes a significant amount of time, and communication with donors and beneficiaries often lacks a digital platform for streamlined

interaction. These challenges limit the NGO's capacity to scale its efforts and reach more women in need (Bharadwaj et al., n.d.; Thomas, 2023).

While existing studies explore the benefits of digital tools in enhancing NGO operations, few focus on technology-based systems that close the gap between training and sustained livelihood. Most livelihood programs prioritize the delivery of training but lack mechanisms for ongoing monitoring and structured resource support. Auti et al. (2023) propose a web-based system that streamlines NGO operations, strengthens donor engagement, and improves transparency through automation and real-time data access. Likewise, Bharadwaj et al. (n.d.) and Thomas (2023) demonstrate how web-based platforms enhance operational efficiency, streamline administrative tasks, and facilitate transparent resource management in NGOs operating in India. These studies show that automation allows staff to focus on core programs, while centralized data collection enhances progress tracking and administrative accuracy.

In response to these challenges, this study proposes the development of a web-based Livelihood Training and Resource Management

Platform. The system aims to enhance training accessibility, streamline resource allocation, and improve the monitoring of donations and sponsorships. It uses a centralized database to manage donor profiles, training records, and resource distribution. Features such as real-time updates, beneficiary tracking, role-based access, and automated validation support performance, accuracy, and efficient resource delivery. The portal also enables beneficiaries—such as unemployed women, single mothers, and community groups—to apply for assistance and monitor their development beyond initial training.

This study examines how a web-based platform enhances the monitoring of training progress and resource allocation in livelihood programs. It focuses on key features such as real-time tracking, centralized management, and post-training feedback tools to optimize delivery and impact. The research also examines how digital monitoring enables the NGO to assess both short-term and long-term outcomes for beneficiaries, while enhancing program efficiency and effectiveness.

Through this digital solution, the NGO can modernize its operations, streamline administrative tasks, and enhance transparency

in resource allocation. By adopting technology, the organization bridges the gap between skills development and sustainable livelihoods, thereby contributing to the economic empowerment of women and their long-term well-being in Nueva Ecija. Ultimately, this system enhances the NGO's outreach and enables it to provide consistent, meaningful support to more women, underscoring the importance of integrating digital tools into social development efforts.

Review of Related Literature and Systems

The Importance of Technology in Women's Empowerment. Information and Communication Technologies (ICTs) played a significant role in improving the lives of women by enhancing access to education and skill development through online learning platforms and digital resources (Lakshmi Priya, 2023; Mackey & Petrucka, 2021; Noor et al., 2021). These technologies also facilitated economic opportunities by enabling women to participate in online marketplaces and engage in mobile-based businesses, thereby promoting financial inclusion and increasing their economic participation (Lakshmi Priya, 2023; Noor et al., 2021). In addition, ICTs strengthened women's

social and political engagement by supporting communication, information exchange, and advocacy efforts, which allowed women to assert their rights and challenge existing gender norms (Lakshmi Priya, 2023; Noor et al., 2021).

Despite these advancements, the digital gender divide remained a pressing concern. Many women, particularly those in marginalized communities, continued to experience limited access to ICTs due to infrastructural, educational, and socio-cultural barriers (Mackey & Petrucka, 2021; Noor et al., 2021). Researchers emphasized the necessity of a multi-dimensional approach to bridge this gap, advocating for improvements in infrastructure, targeted digital literacy programs, and the implementation of supportive policies (Mackey & Petrucka, 2021; Sicat et al., 2020).

Several initiatives successfully demonstrated the empowering potential of ICTs. Programs such as Swift Jyoti, Sewa, and Ujjas Innovation effectively utilized digital technologies to enhance women's socio-economic empowerment in various contexts (Lakshmi Priya, 2023). Furthermore, empirical studies using panel data modeling revealed a positive relationship between access to ICTs and increased participation of women in the labor

force (Noor et al., 2021).

Finally, conceptual frameworks, such as those proposed in "Leveraging ICT Technologies in Closing the Gender Gap," underscore the importance of integrating gender perspectives into electronic platforms. These frameworks underscore the importance of considering the interconnections among various sectors within the e-commerce ecosystem to fully leverage the transformative potential of ICTs for women's empowerment (Sicat et al., 2020).

Challenges faced by the NGO. Non-governmental organizations (NGOs) faced a wide range of interconnected challenges that impacted their ability to deliver services effectively. A major concern was financial instability, as many NGOs relied on uncertain funding sources, such as grants and donations, which were often inadequate to sustain operations, particularly during global crises like the COVID-19 pandemic (Bodi, 2024; Ijon et al., 2023; Pandey et al., 2024). This lack of consistent funding limited their ability to hire and retain qualified personnel, resulting in increased workloads, staff burnout, and difficulties in attracting volunteers with specialized expertise (Bodi, 2024; Ijon et al.,

2023; Pandey et al., 2024).

In addition to financial constraints, NGOs faced significant challenges in communication and outreach. Limited resources and technical expertise hindered their efforts to raise awareness, engage with target communities, and effectively utilize social media platforms, which had become increasingly important in public engagement (Bodi, 2024; Ijon et al., 2023; Pandey et al., 2024). Operational and regulatory complexities further strained their resources. NGOs needed to comply with evolving bureaucratic requirements, manage international partnerships, and uphold transparency and accountability standards, all of which demanded substantial administrative effort (Bodi, 2024; Ijon et al., 2023).

Ensuring long-term sustainability required NGOs to adopt a holistic and strategic approach. This included the development of innovative fundraising mechanisms, effective human resource management, structured communication strategies, and sound organizational frameworks (Bodi, 2024; Ijon et al., 2023; Pandey et al., 2024). The literature has emphasized specific operational barriers, including technological limitations related to platform integration (Pandey et al., 2024), competition among NGOs

due to overlapping services and beneficiaries (Ijon et al., 2023), and the difficulty of reaching communities with limited internet access (Bodi, 2024).

Despite these challenges, NGOs implemented various strategies to enhance their resilience and service delivery. Some organizations have developed digital platforms to efficiently connect donors and beneficiaries (Pandey et al., 2024), while others have established partnerships to leverage shared resources and expertise (Bodi, 2024; Ijon et al., 2023). Community-centered programs also played a crucial role in expanding local engagement and improving project visibility and impact (Bodi, 2024). These adaptive measures demonstrated the importance of innovation, collaboration, and strategic planning in strengthening NGO operations. Ultimately, the success of NGOs depended on their ability to remain agile, embrace technology, and foster meaningful relationships with stakeholders.

Importance of Website Platforms for NGOs.

This review of related literature examined the significance of web-based platforms in empowering women, facilitating donations, and supporting volunteerism, in alignment with the

research questions posed in the study. The reviewed literature primarily focused on the development and implementation of web systems for non-governmental organizations (NGOs), emphasizing their influence on operational efficiency, accountability, and stakeholder engagement. Although the studies did not directly address women's empowerment, their findings indicated that technological innovations in NGO operations indirectly contributed to enhancing economic opportunities for women.

Web platforms provided women with improved access to skills training, business resources, and networking opportunities, thereby contributing to their economic empowerment (Bhatt, 2021). These platforms also enabled broader market access for women's products and services, supporting their transition toward financial independence. Additionally, web-based systems simplified the donation process by promoting transparency and accountability, streamlining contributions, and enabling targeted resource allocation (Bhatt, 2021). In terms of volunteer management, web platforms improved communication, facilitated recruitment, and strengthened volunteer engagement.

The development of a cross-platform NGO management portal addressed these operational needs. It aligned with core research concerns such as data security, system reliability, and operational efficiency. Furthermore, the system design considered the ISO/IEC 25010:2023 software quality standards, which ensured usability, performance, and maintainability (Auti, 2023; Dinata et al., 2021; Kumar et al., 2022; Yadav, 2019).

In conclusion, the literature provides strong support for the development of the proposed system. By addressing key factors such as security, reliability, and user experience, the portal aimed to enhance NGO operations and expand their capacity to empower women through sustained digital support and efficient resource management.

The Importance of NGOs in Women's Empowerment and Livelihood Training. Non-governmental organizations (NGOs) played a significant role in empowering women through livelihood training and skill development, as documented in various studies. In India, organizations such as Aanchal Dairy and Swati Gramodhyog Sansthan enabled women to engage in the production and sale of dairy products and

traditional crafts, thereby promoting income generation and economic participation (Rawat & Rawat, 2024). In Nepal, NGOs offered vocational training in tailoring, handicrafts, agriculture, and small-scale entrepreneurship, which contributed to increased income levels and financial stability among women beneficiaries (Acharya, 2024). Similarly, in Nigeria, NGOs supported rural women by providing basic assistance, technical advice, education, and job training, all of which enhanced their self-sufficiency and resilience (Al Hakim et al., 2022).

These interventions not only improved women's economic independence but also fostered greater self-confidence and enhanced their capacity for decision-making (Acharya, 2024; Al Hakim et al., 2022; Rawat & Rawat, 2024). However, the reviewed literature also underscored the importance of adopting a holistic approach to women's empowerment. Addressing systemic barriers, alongside skill development, was deemed essential to achieving sustainable and transformative outcomes (Al Hakim et al., 2022). This approach included challenging restrictive social norms, advocating for inclusive policy reforms, and promoting collaboration with government

institutions and local communities to institutionalize empowerment efforts (Cardoza & Vidya, 2023).

Furthermore, the utilization of various communication channels—particularly community radio—was identified as a vital strategy for disseminating information and raising awareness about women's rights and available resources (Al Hakim et al., 2022). These findings suggest that sustainable empowerment requires a multidimensional framework that combines skills training with advocacy, community engagement, and inclusive communication strategies.

Challenges faced by women. This review of related literature examined the challenges encountered by women in marginalized communities in accessing resources, training, and opportunities for economic empowerment. It drew upon two studies—one focusing on rural women in Tanzania (Malima, 2024) and the other on rural women in India (Kanyagui et al., 2023). Both studies revealed significant barriers to women's economic advancement, particularly in rural contexts.

In the Tanzanian setting, Malima (2024) found that rural women experienced restricted access to financial resources due to

infrastructural limitations, insufficient financial literacy, and overall economic hardship. The fragmented structure of empowerment organizations and the lack of specific legislation supporting women's economic rights further compounded these issues. Likewise, Kanyagui et al. (2023) identified limited access to land ownership, financial capital, and entrepreneurial training as key obstacles hindering the economic empowerment of rural women in India.

Both studies emphasized that inadequate access to training and skills development remained a major impediment. Poor infrastructure and a lack of awareness about available opportunities further limited women's participation in economic activities. Additionally, the limited availability of livelihood programs in rural areas—many of which remained concentrated in urban centers—reflected systemic inequalities in the country. This urban-rural divide contributed to the low labor force participation rates among rural women in India, as observed by Kanyagui et al. (2023).

Addressing these challenges required a multifaceted approach involving coordinated government efforts, partnerships with civil

society organizations, and the development of inclusive, needs-based programs. Kanyagui et al. (2023) proposed the use of participatory action research as a viable methodology for ensuring that interventions were both sustainable and responsive to community needs.

Disadvantages of Manual Operation of NGOs.

Several studies underscored the limitations of manual operations in non-governmental organizations (NGOs). Parhad et al. (2024) highlighted the inefficiencies associated with paper-based documentation and face-to-face coordination, which hindered the effective tracking of beneficiary progress, resource management, and assessment of program outcomes. Bhatt (2021) examined the implementation of an enterprise resource planning (ERP) system in NGOs, revealing how digitalization addressed challenges such as volunteer tracking, financial transparency, and budget allocation. This study highlights the limitations of manual systems and the benefits provided by digital tools in streamlining organizational processes.

Sangeetha and Minimol (2024) conducted a case study on the Atma Foundation, which explored both the challenges and potential of technology adoption in NGO operations. Their

findings indicated that while initial costs, employee upskilling, and ethical considerations—particularly those related to artificial intelligence—posed barriers, digital transformation significantly improved operational efficiency and long-term sustainability. Collectively, these studies supported the necessity for NGOs to transition from manual to digital systems to enhance organizational effectiveness.

Furthermore, Parhad et al. (2024) proposed the development of a centralized and integrated platform to streamline processes, strengthen collaboration, and promote accountability within NGOs. These insights emphasized that embracing technological innovations was crucial for modern NGOs seeking to enhance service delivery and achieve greater impact.

Advantages of using a Web-Based Platform for NGOs. The study *Web-Based Technology Interventions for Social Change: Insights from NGOs in Assam* examined the use of web-based technologies by non-governmental organizations (NGOs) in Assam, India, to enhance their operations and social impact (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b). Employing a qualitative approach that involved interviews,

surveys, and content analysis, the researchers found that NGOs adopted web-based tools for various functions, including advocacy, fundraising, community mobilization, and capacity building (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b). These technological interventions expanded the reach of social programs, promoted transparency, and increased stakeholder engagement.

Despite the benefits, the study also identified persistent challenges, including limited digital literacy, inadequate resources, and unreliable internet connectivity. These constraints underscore the necessity for tailored strategies that consider the specific context of marginalized regions (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b). The literature highlighted the growing global trend of web-based technology adoption among NGOs and acknowledged its effectiveness in enhancing outreach, improving donor relations, and fostering community participation.

Furthermore, the researchers emphasized the significance of transparency and accountability in building trust among stakeholders. They noted that web-based platforms played a critical role in supporting these principles by enabling real-time

information sharing and reporting (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b). The study also explored the role of digital technologies in promoting digital inclusion and empowering underserved communities, especially in remote or rural areas.

In addition to facilitating online fundraising, web-based systems supported evidence-based decision-making by providing centralized data and performance metrics. The discussion highlighted the transformative potential of digital tools in addressing social challenges, while also acknowledging the digital divide as a significant barrier to equitable access. The study recommended inclusive implementation models that bridge technological gaps and ensure equitable outcomes (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b).

Case studies of organizations such as EmpowerEd and Rural Health demonstrated practical applications of web-based interventions in the fields of education and healthcare. These examples illustrated how digital platforms improved program delivery, resource management, and service accessibility (Thomas, 2023a). In conclusion, the research called for context-specific strategies and

continued scholarly investigation to better understand the complexities of digital adoption within the nonprofit sector in regions like Assam (Bharadwaj et al., n.d.; Thomas, 2023a, 2023b).

The Role of Livelihood Programs for Women in NGOs. Sustainable livelihood programs implemented by NGOs played a crucial role in empowering women, fostering economic independence, and promoting social development. Studies conducted in Nepal have demonstrated the effectiveness of vocational training in fields such as tailoring, handicrafts, and small-scale businesses, leading to increased financial independence and self-confidence among women (Acharya, 2024). In India, initiatives such as Mahila EHaat, Mudra Yojana, and Stand-Up India, supported by NGOs, have provided financial assistance and skill development, thereby empowering women entrepreneurs (Panda & Gope, 2023). Access to credit and microfinance, facilitated by NGOs, emerged as a key factor in breaking the cycle of poverty and achieving economic independence in both Nepal and India (Acharya, 2024).

However, the success of these programs depended on addressing systemic barriers, such

as cultural norms and gender inequality, through advocacy and policy reform (Panda & Gope, 2023; Rawat & Rawat, 2024). NGOs in various countries actively challenged harmful gender norms and advocated for legal reforms to protect women's rights, including efforts to combat gender-based violence and promote reproductive health rights (Acharya, 2024; Rawat & Rawat, 2024). Context-specific approaches were identified as vital, with NGOs needing to understand local needs and tailor interventions accordingly (Al Hakim et al., 2022).

Despite facing challenges, including funding constraints and resistance from conservative communities, the overall impact of NGO-led livelihood programs on women's empowerment remained significant and warranted continued support and further research (Panda & Gope, 2023; Al Hakim et al., 2022).

The Use of a convergent parallel mixed-method research design. The convergent parallel mixed methods design, originally introduced in the 1970s and initially referred to as a triangulation design, involved the independent collection and analysis of quantitative and qualitative data, followed by the integration of results for comparison (Adhikari & Timsina,

2024). This design aimed to provide a comprehensive understanding of a research problem by combining the strengths of both approaches: the large sample size, objectivity, and generalizability of quantitative methods, alongside the subjective interpretation and depth offered by qualitative methods (Adhikari & Timsina, 2024).

The integration phase entailed a comparative analysis to identify areas of convergence, divergence, and contrast between the datasets, often utilizing a triangulation approach to facilitate a cohesive analysis (Adhikari & Timsina, 2024). Data collection occurred concurrently for both types of data, typically in a single visit, which enhanced efficiency (Adhikari & Timsina, 2024).

However, this design presented several challenges, including the management of differing sample sizes and the integration of diverse data types—textual and numerical (Adhikari & Timsina, 2024). Addressing contradictions between quantitative and qualitative findings often required additional data collection or reanalysis (Adhikari & Timsina, 2024). The adoption of this design was frequently justified by time constraints, the researcher's expertise in both quantitative and

qualitative methods, or the composition of a mixed-methods research team (Adhikari & Timsina, 2024).

Philosophically, pragmatism was often viewed as an appropriate framework for this design, facilitating the integration of diverse perspectives (Adhikari & Timsina, 2024). The convergent parallel design provided simplicity, efficiency, and the capacity to compare different perspectives, but successful implementation required careful attention to potential challenges (Adhikari & Timsina, 2024).

Synthesis

The reviewed literature underscores the crucial role that technology, particularly Information and Communication Technologies (ICTs), plays in empowering women and facilitating economic development. ICTs have expanded access to education, skill development, and economic opportunities, enabling women to engage in online markets, mobile businesses, and digital advocacy. However, significant barriers to digital inclusion persist, particularly for marginalized women in rural areas, underscoring the need for targeted interventions, including

improved infrastructure, digital literacy programs, and supportive policies. Successful initiatives have demonstrated the potential of ICTs to increase women's economic participation and social engagement, but further efforts are needed to bridge the gender digital divide.

Non-governmental organizations (NGOs) also contribute significantly to women's empowerment through livelihood training and resource management. These organizations offer skill development, vocational training, and financial resources to women, thereby enhancing their economic independence and decision-making capabilities. In countries like India, Nepal, and Nigeria, NGOs have facilitated income-generating activities, such as dairy farming, tailoring, and small-scale entrepreneurship, significantly improving the financial stability and self-sufficiency of women. However, these programs must address systemic barriers, including restrictive social norms and limited access to resources, to achieve sustainable empowerment outcomes.

Despite the positive contributions of NGOs, they face numerous operational challenges, particularly financial instability, resource constraints, and communication difficulties. These challenges hinder their

ability to retain staff, attract volunteers, and effectively reach their target communities. NGOs have adopted digital platforms to streamline operations, improve transparency, and enhance stakeholder engagement, which have proven effective in overcoming some of these hurdles. The use of web-based systems has also been shown to support efficient resource management, enhance accountability, and facilitate online donations, all of which contribute to empowering women.

The manual operation of NGOs has been identified as a key limitation, with inefficiencies in resource tracking and communication. The transition to digital systems, including integrated web platforms, is crucial for enhancing organizational effectiveness and fostering long-term sustainability. By embracing technology, NGOs can enhance their operational efficiency, improve service delivery, and expand their impact, particularly in empowering women.

In summary, the literature highlights the transformative potential of ICTs and web-based platforms in supporting women's economic empowerment and improving NGO operations. These technologies enable better resource management, enhance communication, and foster greater

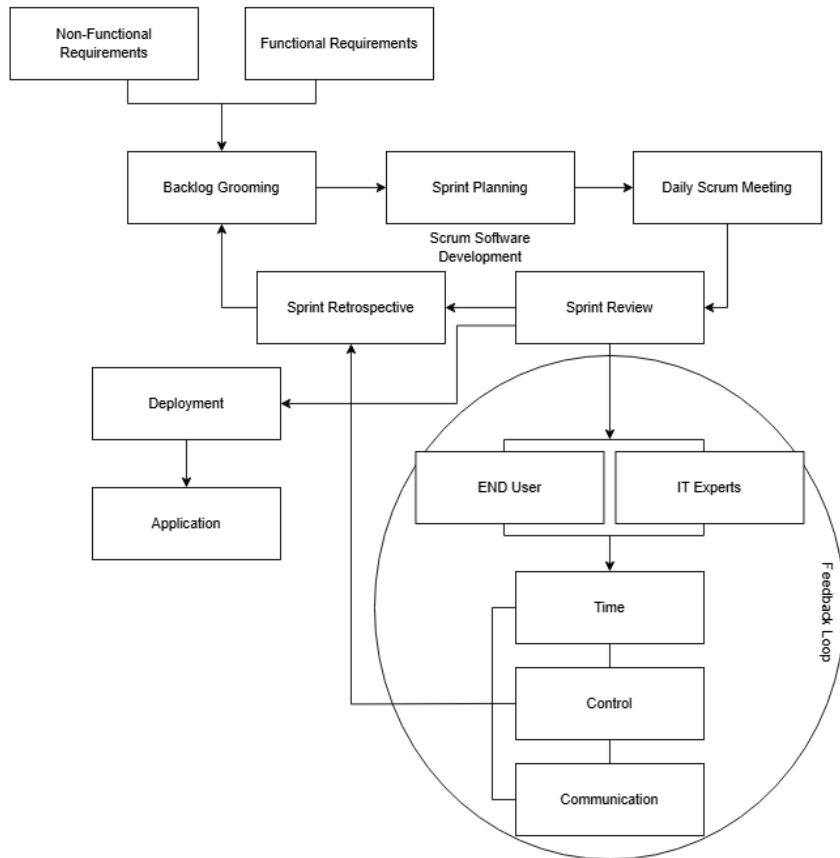
engagement with communities and stakeholders. To maximize their effectiveness, it is essential to address the challenges of digital inclusion, ensure infrastructure development, and implement context-specific strategies that take into account the unique needs of marginalized women. The role of livelihood programs, supported by NGOs, is vital in fostering economic independence, but their success depends on addressing systemic barriers and ensuring access to relevant resources and training.

Conceptual Framework

This study adopts the Agile methodology, specifically the Scrum framework, to develop a cross-platform livelihood training and resource management portal for women-focused non-government organizations (NGOs). This approach supports flexibility and iterative development, allowing continuous refinements based on user feedback.

Figure 1

Conceptual Framework for the Design and Development of Project Hera



The development process began with Backlog Grooming, where the platform requirements were organized to create a clear guide for its development. These included both functional and non-functional requirements. Functional requirements covered registration, fundraising, and resource distribution tracking, while non-functional requirements involved data security, platform performance, and usability. By

continuously refining and optimizing the backlog, the development remained focused on prioritizing features that aligned with the project's objectives.

Following backlog grooming, Sprint Planning was conducted, which involved selecting and prioritizing specific platform features for development within different sprint cycles. Each sprint focused on building key components of the platform. The first sprint established the core structure, user authentication and authorization mechanisms, and the initial database. The second sprint implemented the program information display to ensure that schedules and locations were clearly presented. The next sprint showcased the NGO programs and established the foundation for donation processing. The fourth sprint integrated payment gateways such as GCash and debit cards to ensure secure transactions and efficient tracking of funds. Lastly, the fifth sprint enhanced the user interface, improving usability while also developing the administrative dashboard to support NGO operations.

Throughout the development, Daily Scrum Meetings served as checkpoints where the team reviewed progress, identified challenges, and

discussed next steps. These discussions helped keep the project aligned with its goals by addressing drawbacks and making necessary adjustments. Sprint Reviews were conducted with the developers, NGO administrators, and IT experts to assess the platform's progress. Functionality was evaluated by checking whether the platform accurately displayed program information and processed donations correctly. Security measures were examined to confirm the safety of donation transactions, data encryption, and compliance with data protection standards. Usability testing assessed the intuitiveness of the user interface and smoothness of navigation, while accessibility was measured against the Web Content Accessibility Guidelines (WCAG).

A Sprint Retrospective was conducted to assess successes, challenges, and areas for improvement, ensuring continuous workflow efficiency and better user experience. Once development was completed, the platform entered the Deployment Phase, where the NGO began real-world usage. To ensure seamless integration, multiple tests were conducted to verify the system's performance. Beta testing was initially conducted with selected NGO users to assess the clarity of program information and

the donation process. User Acceptance Testing (UAT) was then conducted to verify that the platform met user requirements. Security testing ensured that donation transactions and user data were protected, while performance testing evaluated page load times and platform responsiveness to achieve an optimal user experience.

By integrating the Scrum framework, the development process remained structured yet flexible, allowing iterative refinements and continuous improvements. This approach ensured that the final product was of high quality, scalable, and tailored to the needs of the NGOs, thereby enhancing their ability to deliver sustainable livelihood opportunities effectively.

Statement of the Problem

The study generally aimed to design, develop, and assess a cross-platform livelihood training and resource management portal for women's non-government organizations. Specifically, it aimed to address the following problems:

1. How may the challenges and pain points for the manual process of record keeping be described in terms of:

- 1.1. Time Consumptions;
 - 1.2. Data Security;
 - 1.3. Record Loss; and
 - 1.4. Transparency?
2. What are the key features preferred by users for the development of a Web Platform to improve the livelihood and resource management of the women's NGO?
3. How will the system be developed based on the following stages of the Agile Scrum Model:
 - 3.1. Planning and Requirement Gathering;
 - 3.2. Design and Prototyping;
 - 3.3. Development and Implementation; and
 - 3.4. Testing and Deployment?
4. How can the elected IT consultants evaluate the developed project in terms of the following characteristics:
 - 4.1. Functional Suitability;
 - 4.2. Performance Efficiency;
 - 4.3. Compatibility;
 - 4.4. Usability;
 - 4.5. Reliability; and
 - 4.6. Security?
5. What are the end-users' evaluations of the developed application based on the System

Usability Scale (SUS) in terms of the following usability characteristics?

- 5.1. Ease of Use;
- 5.2. Efficiency;
- 5.3. Learnability; and
- 5.4. User Satisfaction?

Scope and Delimitations

This study focused on developing a web-based Livelihood Training and Resource Management Portal for Angat Nueva Ecija NGO to address inefficiencies in training management, resource allocation, and donation tracking. The organization previously relied on manual processes, which made it difficult to monitor available resources, track financial contributions, and manage operations efficiently.

To improve these operations, the system introduced automated functionalities designed to enhance efficiency and transparency. A training program management feature was implemented to track training schedules and ensure better coordination of programs. Additionally, a donation and sponsorship monitoring module provided a structured dashboard that allowed donors and administrators to track financial contributions

and resource allocation. To safeguard user information and maintain data security, the system included user role management, which restricted access based on predefined roles for NGO administrators and donors.

While the system aimed to streamline the NGO's operations, its scope was limited to Angat Nueva Ecija NGO and did not extend to other organizations or regions. The system did not facilitate direct financial aid but instead focused on training accessibility, resource tracking, and donation monitoring. Furthermore, the study developed a secure web-based platform, but mobile platform development was not included in the scope.

The system underwent testing with NGO administrators and donors to assess its effectiveness, usability, and security. Usability testing evaluated navigation and functionality, while feedback surveys and system performance analysis measured speed, uptime, and security.

To ensure data protection, the platform complied with the Data Privacy Act of 2012 and the GDPR, which require user consent for data collection, restrict usage to legitimate purposes, and allow users to update or delete their information. Security measures, including

encryption, authentication, and access controls, prevented unauthorized access.

Development followed the Scrum Agile methodology, enabling continuous improvements based on user feedback. Sprint-based refinements ensured flexibility and alignment with the needs of NGOs. The research took place in Nueva Ecija from January to December 2025, adhering to the established study timeline.

Significance of the Study

This study will develop a web-based Livelihood Training and Resource Management Portal for Angat Nueva Ecija NGO to improve training program management, donation tracking, and resource allocation. The NGO currently relies on manual processes, which make it difficult to monitor donations, allocate resources effectively, and provide accessible training to women beneficiaries. The digital system will streamline operations, increase transparency, and ensure the effective delivery of support services.

Beneficiaries. Women will gain better access to livelihood training programs and post-training support, including tools, equipment, and financial aid. Automating donation tracking and resource allocation will enhance

efficiency, reduce administrative workload, and ensure a fair distribution of resources.

NGO and Partner Organizations. The system will help automate registration, track training progress, and manage resource distribution. It will enhance coordination with partners and donors, reduce administrative burdens, and ensure the equitable delivery of support.

Donors and Funding Organizations. The portal will enable donors to track their contributions, monitor resource allocation, and assess the impact of their programs. Transparent reporting will encourage continued support and sponsorships, helping the NGO expand its reach. Community and Society. By providing essential skills and resources, the platform will promote financial independence, self-sufficiency, and community growth. Supporting women in starting small businesses will improve family well-being and strengthen the local economy.

Researchers. The study will provide insights into how digital solutions enhance NGO operations, particularly in terms of training accessibility, resource management, and donor transparency. It will contribute to limited research on technology in NGO management and serve as a reference for future studies.

Future Researchers. The findings will

provide a foundation for exploring digital solutions in livelihood training and resource management, including innovations such as analytics and mobile accessibility, to further enhance NGO efficiency and program delivery.

Definition of Terms

The following definitions may help you comprehend the concepts used in the study:

Community-based resource sharing refers to a cooperative system in which local groups manage and distribute resources collectively to enhance economic sustainability. This approach ensures that community members can maximize shared assets and support livelihood programs.

Digital Literacy is the ability to effectively use digital tools, such as computers and mobile devices, for communication, learning, and business purposes. In this study, digital literacy is essential for women engaging in online training and entrepreneurship.

Economic Empowerment is the process of equipping individuals, particularly women, with financial resources and skills to achieve economic independence. This includes access to employment opportunities, entrepreneurship, and financial management training.

E-Learning Platforms refer to web-based

training systems that provide accessible educational resources for livelihood development. These platforms enable women to acquire skills remotely through structured online courses and interactive learning modules.

Financial Sustainability is the ability of a program or organization to maintain stable funding and continue operations over time without relying on external aid. In this study, financial sustainability is a crucial factor in ensuring the long-term viability of livelihood training initiatives.

Livelihood Programs are structured initiatives designed to equip individuals with the skills necessary for income generation. These programs, often implemented by NGOs, focus on trades such as sewing, baking, and cosmetology to help women achieve financial independence.

Livelihood Training is the process of providing hands-on education in specific vocational skills. This training prepares women for employment or entrepreneurship by enhancing their capabilities in various industries.

Micro-entrepreneurship refers to the establishment and operation of small businesses with minimal capital investment. Women engaged

in micro-entrepreneurship often use skills acquired through livelihood training to start sustainable businesses.

[A] Non-Governmental Organization (NGO) is a non-profit organization that operates independently from the government to provide social services, including livelihood training and financial assistance. NGOs play a critical role in supporting marginalized communities.

Operational Efficiency is the ability of an organization to maximize productivity and minimize resource wastage in delivering its services. Improving operational efficiency in NGOs can enhance the effectiveness of training programs.

Sustainable Livelihood is the income-generating activities that provide long-term financial stability while minimizing environmental and social risks. Sustainable livelihood strategies enable women to achieve economic security without relying on external aid.

Chapter 2

RESEARCH METHODOLOGY

This chapter describes the study's research design, location, respondents, instruments, data collection procedures, data analysis, and ethical considerations in examining the efficacy and feasibility of the Livelihood Training and Resource Management Portal.

Research Design

The convergent parallel mixed-method research design is a well-established approach that combines quantitative and qualitative data to provide a comprehensive understanding of complex research problems. According to Creswell and Clark (2021), this design is particularly effective for addressing multifaceted issues, such as the development of web-based platforms, as it allows researchers to gather both numerical data and in-depth insights simultaneously. By integrating these two data types, researchers can identify areas of convergence, divergence, or complementarity, leading to more robust and actionable findings (Tashakkori et al., 2021). This approach has been widely adopted in technology development projects, where understanding user needs and

preferences is critical for creating effective and user-centered solutions (Smith et al., 2022).

In the context of NGO-led initiatives, qualitative research plays a vital role in capturing the lived experiences and perspectives of stakeholders. For example, Johnson and Lee (2023) emphasize the importance of structured interviews and content analysis for exploring the challenges and strategies of livelihood programs, particularly those targeting women. Their study highlights how qualitative data provides rich, contextual insights that inform the design of technology-driven solutions, ensuring they are aligned with the community's needs. Similarly, Garcia et al. (2024) argue that incorporating stakeholder feedback through qualitative methods is essential for addressing gaps and inefficiencies in resource management and training coordination.

On the quantitative side, usability metrics and user feedback surveys are critical for evaluating the effectiveness of web-based platforms. Brown et al. (2023) demonstrate how quantitative data, such as task completion rates and user satisfaction scores, help measure a platform's success in meeting its objectives.

Their findings underscore the importance of iterative development, where continuous feedback and refinement are integral to optimizing usability and functionality. This aligns with the Agile Scrum framework, which has been widely recognized for its effective management of iterative development processes (Martinez et al., 2025). By combining quantitative metrics with qualitative insights, developers can create platforms that are not only technically sound but also deeply responsive to user needs.

Recent studies have also highlighted the value of mixed-methods research in addressing gender-specific challenges in the development of technology. For instance, Smith et al. (2022) found that women, who often constitute the majority of participants in NGO-led livelihood programs, face unique barriers that require tailored solutions. By integrating quantitative data on program outcomes with qualitative insights from participants, researchers can design platforms that empower women and promote sustainable livelihoods. This approach is further supported by Garcia et al. (2024), who advocate for a user-centered design process prioritizing the needs of marginalized groups.

The convergent parallel mixed method research design offers a powerful framework for developing web-based platforms that address the complex needs of NGOs and their beneficiaries. By combining quantitative and qualitative data, researchers can gain a holistic understanding of user needs, preferences, and challenges, leading to more effective and sustainable solutions. This approach is supported by a growing body of literature, including studies by Creswell and Plano Clark (2021), Tashakkori et al. (2021), and Martinez et al. (2025), which highlight the importance of integrating multiple data sources into technology development projects.

The convergent parallel mixed-method research design was chosen for this study because it provides a comprehensive approach to understanding the challenges associated with developing a livelihood training and resource management portal for women's NGO. Given the complexity of the project, this method enables the simultaneous collection of quantitative data, such as usability metrics and user feedback, alongside qualitative insights from stakeholders, thereby ensuring a comprehensive analysis. Additionally, this approach aligns with the participatory nature of NGO-led

initiatives, where stakeholder involvement is crucial for designing user-centered solutions. The combination of numerical data and in-depth perspectives ensures that the platform is both functionally efficient to the real-world needs of women beneficiaries. This research design supports the development of an evidence-based, user-driven platform that effectively addresses the needs of Angat Nueva Ecija NGO.

Research Locale

Figure 2 below presents the locator map that highlights the area of the research site.

Figure 2

Research Locale



This study was conducted in Nueva Ecija, Philippines, focusing on the Angat Nueva Ecija NGO, which was located in the province. It was

chosen as the research site because of its active livelihood training programs for women and the NGO's role in providing skills development and resource support.

However, challenges in resource distribution and program management highlighted the need for a centralized digital system, which this study sought to address through the development of a Livelihood Training and Resource Management Portal.

Research Respondents

The selected respondents for this study included NGO staff, IT professionals, donors, women beneficiaries, and youth. These respondents played a crucial role in evaluating and assessing the effectiveness of the proposed web platform. The inclusion of various stakeholders ensured a comprehensive understanding of the system's usability and impact.

In terms of familiarity with technology, IT consultants possessed advanced technical knowledge, while NGO staff and donors had varying degrees of digital proficiency. Women beneficiaries, as the primary end-users, provided valuable insights regarding accessibility and usability. Ensuring that a

diverse set of respondents was involved aligned with the study's objective of creating an inclusive and effective platform.

Table 1 presents the total number of respondents classified according to their roles in the study.

Table 1. Distribution of the Respondents

Respondents	Frequency	Percentage
IT Consultants	30	48.39%
End-Users	32	51.61%
<i>Admin</i>	2	3.23%
<i>Donors</i>	7	11.29%
<i>Women</i>	12	19.35%
<i>Beneficiaries</i>		
<i>Youth</i>	11	17.74%
Total	62	100%

The table above represents a sample drawn from the broader population relevant to the study. It is important to note that the table does not claim to represent the entire demographic of each respondent category. Instead, the selection process follows established methodologies aligned with the study's objectives and logistical considerations.

Ensuring the selected respondents adequately represent the broader population is crucial for maintaining the validity and credibility of the research findings. This

involves considering demographic diversity, geographic distribution, and alignment with the research objectives to capture the study's scope effectively. The inclusion of IT professionals ensures technical accuracy, while feedback from NGO staff, donors, women beneficiaries, and youth contributes to a user-centered evaluation of the proposed web platform.

It is essential to note that Admin, Youth, Donors, and Women Beneficiaries are subcategories under the broader category of Candidate End Users. This classification reflects the diverse stakeholder groups that are integral to the study's focus on user-centered design and evaluation. By categorizing these groups under Candidate End Users, the study ensures a comprehensive representation of the end-user perspective, which is critical for the successful implementation and adoption of the proposed web platform. This approach highlights the importance of gathering diverse insights from various segments of the target population to inform the development process effectively.

Sample and Sampling Procedures

The researchers employed a purposive sampling technique to select the respondents for this study, ensuring that the chosen individuals

had direct relevance to the research objectives. The respondents consisted of five distinct groups: NGO staff, IT professionals, donors, women beneficiaries, and youth.

The first group of respondents included NGO staff involved in the design, implementation, and management of livelihood training programs. They were selected based on the following criteria: (a) actively engaged in the NGO's training and resource management initiatives; (b) possessing direct experience in program coordination, training facilitation, or resource distribution.

The second group of respondents comprised IT consultants who provided insights into the technical feasibility and efficiency of the system. Their selection was based on the following criteria: (a) having expertise in web development, user experience design, or system evaluation; (b) having at least two years of work experience in the IT field.

The third group of respondents included donors who provided financial contributions to the NGO's initiatives. They were selected based on the following criteria: (a) having previously donated or committed to supporting NGO programs; (b) having an interest in tracking donation allocation and program impact through digital

systems.

The fourth group of respondents consisted of women beneficiaries enrolled in the NGO's livelihood training programs. They were chosen based on the following criteria: (a) Actively participating in at least one of the organization's livelihood programs; (b) Having prior experience engaging with training resources, whether online or in person.

The fifth group of respondents included youth who were beneficiaries or participants in the NGO's programs. They were selected based on the following criteria: (a) actively participating in at least one of the NGO's youth-focused programs, such as skill development, education, or community engagement initiatives; (b) possessing basic digital literacy skills to interact with the proposed web platform effectively.

By employing purposive sampling, the study ensured that the selected respondents were directly relevant to the research objectives. This method allowed for a comprehensive assessment of the web platform's usability, effectiveness, and overall impact in facilitating livelihood training and resource management for women beneficiaries.

Research Instruments

To evaluate the Livelihood Training and Resource Management Platform effectively, a set of research instruments was carefully designed and employed. These instruments were crucial for gathering both quantitative and qualitative data from various stakeholders, including NGO administrators, women beneficiaries, youth volunteers, donors, and IT experts. The five main research instruments used are:

Survey Questionnaires

The survey was structured based on recognized ISO/IEC 25010 software quality standards and principles of usability evaluation. This model is recognized for evaluating key software characteristics, including security, efficiency, and performance. This instrument was self-made, although it was derived from established quality standards and best practices in software evaluation. The questionnaire included sections such as demographic information, system usability, security, efficiency, data transparency, and more, tailored to measure different system characteristics.

Interview Guide

The interview guide was developed using principles of thematic analysis, a qualitative research approach that helps uncover common themes and insights from open-ended responses. This instrument was also self-made, drawing from qualitative research methodologies, and focused on extracting deeper insights from participants about their experience with the system. The interviews asked questions about user experience, challenges faced, feature preferences, and recommendations for improving the platform. These topics helped to understand the subjective experiences of users.

System Performance Evaluation

This checklist was adapted from ISO/IEC 25010:2023, the software quality standards, which are globally recognized guidelines for assessing software quality in various aspects, including functionality, performance, and security. This instrument was developed from existing software evaluation frameworks, ensuring that the system meets international quality standards. The checklist evaluated specific characteristics like functional suitability, performance efficiency, compatibility, usability, and reliability of

the system. This tool was used by IT consultants and system testers, who assessed the system's technical performance.

User Preference Analysis Tool

This tool combined quantitative survey questions with qualitative themes derived from the interviews. It was designed to understand user preferences in relation to the platform's features and functionality. The tool was adopted from established user experience evaluation models that focus on how users interact with digital platforms and their needs. It included questions on feature preferences, accessibility, security concerns, and data management, all of which are critical to improving the platform's usability and user experience.

The researchers employed a combination of qualitative and quantitative research methods to systematically gather information, analyze, and interpret the data in accordance with the research objectives. A survey tool was used to identify problems with the manual operation and process in the NGO, focusing on four major dimensions: time consumption, data security, record loss, and transparency.

The researchers utilized Google Forms,

questionnaires, and open-ended questions as primary data collection tools to develop and assess the system. Inquiries about implemented recommendations to improve procedures and project oversight were also included.

For the questionnaire component, data were gathered through the use of Google Forms, questionnaires, and interviews, not only to gather closed-ended responses but also to obtain participants' insights about the system they will use. Respondents were allowed to evaluate the platform following a demonstration of how the system operates.

Data Gathering Procedure

The following phases formed the basis for the procedures of this study.

Phase I: Planning

The researchers began the study by carefully formulating its objectives, ensuring that each goal aligned with the overall purpose of the project. They proceeded to define the scope of the research, clearly outlining its boundaries to maintain focus and relevance. In preparation for data gathering, they thoughtfully selected the most appropriate research methodology to guide the study's

direction. As part of their groundwork, the researchers also identified the key stakeholders within the target organization, recognizing their essential roles in providing valuable insights through planned interviews and surveys. This initial phase made a solid foundation for the effective execution of the research process.

Phase II: Instrument Preparation and Validation

The research instruments underwent a development and validation process. The questionnaire incorporated validated measurement scales from previous NGO efficiency studies, combining quantitative Likert-scale items with qualitative open-ended questions. Upon approval, these instruments were subjected to expert validation by the research adviser, who assessed content validity and construct alignment. Final revisions were implemented based on this comprehensive validation process before deploying the digital survey via Google Forms, printed questionnaires, and preparing semi-structured interview protocols for in-depth follow-up discussions.

Phase III: Pre-Interview with the NGO Director

After finalizing the survey questionnaire and open-ended interview questions, the researchers proceeded to the office of the NGO director to initiate a preliminary consultation and present the instruments. Their objective was to identify suitable participants for the study, including beneficiaries who met the criteria for the interviews. During the discussion, they carefully outlined their research intentions and sought the director's insights on which individuals within the organization could meaningfully contribute to the data-gathering process. They also inquired about the most appropriate schedule for conducting the surveys and interviews in person, ensuring that the selected participants would be available and that the process would align with the organization's operations. This meeting served as a crucial step in coordinating the research activities and establishing collaboration with the NGO.

Phase IV: Manual Operation Process Evaluation

Once the participants' schedules had been finalized, the researchers sought and obtained permission and approval from their adviser to proceed with distributing the research

instruments. The respondents included key stakeholders of the Angat Nueva Ecija NGO—specifically the administrator, youth volunteers, women beneficiaries, and donors. To ensure accessibility and convenience, the questionnaire was made available through Google Forms, which was then distributed to youth volunteers and donors via email and Messenger. This strategic approach facilitated efficient data collection while accommodating the respondents' availability and communication preferences.

Phase V: Interview with the Beneficiaries

The researchers personally delivered the printed surveys to the beneficiaries in Rizal, Nueva Ecija, ensuring that each respondent received a clear explanation of the project's purpose and objectives. This direct approach not only fostered better understanding but also encouraged more accurate and thoughtful responses, as the beneficiaries were given the opportunity to ask questions and clarify any concerns before completing the survey.

Phase VI: Data Analysis and Interpretation

The researchers conducted a comprehensive analysis of the data gathered from

questionnaires and interviews to evaluate the effectiveness of the Livelihood Training and Resource Management Platform. Quantitative data collected through Google Forms and printed surveys were analyzed using statistical methods to identify trends related to time efficiency, data security, record loss, and transparency. Simultaneously, qualitative responses from open-ended interviews were examined through thematic analysis to extract key insights and recommendations for improvement. Follow-up validations were carried out to ensure the reliability and accuracy of the findings. The interpreted results provided a clear understanding of the platform's overall impact and guided the implementation of necessary refinements to enhance its usability, security, and operational efficiency.

Data Analysis Techniques

This study utilized a Convergent Parallel Design, wherein quantitative and qualitative data were collected simultaneously, analyzed independently, and then integrated to provide a comprehensive and cohesive interpretation of the findings.

1. To analyze time consumption, data security, record loss, and transparency

under the first research question, the researchers employed descriptive statistical methods, including frequency distribution and weighted mean, to summarize the responses from NGO administrators, youth volunteers, beneficiaries, and donors. These statistical techniques provided a quantitative overview of the system's performance in the aforementioned areas. Furthermore, IT experts conducted an assessment of the system's quality based on the ISO/IEC 25010 software standards, evaluating critical attributes such as functional suitability, performance efficiency, compatibility, and usability. This rigorous and comprehensive evaluation facilitated a thorough understanding of the system's overall effectiveness and provided valuable insights for potential improvements.

The scoring rubric presented in Table 2 will serve as the basis for accurately interpreting the results.

Table 2*Scoring Rubric for Level of Agreement*

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	A weighted mean range of 3.26 to 4.00 is classified as "Strongly Agree" in the qualitative description. This signifies a high level of agreement among respondents, reflecting strong support, approval, or endorsement of the given statement, concept, or idea.
2.51 - 3.25	Agree	A weighted mean range of 2.51 to 3.25 is classified as "Agree," indicating a positive but moderate level of agreement with the statement. While

Range	Qualitative Remarks	Verbal Description
		respondents generally support the idea, their approval is not as strong as that of those in the "Strongly Agree" category.
1.76 - 2.50	Disagree	A weighted mean range of 1.76 to 2.50 is classified as "Disagree," suggesting that respondents hold an unfavorable perception of the statement. This suggests a lack of support or agreement with the presented idea, concept, or question.
1.00 - 1.75	Strongly Disagree	A weighted mean range of 1.00 to 1.75 is classified as "Strongly Disagree," indicating an

Range	Qualitative Remarks	Verbal Description
		unfavorable perception of the statement. This reflects a clear rejection or strong disapproval of the idea, concept, or question presented.

Table 2 presents the scoring rubric for the level of agreement, outlining the range of values, qualitative remarks, and corresponding verbal descriptions within the four-point scale.

2. To address the second statement of the problem, the researchers employed a mixed-method approach, integrating both quantitative surveys and qualitative face-to-face interviews with key stakeholders, including NGO administrators, women beneficiaries, volunteers, and donors. The analysis focused on critical aspects, including user interface design, accessibility, security, data management, collaboration, and the platform's

integration with existing NGO workflows. To interpret the collected data, a frequency distribution was used for the quantitative responses. At the same time, a thematic analysis was applied to the qualitative data, enabling the identification of high-priority features based on stakeholder preferences. The insights gained from this comprehensive analysis served as a foundational basis for the design and development of a web platform aimed at enhancing the efficiency, usability, and overall effectiveness of livelihood and resource management for NGOs.

3. For the third research question, the system was developed using the Agile Scrum Model, which emphasized adaptability and continuous improvement throughout the development process. The project began with the gathering of project objectives and stakeholder requirements, followed by the creation of a product backlog that outlined the features and tasks to be accomplished. These tasks were then prioritized and assigned to iterative sprints, ensuring incremental progress.

The study used the Agile Scrum model

as a guide for the system's development.
The model's phases were as follows:

Initiation. The project objectives and stakeholder requirements were gathered, and a product backlog was created.

Planning and Estimation. Development tasks were planned, prioritized, and assigned to iterative sprints based on stakeholder feedback and requirements.

Implementation. The development team worked on core system features, including user authentication, data security, and reporting functionalities, with each feature being developed, tested, and refined in short cycles.

Reviewing. Sessions were held to assess progress, and stakeholder feedback was integrated to ensure continuous improvement and alignment with project goals.

Releasing. The system was deployed, accompanied by user training and comprehensive documentation. Post-release maintenance followed to address any emerging issues or improvements. This iterative approach ensured efficiency, adaptability, and adherence to ISO/IEC 25010 software quality standards, facilitating a responsive

and user-centered development process.

4. To ensure that the project met industry standards for software quality, IT consultants conducted a comprehensive evaluation based on the ISO/IEC 25010:2023 product quality model. This assessment focused on key software characteristics, ensuring that the system aligned with established benchmarks for functionality, performance, compatibility, usability, reliability, security, maintainability, and portability.

Functional suitability was measured by how well the system met the specified requirements. Performance efficiency was assessed by evaluating the system's speed, responsiveness, and stability under various conditions. Compatibility evaluated the system's ability to operate in different environments without issues. Usability was evaluated to measure the ease of navigation, accessibility, and overall user experience.

The researchers emphasized the use of IT consultants' evaluation of the system's quality, based on the software characteristics guidelines, to interpret

the results. Each factor was assessed using a four-point scale, and the weighted mean was calculated for each category. The evaluation results provided valuable insights into the system's strengths and areas for improvement.

Table 3 shows the scoring rubric for Functional Suitability.

Table 3.

Scoring Rubric for Functional Suitability

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Highly suitable	The system completely fulfils all functional needs with outstanding precision and efficiency.
2.51 - 3.25	Suitable	The system satisfactorily addresses most functional needs with reliable precision and efficiency.
1.76 - 2.50	Unsuitable	The system meets only some functional needs, with

Range	Qualitative Remarks	Verbal Description
1.00 - 1.75	Highly unsuitable	limited precision and efficiency. The system fails to fulfil functional needs, showing significant deficiencies in precision and efficiency.

As shown in Table 3 above, the scoring rubric will be employed to assess the system's functionality under specific conditions.

Table 4 shows the scoring rubric for Performance Efficiency.

Table 4

Scoring Rubric for Performance Efficiency

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Excellent	The system's performance delivers top- tier performance with zero disruptions across all scenarios.
2.51 - 3.25	Good	The system's performance

Range	Qualitative Remarks	Verbal Description
		consistently delivers strong results with minimal disruptions in typical scenarios.
1.76 - 2.50	Fair	The system's performance offers moderate performance with noticeable disruptions, requiring improvement.
1.00 - 1.75	Poor	The system exhibits weak performance with frequent disruptions, hindering effectiveness.

Table 4 is used to evaluate the system's ability to perform its intended function within a specified timeframe.

Table 5 shows the scoring rubric for Compatibility.

Table 5*Scoring Rubric for Compatibility*

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Highly compatible	The system or software integrates effortlessly with other systems or hardware without issues.
2.51 - 3.25	Compatible	The system or software operates well with minor tweaks to the system or hardware.
1.76 - 2.50	Somewhat compatible	The system or software can function with substantial modifications to systems or hardware.
1.00 - 1.75	Not compatible	The system or software is inoperable with existing systems or hardware and requires extensive repair.

Table 5 above is the rubric that will be used to evaluate the system's ability.

Table 6 provides the scoring rubric for Usability.

Table 6

Scoring Rubric for Usability

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Excellent	The system is exceptionally user-friendly and efficient, delivering seamless task completion and satisfaction.
2.51 - 3.25	Good	The system is mostly intuitive, with minor flaws, and consistently maintains solid user satisfaction.
1.76 - 2.50	Fair	The system is hindered by noticeable usability flaws, leading to user frustration. The system may need to be

Range	Qualitative Remarks	Verbal Description
1.00 - 1.75	Poor	better designed or require significant effort to improve. The system is severely flawed and impractical, blocking effective task execution. The system may be unusable due to significant errors.

Table 6 above depicts the visual rubric that will be used to evaluate the system's usability, focusing on its ability to facilitate communication and data transfer between the user and the system through the user interface.

In Table 7, the scoring rubric for Reliability is shown.

Table 7*Scoring Rubric for Reliability*

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Highly Reliable	A highly reliable system dependently produces precise outcomes with no risk of failure.
2.51 - 3.25	Moderate Reliable	A moderately reliable system usually works, but is mostly consistent, with minor errors and a slight risk of failure.
1.76 - 2.50	Slightly Reliable	A slightly reliable system is frequently error-prone, with an elevated failure risk that requires attention. It is likely to fail, and its overall performance could be better.

Range	Qualitative Remarks	Verbal Description
1.00 - 1.75	Non-Reliable	An unreliable system is regularly inaccurate and undependable, with a high risk of failure.

As displayed in Table 7 above, the rubric for system reliability will be applied to assess its capability to perform tasks without disruptions or failures.

In addition to analyzing the numerical data, the researchers carefully examined the qualitative responses from open-ended questions and interviews using content analysis. This approach allowed them to uncover recurring themes and extract key insights. The analysis primarily focused on the perspectives of various stakeholders, with particular attention given to the usability of the system for beneficiaries, the efficiency and reliability for donors, the feasibility and compliance for administrators, and the security and functionality for IT experts. By examining these areas, the researchers aimed to gain a comprehensive understanding of the

system's impact and its alignment with the needs and expectations of all parties involved.

Following the separate analyses of the quantitative and qualitative data, the researchers integrated the findings to present a thorough evaluation of the Livelihood Training and Resource Management Platform. By employing a mixed-methods approach, they provided a comprehensive assessment that captured not only the measurable outcomes derived from the quantitative data but also the in-depth insights and perspectives gathered from stakeholders through qualitative responses. This combined analysis offered a holistic view of the platform's effectiveness, ensuring that both statistical evidence and personal experiences were considered in the evaluation.

Ethical Considerations

Throughout the study, the researchers adhered to strict ethical guidelines to safeguard the rights, privacy, and well-being of all participants. One of the foundational principles they followed was informed consent.

Prior to their involvement, participants were provided with clear, comprehensive information regarding the study's purpose, their role within it, and any potential risks. This transparency enabled participants to make informed decisions, fostering trust in the research process. To ensure the protection of personal data, the researchers implemented robust security measures, including the safe storage and encryption of sensitive information, such as donor details, NGO staff, and women's beneficiaries. Access to this data was restricted to authorized personnel only, maintaining confidentiality and strengthening the trust placed in the study.

In line with ethical best practices, transparency remained a core focus throughout the research. The researchers maintained open communication about the methods used for data collection and analysis, clearly explaining how participants' information would be utilized. This transparency enhanced the credibility of the study and provided participants with a sense of assurance regarding the fairness of the research process.

Fairness was another fundamental value embedded in the study. The web platform, a central element of the research, was designed

with accessibility in mind to ensure that all users—women beneficiaries, NGO staff, and donors—had equal opportunities to engage with and benefit from its features. By preventing any form of discrimination, the researchers promoted inclusivity, reinforcing a commitment to fairness for all stakeholders involved.

Lastly, the researchers acknowledged the broader societal responsibility associated with the platform's purpose. To support livelihood training and resource management, the study carefully considered the potential positive impact on marginalized communities. In its ethical engagement with these groups, the researchers aimed to promote sustainable development, empowerment, and long-term positive change, thereby contributing to the overall well-being of the communities they served.

Chapter 3

RESULTS AND DISCUSSION

This chapter presents the design and development process of Project Hera, including the results, interpretation, and analysis of the data gathered from the respondents. The discussion focuses on how the findings address the study's objectives and support the overall development and functionality of the system.

1. Challenges and Pain Points for the Manual Process of Record-Keeping in Angat, Nueva Ecija, NGO.

This section presents the challenges experienced by the Angat Nueva Ecija NGO in its manual record-keeping process. The purpose of this evaluation is to identify the challenges and pain points that the organization experiences in its manual record-keeping process, which affect efficiency, accuracy, and data accessibility. These issues serve as the basis for developing Project Hera.

1.1 Time Consumption

The time consumptions ensure that the respondents' perceptions are accurately reflected in identifying how long it takes to complete record-keeping tasks.

Table 8 presents the results of the data collected from respondents on how the manual system affects the timeliness and efficiency of recording, storing, and retrieving data within the organization.

Table 8

Results of the Challenges and Pain Points for the Manual Record-Keeping Process in Angat, Nueva Ecija, an NGO, in Terms of Time Consumption.

Item Statements	Weighted Mean	Verbal Description
The manual process is too time-consuming.	3.00	Agree
Participants experience delays in training registration.	2.66	Agree
Generating reports manually is time-consuming.	3.28	Strongly Agree
Resource tracking is inefficient.	2.50	Agree
Updating information requires excessive effort.	2.69	Agree
Grand Mean	2.83	Agree

As presented in Table 8, the respondents generally agreed that the manual process of

record-keeping in Angat, Nueva Ecija, NGO is time-consuming, with a grand mean of 2.83, which is verbally described as 'Agree'. Among the given statements, "Generating reports manually is time-consuming" received the highest weighted mean of 3.28, interpreted as *Strongly Agree*. This indicates that manual report generation is the most significant time-related challenge encountered by the respondents. Meanwhile, "Resource tracking is inefficient" obtained the lowest weighted mean of 2.50, although it is still interpreted as *Agree*, implying that inefficiencies also exist in monitoring and managing resources. The results reveal that the manual processes cause delays and require considerable effort to complete daily record-keeping tasks.

The results show that the current manual process slows down the organization's work. The amount of time spent on repetitive processes, such as report generation, updating information, and resource tracking, highlights the need for automation. Implementing Project Hera would significantly reduce time consumption and streamline the organization's operations. These challenges can be addressed through digital transformation by using multi-criteria algorithms to optimize workstations

(Turk et al., 2022, p. 112). Many industries are actively seeking ways to digitize and automate manual processes to reduce errors and increase efficiency (Adrita et al., 2021, p. 45).

1.2 Data Security

Data security ensures that private and sensitive data are secure and protected against unauthorized access, loss, or misuse. It is the process of protecting data from loss, alteration, or unauthorized access during its entire lifecycle.

Table 9

Results of the Challenges and Pain Points for the Manual Record-Keeping Process in Angat, Nueva Ecija, an NGO, in Terms of Data Security.

Item Statements	Weighted Mean	Verbal Description
Records are safe from unauthorized access.	1.94	Disagree
Important documents are not at risk of being misplaced.	2.03	Disagree
Financial records are protected.	2.55	Agree
There is a backup system for important files.	2.16	Disagree
Data	1.66	Disagree

Item Statements	Weighted Mean	Verbal Description
confidentiality is maintained.		
Grand Mean	2.07	Disagree

As shown in Table 9, the respondents generally disagreed that the current manual process ensures data security, with a grand mean of 2.07, indicating a "Disagree" response. Among the statements, "Financial records are protected" received the highest weighted mean of 2.55, which indicates that respondents somewhat agree that financial data are kept safe. However, most respondents disagreed with statements such as "Records are safe from unauthorized access" (1.94), "Important documents are not at risk of being misplaced" (2.03), and "Data confidentiality is maintained" (1.66). These findings show that respondents believe the existing manual process lacks reliable measures to protect important records and maintain privacy.

The results suggest that the manual system employed by Angat Nueva Ecija NGO has inadequate data security, which increases the risk of unauthorized access, data loss, or exposure of sensitive information. Without secure backup and confidentiality measures, important files may be easily misplaced or compromised. This

highlights the need for a secure, automated record-keeping system that includes user authentication, data encryption, and regular backup features to ensure the protection and confidentiality of the organization's records. According to Deepa and Dhiipan (2022), data security tools and technology increase transparency into an organization's data and its usage. These tools can protect data by employing methods including encryption and data masking of personally identifiable information. The evidence demonstrates the multifaceted importance of data security across various sectors. Jevtić et al. (2023) reveal that effective data protection is key to maintaining client trust, reducing financial losses, and preserving organizational reputation, while Pant et al. (2023) emphasize the necessity of safeguarding personal information against unauthorized access.

1.3 Record Loss

This presents the respondents' evaluation of the issue of record loss within the manual record-keeping process. This factor measures the frequency and impact of misplaced, damaged, or missing records that the organization encounters.

Table 10

Results of the Challenges and Pain Points for the Manual Process of Record-Keeping in Angat, Nueva Ecija, NGO, in terms of Record Loss.

Item Statements	Weighted Mean	Verbal Description
Training records have been lost or misplaced.	3.59	Strongly Agree
Important financial records have been damaged.	3.50	Strongly Agree
There are no duplicate copies of critical files.	3.59	Strongly Agree
Resource inventory tracking is unreliable.	3.47	Strongly Agree
The NGO has experienced a loss of essential documents.	3.63	Strongly Agree
Grand Mean	3.56	Strongly Agree

As presented in Table 10, the respondents strongly agreed that the manual record-keeping process in Angat Nueva Ecija NGO often leads to record loss, with a grand mean of 3.56 described as *Strongly Agree*. The statement "The NGO has experienced loss of essential documents" received the highest weighted mean of 3.63,

showing that this issue is a major concern among respondents. Similarly, high ratings were given to statements such as "Training records have been lost or misplaced" (3.59) and "There are no duplicate copies for critical files" (3.59). These results indicate that record loss, misplacement, and lack of backup copies are common problems that affect the organization's ability to maintain accurate and reliable data.

The findings suggest that inadequate records management in non-profit and public organizations poses significant risks, including data loss, lack of accountability, and operational inefficiency. Glassford (2024) argues that many NGOs operate on tight budgets, relying on volunteers to manage archives, and often lose historical records during office relocations, threatening their institutional memory. Losing or damaging important files can disrupt activities, delay services, and reduce accountability. This highlights the need for more strategic and integrated approaches to managing records across NGO projects, stressing that fragmented record systems and inadequate infrastructure contribute to long-term data loss (Gomes et al., 2024). Through a digital system, the organization can protect its essential documents, prevent data loss, and

ensure that records remain accurate and accessible when needed.

1.4 Transparency

This shows the respondents' assessment of transparency issues experienced in the manual record-keeping process. Transparency refers to the openness and accessibility of information for authorized personnel and stakeholders.

Table 11

Results of the Challenges and Pain Points for the Manual Process of the Record-Keeping in Angat, Nueva Ecija, NGO in terms of Transparency.

Item Statements	Weighted Mean	Verbal Description
The manual system ensures fairness in resource allocation.	3.16	Agree
Financial transactions are transparent and traceable.	2.18	Disagree
Members feel confident about data integrity.	2.22	Disagree
There is accountability in training processes.	2.20	Disagree

Item Statements	Weighted Mean	Verbal Description
Decision-making processes are well-documented.	2.25	Disagree
Grand Mean	2.40	Disagree

As shown in Table 11, the respondents disagreed that the manual record-keeping system of Angat Nueva Ecija NGO provides sufficient transparency, as reflected in the grand mean of 2.20. This indicates that the end-users perceive the current processes as lacking clarity, openness, and reliable documentation. The item *"Decision-making processes are well-documented"* received the highest weighted mean of 2.25, yet it still falls under the "Disagree" category, suggesting that even the strongest aspect of transparency remains inadequate in meeting user expectations. This implies that members often struggle to understand how decisions are made or who is responsible for them, highlighting gaps in documentation and organizational communication.

The respondents also disagreed that there is accountability in training processes ($\mu = 2.20$) and that data integrity inspires confidence among members ($\mu = 2.22$). These perceptions likely stem from inconsistencies in how records are maintained across various

activities. Manual documentation—such as paper files, scattered logbooks, and separately stored folders—reduces uniformity and complicates the verification of attendance, training outcomes, and accurate data encoding. As a result, members may question whether the information they receive is complete, consistent, and up-to-date.

In addition, respondents disagreed that financial transactions are transparent and traceable ($\mu = 2.18$) and that the system ensures fairness in resource allocation ($\mu = 2.16$). This dissatisfaction reflects common constraints of manual financial recording, including delayed updates, difficulty retrieving logs, and limited visibility for stakeholders. Without a centralized and easily accessible database, members cannot confirm how funds are allocated or whether processes are performed consistently and fairly. Such a lack of visibility weakens trust, especially within NGOs, where accountability is a fundamental expectation.

These findings align with the existing literature, which emphasizes the limitations of manual record-keeping systems. NGOs often struggle to maintain transparency and accountability due to fragmented or inconsistent documentation practices (Dery et.

al., 2023, p. 12). Similarly, Ortega-Rodríguez, Licerán-Gutiérrez, and Moreno-Albarracín (2020) highlight that the lack of systematic and accessible records reduces stakeholder confidence in organizational processes. Wanyama, Ndiku, and Kamau (2023) assert that digitalization of records improves accessibility, accuracy, and traceability, addressing many of the deficiencies identified in manual systems. Traditional paper-based documentation is vulnerable to errors and tampering, which limits accountability and trust. In contrast, Mulyana, Holiawati, and Suropto (2024) suggest that integrating technology into governance and accountability structures enhances transparency and overall NGO performance.

The results underscore the urgent need for a digital record-keeping system within the Angat Nueva Ecija NGO. By automating documentation, enabling real-time tracking, and providing centralized access to records, a digital system would improve transparency, strengthen accountability, and enhance stakeholder confidence. Implementing such a system would ensure fair resource allocation, accurate financial tracking, and reliable documentation of decisions and training processes, addressing

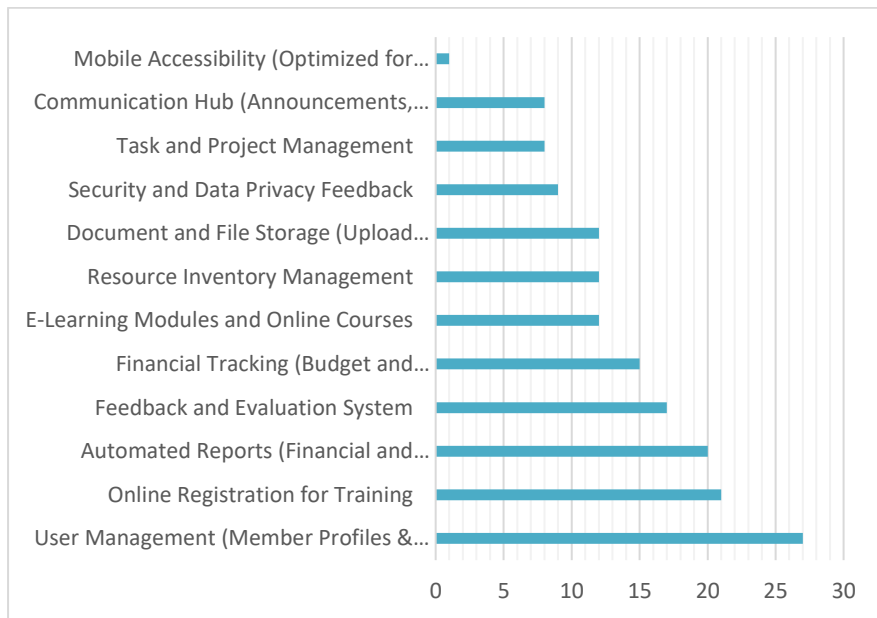
the gaps identified by end-users in the manual process.

2. Key Features Preferred by Users for the Development of a Web Platform to Improve the Livelihood and Resource Management of the Angat Nueva Ecija NGO

This section presents the data gathered from respondents regarding the features they would like to include in the proposed web platform. Identifying these preferences is essential to ensure that the system being developed aligns with the users' expectations and functional requirements. The data were collected through a survey that allowed participants to select multiple features they deemed important for improving accessibility, management, and efficiency in organizational operations.

Figure 3

Results of the Key Features Preferred by Users for the Development of a Web Platform to Improve the Livelihood and Resource Management of the Angat Nueva Ecija NGO.



As shown in Figure 3, User Management is the most preferred feature, selected by 27 of the 32 respondents (84.4%). This high preference underscores the importance of an organized system for managing user accounts, membership profiles, and role-based access control. Such functionality promotes accountability and proper governance, which is particularly crucial for NGOs with multiple stakeholders and varying permission needs (Vogelsang, Packmohr, & Brink, 2021; Radu, Jonathan, & Perjons, 2025).

These studies highlight that clear role definitions and access control structures are essential to overcome barriers to digital transformation in nonprofit organizations.

Mobile Accessibility is the second most favored feature (65.6%), reflecting user demand for flexibility and convenience. In resource-limited or geographically dispersed environments, mobile-friendly platforms are more likely to be adopted and used effectively (Jong & Ganzaroli, 2024). Users value the ability to access the platform from their mobile devices at any time, which enhances engagement, responsiveness, and real-time participation. Other highly prioritized features include Financial Tracking (62.5%) and Automated Reports (53.1%). These speak to a strong desire for transparency and administrative efficiency. Real-time financial tracking, coupled with the automatic generation of reports, can help reduce manual errors, improve accountability, and provide valuable insights for informed decision-making (Holzer, Holzer, & Smith, 2024). These features align with research demonstrating that the digitization of financial processes in NGOs strengthens stakeholder trust and supports data-driven governance.

Moderately preferred features include Online Training Registration (46.9%), E-Learning Modules and Online Courses (37.5%), Feedback and Evaluation System (37.5%), and Document and File Storage (37.5%). These functions support continuous learning, capacity-building, and information management within the NGO. They allow members to sign up for training, access learning materials, provide feedback, and securely store important documents – all of which are essential for a dynamic, mission-driven organization (Radu, Jonathan, & Perjons, 2024; Jong & Ganzaroli, 2024). These studies emphasize that digital tools supporting education and training improve efficiency and organizational learning outcomes in nonprofit settings. Resource Inventory Management (28.1%), Communication Hub (25%), and Task and Project Management (25%) received lower levels of preference. Although these are useful modules, users may perceive them as less critical than features tied directly to user control, financial oversight, and learning. Finally, Security and Data Privacy Feedback was the least preferred (3.1%), possibly because respondents assumed that modern web platforms already integrated baseline security or may not have fully understood the need for explicit

feedback loops on data privacy.

The preferences captured in Figure 3 suggest that users place the greatest value on user management, mobile access, financial transparency, and automated reporting, indicating a strong demand for features that enhance control, trust, and operational efficiency. These priorities are consistent with recent literature on nonprofit digital transformation, which emphasizes that role-based access, mobile usability, financial automation, and training support are crucial components for effective technology adoption in NGOs (Vogelsang et al., 2024).

The result indicated that the proposed web platform should prioritize features that enhance user management, mobile accessibility, financial transparency, and automated reporting, as these directly address the operational needs and preferences of end-users. Implementing these features can significantly improve organizational efficiency, accountability, and decision-making within the Angat Nueva Ecija NGO. A platform with robust user management ensures clear roles and access control, thereby reducing errors and data misuse, while mobile accessibility facilitates real-time engagement and flexibility for

geographically dispersed members (Vogelsang, Packmohr, & Brink, 2021; Jong & Ganzaroli, 2024), automated financial tracking and reporting increase transparency, strengthen stakeholder trust, and minimize administrative workload (Holzer, Holzer, & Smith, 2024). Incorporating online training modules, feedback systems, and document storage supports continuous learning and knowledge sharing, which are essential for building the capacity of both staff and beneficiaries (Radu, Jonathan, & Perjons, 2024; Jong & Ganzaroli, 2024). These findings are consistent with the literature on digital transformation in nonprofit organizations, which emphasizes that integrating user-centered, accessible, and transparent technological solutions enhances organizational governance, operational efficiency, and stakeholder trust (Radu et. al., 2025; Vogelsang et. al., 2021). Overall, these insights provide a strategic guide for developers to design a user-centered, efficient system that aligns with the NGO's mission of effective resource and livelihood management.

3. Design and development of the Livelihood and Resource Management System of the Angat Nueva Ecija NGO

The development of the Livelihood and Resource Management System by the Angat Nueva Ecija NGO underwent four phases of the Agile Scrum Method, as illustrated in the figures below.

3.1 Planning and Requirement Gathering

The Agile Scrum methodology emphasizes flexibility and collaboration, allowing the project to adapt to changes throughout its development. In the context of the Livelihood and Resource Management System of the Angat Nueva Ecija NGO, the researchers began the study by carefully formulating its objectives, ensuring that each goal aligned with the overall purpose of the project. The researchers defined the scope of the research, clearly outlining its boundaries to maintain focus and relevance. In preparation for data gathering, they thoughtfully selected the most appropriate research methodology to guide the study's direction. These activities aimed to assess the existing challenges in managing livelihood programs, financial resources, and member records. After the assessment, the researchers

proceeded to plan the system's design and development by selecting appropriate tools and technologies, including the programming language, database, and development framework. This initial phase made a solid foundation for the effective execution of the research process.

To further align the project with the principles of Agile, the researchers conducted requirement gathering through user stories, which allowed stakeholders to express their specific needs and expectations for the system. By adopting this approach, the researchers ensured that the system design would be user-centered, practical, and responsive to the organization's daily operations. According to Schwaber and Sutherland (2021), Agile Scrum facilitates iterative development and continuous stakeholder engagement, thereby enhancing system adaptability and ensuring alignment with organizational goals. Scrum practices, such as user stories and sprint planning, promote active collaboration between developers and end-users, resulting in systems that better meet user requirements and improve usability (Dingsøyr et. al., 2022). Applying Agile in nonprofit or resource-constrained settings enhances project responsiveness, minimizes development risks, and strengthens

user satisfaction by incorporating feedback throughout the development lifecycle (Misra et. al., 2023).

Table 12

Sample User Stories for the Development of the Livelihood and Resource Management of the Angat Nueva Ecija NGO.

User Story	
As a donor, I want to create a profile so that I can manage my donations and track the projects I support.	As a donor, I want to select a specific NGO and project to donate to so that I know exactly where my contribution is going.
As a donor, I want to choose between in-kind and monetary donations so that I can give according to my capacity.	As a donor, I want to select from predefined donation amounts (e.g., ₱500, ₱ 1,000) or enter a custom amount, so I don't feel pressured to give more than I can afford.
As a donor, I want the system to notify me of my donation history so I can keep track over time.	As a donor, I would like to receive updates or confirmation after my donation is processed, so I know it was successful.
As an admin, I want to generate reports by project or donor so I can present these to funders, partners, or auditors.	As an admin, I want to archive failed or inactive projects so we can focus on successful, ongoing initiatives.
As a beneficiary, I	As a beneficiary, I want to

User Story	
want to create my own account so that I can track my progress and access additional resources provided by the organization.	join one program at a time so that I can focus on fully participating and completing it before moving to another.
As a beneficiary, I would like the program to suggest external training or job opportunities after I complete the program, so that I can continue developing my skills and advancing my career.	As a user, I want to see clear descriptions and photos of each project so I can understand where my help is needed most.

3.2 Design and Prototyping

Once the researchers collected the necessary requirements for developing the application, they transitioned to the subsequent phase of the Agile Scrum Method: the design and prototyping phase. This phase aimed to present diagrams that represent the different components of the system and their interactions. The diagrams included in this section are the Use Case Diagram, Context Diagram, Data Flow Diagram, Entity-Relationship Diagram, Database Normalization, Class Diagram, and Data Dictionary. Iterative prototyping in Agile enables teams to validate system designs early

with stakeholders, thereby reducing errors, enhancing usability, and ensuring that the final product meets user expectations (Ambler et al., 2021). Incorporating structured diagrams and prototypes within Agile frameworks improves communication between developers and end-users, fosters a shared understanding of the system, and increases overall project success in software development projects (Fernandes, Mendes, & Silva, 2023).

Use Case Diagram

A Use Case Diagram was developed to visually represent how users interact with the system. It illustrates the different types of users, or actors, and the various functions or services, known as use cases, that the system provides.

Figure 4

*Use Case Diagram for the Development of
Livelihood and Resource Management at Angat,
Nueva Ecija, NGO.*

The use case diagram illustrates how the three main actors, Beneficiaries, Donors, and Administrators, interact with the Project Hera. Each actor is assigned specific functions that match their roles and responsibilities within the NGO's operations. Beneficiaries can register and sign in, view available programs, enroll in training, track their program history, view NGO operation summaries, monitor donations received, and update their profiles. Donors, on the other hand, can register and sign in, view programs, donate, review donation summaries and histories, and manage their profiles. Administrators are granted full system oversight, encompassing the management of user accounts, donations, programs, and newsroom content. They also generate reports, with the option to print them for documentation and decision-making.

The diagram clearly separates user roles to avoid overlap, ensuring that each group accesses only the features relevant to their needs. It also centralizes administrative tools to strengthen operational control and accountability. Beneficiaries and donors receive simplified access to essential services, making the platform user-friendly and efficient.

The diagram presents a well-structured, user-centered model that supports organized record-keeping, transparent donation management, and streamlined program administration. This approach is consistent with recent software-engineering findings, which emphasize that use case modeling helps clarify system requirements, enhances communication with stakeholders, and supports early validation of system architecture (Vranić et al., 2024).

Context Diagram

The context diagram provides an overview of the Livelihood Training and Resource Management Portal, illustrating how the system interacts with external entities, including beneficiaries, donors, and administrators. It illustrates the system's overall boundary, emphasizing the flow of information between users and the system's main processes.

Figure 5

Context Diagram for the Development of the Livelihood and Resource Management of the Angat Nueva Ecija NGO.

The context diagram illustrates the interactions between the Admin, Beneficiary, and Donor and the Project Hera through clearly defined data flows. Each user communicates with the system by sending specific inputs and receiving corresponding outputs, ensuring a secure and organized exchange of information.

Administrators manage the backbone of the platform by providing account details, program information, donation records, and newsroom content. In return, they receive dashboard displays, system reports, and generated report data that support decision-making and organizational oversight. Beneficiaries interact with the system through registration data, login credentials, enrolment requests, program list requests, and profile updates. The platform responds with registration confirmation, authentication status, account verification results, enrolment updates, and report data. Donors request program lists, submit donations, update their profiles, and generate reports. The system returns program details, donation receipts, history lists, confirmation of profile changes, and dashboard data.

The design of a centralized system with clear roles for administrators, beneficiaries,

and donors reflects modern governance and transparency practices in non-profit technology. For instance, they have been shown to improve trust and accountability by enabling donors to track contributions in real-time and providing immutable audit trails (Ahmed et al., 2024). Such technologies support the kind of transparent, role-based system architecture that enhances collaboration and trust among users.

Data Flow Diagram

A Data Flow Diagram (DFD) visually represents how data moves through a system, showing the processes, data stores, and external entities involved. It helps in understanding the flow of information, system functions, and how users interact with the system's components.

Figure 6

Level 1 Data Flow Diagram for the Development of Project Hera.

The Data Flow Diagram at Level 1 illustrates how Project Hera manages information between beneficiaries, donors, administrators, and the system's internal

processes. It illustrates that users begin by registering or logging in, with their data validated and stored in the User Accounts and User Profile data stores. Beneficiaries and donors can then access their dashboards, view programs, enroll in training sessions, update their profiles, and track their activities. Donors can also make contributions, which the system verifies and records in the Donations data store.

Administrators oversee the entire platform by accessing dashboards, generating reports, and managing accounts, programs, donations, and news articles. Every administrative action updates the appropriate data store, ensuring accurate and up-to-date records. The diagram illustrates a well-organized, centralized system where data flows smoothly between users, processes, and data stores, supporting transparency, efficiency, and the structured management of livelihood programs and community resources.

Jiang et al. (2021) demonstrated how data flow diagrams and structured modules in donation systems improve process clarity and operational efficiency. Luciano (2025) found that centralized platforms with dashboards, automated reporting, and structured data stores

enhance decision-making, accuracy, and stakeholder trust. These studies confirm that mapping processes and data interactions through diagrams such as DFDs is a critical step in developing efficient, transparent, and user-centered donation management systems (Jiang et al., 2021; Luciano, 2025).

Entity Relationship Diagram

An Entity Relationship Diagram (ERD) is a graphical tool used to illustrate the relationships among entities within a system. It serves as a blueprint for database design, allowing developers and researchers to visualize the structure, attributes, and interconnections of data essential to the system's functionality.

Figure 7

*Entity Relationship Diagram for the
Development of the Livelihood and Resource
Management of the Angat Nueva Ecija NGO.*

The figure above presents the logical structure of the database, showing how entities interact within the system. It includes core entities such as users, programs, donations, news articles, and volunteer submissions, each linked through defined relationships. The diagram illustrates how users can have profiles, addresses, documents, and income records, as well as participate in programs as enrollees or graduates. Donors can make contributions recorded through donation and payment entities, while administrators can manage content and news updates. The ERD provides a comprehensive framework that supports data organization, accessibility, and efficient retrieval across different system modules.

This implies that the system is designed with an integrated and relational database structure to ensure efficient data management and system scalability. By clearly defining relationships between entities, the system minimizes redundancy and supports accurate, consistent data handling. This structure enhances accountability, allowing seamless tracking of user participation, donations, and program outcomes. It facilitates easier system maintenance, reporting, and future feature expansion, contributing to the platform's goal

of promoting transparency and effective resource management. Using ERD in relational data modeling helps accurately reflect business requirements and ensures data consistency (Alifi et al., 2022). Meanwhile, Trillo-Montero et al. (2023) demonstrate how a relational database structured using ER diagrams supports scalable, maintainable, and reliable data storage in real-world systems.

Database Normalization

It is a process used to organize data efficiently within a database by reducing redundancy and ensuring data integrity. It involves dividing large tables into smaller, related ones to improve consistency, accuracy, and ease of maintenance.

Figure 8

Database Normalization for the Development of the Livelihood and Resource Management of the Angat Nueva Ecija NGO.

The figure above illustrates the structured organization of data within the Livelihood Training and Resource Management Portal. It shows the division of data into multiple interrelated tables, such as users,

programs, donations, and reports, each with its own specific attributes and unique identifiers. The design eliminates redundancy and ensures that each entity, such as beneficiaries, donors, and administrators, interacts with consistent and non-repetitive data. This structure promotes clarity, accuracy, and consistency, supporting efficient management of records related to training programs, donations, and user profiles.

The normalized database design ensures that the system operates efficiently, securely, and reliably. By minimizing data duplication and ensuring referential integrity, it enhances system performance and simplifies data maintenance. This structure allows easier data retrieval, reduces storage requirements, and ensures the accuracy of stored information. These benefits are supported by recent studies, which emphasize that normalized relational databases improve data integrity, facilitate efficient query processing, and provide a strong foundation for scalable and maintainable information systems (Raut et al., 2021; Alifi et al., 2022; Trillo-Montero, Cosano-Lucena, González-Redondo, Luna-Rodríguez, & Santiago, 2023). Normalization supports the portal's goal of providing a stable and well-organized data

management system for the Angat Nueva Ecija NGO.

Data Dictionary

A data dictionary is a centralized repository that defines and describes all the data elements within a system, including their names, types, formats, and relationships. It serves as a reference guide for developers and users to ensure consistency, accuracy, and clarity in data management and system development.

Recent studies have highlighted the importance of well-structured data dictionaries in maintaining data integrity, facilitating communication among stakeholders, and supporting the development of scalable systems (Alifi et al., 2022; Raut, Deshmukh, & Jadhav, 2021; Trillo-Montero, Cosano-Lucena, González-Redondo et al., 2023). By providing a comprehensive description of all system data elements, a data dictionary enables accurate data entry, prevents redundancy, and ensures that developers, administrators, and end-users have a shared understanding of system information, which is essential for effective database management and system maintainability.

Figure 9

Table contact_submission data dictionary for the Development of the Livelihood and Resource Management of the Angat Nueva Ecija NGO.

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	Unique identifier for each record in the table. It serves as the primary key.
first_name	varchar	100	Not Null	This field stores the first name provided by the user during form submission.
last_name	varchar	100	Not Null	This field stores the user's last name as part of their submitted information.
email	varchar	100	Not Null	This field contains the email address used for communication with the user.
subject	varchar	200	Not Null	This field specifies the category or purpose of the user's message.
message	text		Not Null	This field contains the content of the message submitted by the user.
submission_time	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time the contact form was submitted.

Figure 10*Table donations data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each donation and serves as the primary key.
reference_number	varchar	50	Not Null	This field stores the unique reference number assigned to each donation.
program_id	integers	11	FK	This field identifies the program associated with the donation.
program_name	varchar	225	Not Null	This field stores the name of the program receiving the donation.
amount	decimal	10,2	Not Null	This field records the monetary amount donated.
donor_name	varchar	225	Not Null	This field stores the full name of the donor.
donor_email	varchar	225	Not Null	This field stores the email address of the donor for communication and receipt confirmation.
donor_phone	varchar	20	Null	This field contains the contact number of the donor.
payment_method	varchar	50	Null	This field specifies the payment method used for the donation.
payment_status	enum	pending, success, failed, canceled, qr_pending	Null	This field indicates the current status of the payment.
qr_payment_status	enum	pending, verified, failed	Null	This field indicates the current status of the QR payment process.
paymongo_payment_intetent_id	varchar	225	Null	This field stores the PayMongo payment intent ID linked to the transaction.
paymongo_source_id	varchar	225	Null	This field stores the PayMongo source identifier.
paymongo_payment_id	varchar	225	Null	This field stores the PayMongo payment ID after processing is completed.
gcash_reference	varchar	100	Null	This field records the reference number for GCASH payments.
payment_proof_path	varchar	500	Null	This field stores the file path of the donor's uploaded proof of payment.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the donation record was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the donation record was last updated.

Figure 11*Table inkind_donations data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each in-kind donation.
reference_number	varchar	50	Null	This field stores the unique reference number assigned to the in-kind donation.
program_id	integers	11	FK	This field identifies the program that is receiving the donated items.
item_type	varchar	100	Not Null	This field stores the type of the donated item.
item_description	text		Not Null	This field contains a detailed description of the donated item.
quantity	integers	11	Not Null	This field records the quantity of donated items.
dropoff_method	enum	center, pick-up	Not Null	This field indicates whether the donor will drop off the items or request pickup.
donor_name	varchar	255	Not Null	This field stores the full name of the in-kind donor.
donor_email	varchar	255	Not Null	This field stores the donor's email address.
donor_phone	varchar	255	Not Null	This field contains the donor's contact number.
status	enum	pending, confirmed, received, canceled, to complete, completed	Null	This field indicates the processing status of the in-kind donation.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the in-kind donation record was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the in-kind donation record was last updated.

Figure 12*Table news_article data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each news article and serves as the primary key.
title	varchar	255	Not Null	This field stores the title of the news article.
excerpt	text		Not Null	This field contains a brief summary or introduction to the news article.
content	text		Not Null	This field contains the full written content of the news article.
source	varchar	100	Not Null	This field indicates the original source or reference of the news article.
publish_date	date	yyyy - mm - dd	Not Null	This field records the date when the news article was published.
image_path	varchar	255	Null	This field stores the file path of the image associated with the news article.
tags	varchar	255	Null	This field contains tags or keywords used to categorize the news article.
is_current	tinyint	1	Null	This field indicates whether the news article is marked as current.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the news article was created.
status	enum	draft, published, archived, completed	Null	This field indicates the publication status of the article (draft, published, archived, or completed).
is_featured	tinyint	1	Null	This field indicates whether the news article is marked as featured.
is_archived	tinyint	1	Null	This field indicates whether the news article has been archived.
duration_date	date		Null	This field stores the duration period or expiry date associated with the news article.
is_lifetime	tinyint	1	Null	This field indicates whether the news article is set to remain active indefinitely.
scheduled_publish_date	date	yyyy - mm - dd	Null	This field records the scheduled date when the news article will be automatically published.
video_path	varchar	255	Null	This field stores the file path of the video associated with the news article.

Figure 13*Table programs data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program and serves as the primary key.
title	varchar	255	Not Null	This field stores the official title of the program.
category	varchar	100	Not Null	This field indicates the category under which the program is classified.
trainer	varchar	255	Not Null	This field contains the name of the trainer assigned to handle the program.
max_participants	integers	11	Not Null	NU This field records the maximum number of participants allowed for the program.
start_date	date	yyyy - mm - dd	Not Null	This field indicates the date when the program officially begins.
end_date	date	yyyy - mm - dd	Not Null	This field indicates the date when the program concludes.
main_image	varchar	255	Null	This field stores the file path of the program's main image.
description	text		Null	This field provides a detailed description of the program.
status	enum	draft, published, archived, completed	Null	This field indicates the publishing or completion status of the program.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the program was created.
barangay	varchar	255	Null	This field identifies the barangay where the program is conducted.
municipality	varchar	255	Null	This field identifies the municipality where the program is conducted.
province	varchar	255	Null	This field identifies the province where the program is conducted.
completion_date	timestamp	yyyy - mm - dd hh : mm : ss	Null	This field records the date and time the program was completed.
completion_notes	text		Null	This field contains remarks or notes about the program's completion.

Figure 14

Table program_attendance data dictionary.

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program attendance entry.
program_enrollments_id	integers	11	FK	This field identifies the enrollment record linked to the attendance.
session_date	date	yyyy - mm - dd	Not Null	This field indicates the date of the program session attended.
attended	tinyint	1	Null	This field indicates whether the participant attended the session.
attended_at	timestamp	yyyy - mm - dd hh : mm : ss	Null	This field records the timestamp when attendance was marked.
notes	text		Null	This field contains remarks or notes related to the attendance.

Figure 15

Table program_completion_images data dictionary.

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each uploaded completion image.
program_id	integers	11	FK	This field identifies the program associated with the completion image.
image_path	varchar	255	Not Null	This field stores the file path of the uploaded image.
image_caption	text		Null	This field contains the caption or description of the image.
uploaded_by	integers	11	Null	This field identifies the user who uploaded the completion image.
uploaded_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the image was uploaded.

Figure 16*Table program_content data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program content block.
program_id	integers	11	FK	This field identifies the program to which the content belongs.
mini_heading	varchar	255	Not Null	This field stores the heading for the specific content segment.
mini_text	text		Null	This field contains descriptive text for the content segment.
mini_image	varchar	255	Null	This field stores the file path of an image used in the content segment.
position	integers	11	Null	This field determines the display order of the program content.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the content was created.

Figure 17*Table program_donation data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program donation.
program_id	integers	11	FK	This field identifies the program receiving the donation.
reference_number	varchar	11	Not Null	This field stores the reference number assigned to the donation.
amount	decimal	10,2	Not Null	This field records the monetary amount donated to the program.
donor_name	varchar	225	Not Null	This field stores the name of the donor.
donor_email	varchar	225	Not Null	This field stores the email address of the donor.
donor_phone	varchar	20	Not Null	This field stores the contact number of the donor.
payment_status	enum	pending, success, failed	Null	This field indicates the completion status of the payment.
paymongo_payment_intent_id	varchar	225	Null	This field stores the PayMongo payment intent ID.
paymongo_source_id	varchar	225	Null	This field stores the PayMongo source ID.
payment_proof_path	varchar	500	Null	This field stores the file path of the uploaded payment proof.
gcash_reference	varchar	100	Null	This field records the GCASH reference number.
payment_method	enum	gcash, gcash_qr, card	Null	This field identifies the payment method used for the donation.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time the donation was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time the donation was last updated.

Figure 18*Table program_enrollments data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program enrollment.
user_id	integers	11	FK	This field identifies the user who enrolled in the program.
program-id	integers	11	FK	This field identifies the program in which the user is enrolled.
emergency_contact_name	varchar	100	Not Null	This field stores the emergency contact person's name.
emergency_contact_number	varchar	20	Not Null	This field stores the emergency contact number.
consent_demographics	tinyint	1	Null	This field indicates whether demographic consent was provided.
consent_rules	tinyint	1	Null	This field indicates whether the enrollee agreed to the program rules.
consent_media	tinyint	1	Null	This field indicates whether the enrollee consented to media documentation.
enrollment_status	enum	pending, approved, rejected	Null	This field indicates whether the enrollment is pending, approved, or rejected.
enrolled_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when enrollment was submitted
processed_at	timestamp	yyyy - mm - dd hh : mm : ss	Null	This field records when the enrollment was processed by an admin.
admin_notes	text		Null	This field contains administrative remarks about the enrollment.
graduated	tinyint	1	Null	This field indicates whether the enrollee has completed the program.
graduation_date	timestamp	yyyy - mm - dd hh : mm : ss	Null	This field records the date and time the participant graduated.

Figure 19*Table program_graduates data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program graduate.
user_id	integers	11	FK	This field identifies the user who completed the program.
program-id	integers	11	FK	This field identifies the associated program.
enrollment_id	integers	11	FK	This field identifies the enrollment record linked to the graduation.
graduated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time of graduation.
graduation_status	enum	graduated, dropped, incomplete	Null	This field indicates whether the graduate completed, dropped, or left incomplete.
certificate_issued	tinyint	1	Null	This field indicates whether a certificate was issued. issued
certificate_path	varchar	255	Null	This field stores the file path of the digital certificate.
notes	text		Null	This field contains notes related to the graduate's performance or status.
marked_by	integers	11	Null	This field records the administrator who marked the graduation.

Figure 20*Table program_inkind_donations data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program in-kind donation.
program_id	integers	11	FK	This field identifies the program that received the in-kind donation.
item_type	varchar	100	Not Null	This field stores the type of item donated.
item_description	text		Not Null	This field contains the details of donated items.
quantity	integers	11	Not Null	This field records the number of items donated.
dropoff_method	enum	Center, pickup	Not Null	This field indicates whether the donor dropped off or requested pickup.
donor_name	varchar	255	Not Null	This field stores the name of the donor.
donor_email	varchar	255	Not Null	This field stores the donor's email address.
donor_phone	varchar	255	Not Null	This field stores the contact number of the donor.
status	enum	pending, confirmed, received, canceled, to complete, completed	Null	This field indicates the processing status of the in-kind donation.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the donation was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the donation was last updated.

Figure 21

Table program_schedules data dictionary.

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each program schedule.
program_id	integers	11	FK	This field identifies the program associated with the schedule.
title	varchar	255	Not Null	This field stores the title or name of the scheduled activity.
start_datetime	datetime	yyyy - mm - dd hh : mm : ss	Not Null	This field indicates the start date and time of the schedule.
end_datetime	datetime	yyyy - mm - dd hh : mm : ss	Null	This field indicates the end date and time of the schedule.
location	varchar	255	Null	This field specifies the location of the scheduled activity.
description	text		Null	This field contains additional information about the schedule.
status	enum	scheduled, completed, canceled	Null	This field indicates whether the schedule is upcoming, completed, or canceled.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the schedule was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the schedule was last updated.

Figure 22

Table schedules data dictionary.

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each general schedule entry.
title	varchar	255	Not Null	This field stores the title of the scheduled activity.
start	datetime	yyyy - mm - dd hh : mm : ss	Not Null	This field indicates the start date and time of the schedule.
end	datetime	yyyy - mm - dd hh : mm : ss	Null	This field indicates the end date and time of the schedule.
color	varchar	20	Null	This field indicates the display color associated with the schedule status.

Figure 23*Table users' data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each user and serves as the primary key
first_name	varchar	50	Not Null	This field stores the first name of the user.
last_name	varchar	50	Not Null	This field stores the last name of the user.
username	varchar	255	Not Null	This field stores the username selected or assigned to the user.
email	varchar	100	Null	This field stores the user's email address for communication and login purposes.
country_code	varchar	5	Not Null	This field stores the international dialing code associated with the user's phone number.
phone_number	varchar	20	Not Null	This field stores the user's contact number.
password	varchar	255	Not Null	This field stores the user's encrypted password.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the user account was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time when the user account was last updated.
google_id	varchar	255	FK	This field stores the Google account ID if the user signs up via Google
signup_method	enum	from, google	Null	This field indicates whether the user registered using a form or through Google.

Figure 24*Table user_documents data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each uploaded user document.
user_id	integers	11	FK	This field identifies the user to whom the document belongs.
id_type	enum	Passport, national id, driver's licence, sss, pagibig, philhealth, voter's id, postal id, pwd id, barangay	Null	This field specifies the type of identification document uploaded.
id_number	varchar	100	Null	This field stores the identification number found on the user's ID.
id_front_image	varchar	255	Null	This field stores the file path of the front image of the ID.
id_back_image	varchar	255	Null	This field stores the file path of the back image of the ID.
expiry_date	date	yyyy - mm - dd	Null	This field indicates the expiration date of the uploaded ID, if applicable.
document_status	enum	pending, verified, rejected	Null	This field indicates whether the document is pending, verified, or rejected.
verification_notes	text		Null	This field contains remarks or notes from the admin reviewing the document.
uploaded_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time the document was uploaded.

Figure 25*Table volunteer_submission data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each volunteer submission.
first_name	varchar	100	Not Null	This field stores the first name of the volunteer.
last_name	varchar	100	Not Null	This field stores the last name of the volunteer.
email	varchar	100	Not Null	This field stores the email address of the volunteer.
subject	varchar	200	Not Null	This field specifies the purpose or category of the volunteer's message.
message	text		Not Null	This field contains the message submitted by the volunteer.
ip_address	varchar	45	Not Null	This field stores the volunteer's IP address for tracking and security.
submission_time	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records the date and time the volunteer form was submitted.

Figure 26*Table user_profile data dictionary.*

Field Name	Data Type	Field Length	Constraint	Description
id	integers	11	PK	This field stores the unique identifier for each user profile and serves as the primary key.
user_id	integers	11	FK	This field identifies the user associated with the profile.
profile_picture	varchar	255	Null	This field stores the file path of the user's profile picture.
civil_status	enum	single, married, widowed, divorced	Null	This field indicates the user's civil status.
date_of_birth	date	yyyy - mm - dd	Null	This field stores the user's birth date.
gender	enum	male, female, others	Null	This field specifies the user's gender.
house_number_street	varchar	255	Null	This field stores the user's house number and street address.
barangay	varchar	100	Null	This field stores the barangay where the user resides.
municipality_city	varchar	100	Null	This field stores the municipality or city of residence.
province	varchar	100	Null	This field stores the province of residence.
zip_code	varchar	10	Null	This field stores the ZIP code of the user's address.
living_situation	enum	own, rent, relatives, shelter	Null	This field specifies the user's current living arrangement.
household_size	integers	3	Null	This field records the total number of household members.
number_of_dependents	integers	3	Null	This field records the number of dependents in the household.
is_head_of_family	enum	yes, no	Null	This field indicates whether the user is the head of the family.
monthly_income_range	enum	below 5000, 5000 - 10000, 10001 - 20000, above 20000	Null	This field indicates the user's monthly income bracket.
income_source	enum	livelihood, part-time, remittances, none, others	Null	This field specifies the primary source of the user's income.
employment_status	enum	employed, self-employed, unemployed	Null	This field specifies whether the user is employed, self-employed, or unemployed.
occupation	varchar	100	Null	This field stores the user's occupation.
livelihood_type	varchar	100	Null	This field specifies the type of livelihood the user engages in.
work_income	decimal	10,2	Null	This field indicates the user's income from work.
pwd_status	enum	yes, no	Null	This field indicates whether the user is a person with a disability.
solo_parent	enum	yes, no	Null	This field indicates whether the user is a solo parent.
health_conditions	text		Null	This field contains any health conditions disclosed by the user.
verification_status	enum	pending, verified, rejected	Null	This field indicates whether the user profile is pending, verified, or rejected.
verification_notes	text		Null	This field stores remarks from the admin who verified the profile.
created_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the profile was created.
updated_at	timestamp	yyyy - mm - dd hh : mm : ss	Not Null	This field records when the profile was last updated.

The figures above are the data dictionary of the system's database. It served as a core component of the system's documentation, providing detailed definitions, data types, constraints, and purposes for each data element. According to Wijayanto and Halim (2021), a data dictionary enhances communication between technical and non-technical stakeholders, ensuring that everyone involved shares a common understanding of system components.

3.3 Development and Implementation

This phase, often the longest, involved developers working closely with UX designers to integrate system requirements and user feedback into the code. This phase outlined a structured approach to how the system would be developed and implemented if the project were fully carried out. The plan emphasized collaboration between developers and UX designers to ensure that system requirements were met and user feedback was effectively incorporated. The initial version of the system focused on developing core modules, including user registration, donation management, program scheduling, and account handling. Subsequent iterations progressively introduced additional features based on user needs and system

evaluation results.

A systematic implementation plan also included continuous testing at each development stage to verify that each component aligned with functional and usability requirements. Key aspects of this phase entailed ensuring efficient database connectivity, responsive user interface design, and seamless integration of system modules to simulate real-world functionality. While full implementation was beyond the scope of this capstone, this implementation plan provided a comprehensive roadmap for future execution, ensuring that the donation management system could operate efficiently, reliably, and intuitively in a live environment.

3.4 Testing and Deployment

In preparation for potential system implementation, the researchers developed a comprehensive plan for testing and deployment aligned with the Agile Scrum methodology. Throughout the development cycles, the testing plan was executed iteratively during each sprint. Testers collaborated closely with developers to review user stories, define acceptance criteria, and prepare test cases for key features, including user registration,

donation management, and program enrollment. Testing activities were conducted in conjunction with coding to facilitate the early identification and correction of defects, ensuring that each module met the established functional and quality benchmarks. Regular Scrum meetings were crucial for maintaining team alignment, facilitating transparent communication, and facilitating rapid issue resolution throughout the project.

Although full deployment was beyond the scope of this study, a structured deployment plan was outlined to guide future implementation efforts. After each sprint, a prototype version of the system was deployed to a staging or testing environment, simulating real-world usage conditions. This deployment allowed the team to validate system performance, usability, and functionality while gathering feedback from intended users. In the proposed deployment strategy, the system was rolled out incrementally, ensuring that each release was stable, secure, and aligned with stakeholder requirements outlined during sprint planning.

This plan emphasized continuous deployment, user acceptance testing (UAT), and continuous improvement before production release. By following this staged approach,

risks were minimized, and opportunities for refinement were maximized. Although actual deployment was not included in the current scope, this plan provided a clear roadmap for future implementation, ensuring that the final product could be confidently introduced as a reliable, user-centered, and high-performing livelihood and donation management system when it was ready.

4. Assessment of Elected IT Experts on the technical quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO

The design and development quality of the Project Hera is evaluated based on functional suitability, performance efficiency, compatibility, usability, reliability, and security.

4.1 Functional Suitability

Functional suitability evaluates how well a system performs the required functions to meet specified and implicit needs under given conditions. The assessment results from elected IT experts on the functional suitability of the Project Hera are presented in the following table.

Table 13

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Functional Suitability.

Item Statements	Weighted Mean	Verbal Description
The system covers all specified NGO tasks (training management and donation tracking).	3.04	Suitable
The system can provide correct results with the needed degree of precision. (Functional correctness)	3.08	Suitable
The system facilitates the accomplishment of NGO objectives (efficiency, transparency, security).	3.08	Suitable
Grand Mean	3.07	Suitable

The results showed that the elected IT experts found the system to be functionally suitable, as reflected in the grand mean of 3.07 (WM = 3.07), which corresponds to Suitable. They

agreed that the system covers the key tasks required by the NGO, such as training management and donation tracking, with a weighted mean of 3.04. The experts also rated the system as suitable in terms of providing correct results with the needed level of precision (WM = 3.08) and in supporting the NGO's objectives related to efficiency, transparency, and security (WM = 3.08). These ratings indicate that the system meets the essential functional requirements expected by the evaluators.

These findings suggest that the system is capable of performing the tasks required by the NGO and can deliver accurate results that support its operations. The suitability ratings indicate that the technical functions of the system align well with the organization's goals, helping improve key processes such as training coordination and donation tracking. Since the ratings fall within the "Suitable" range rather than "Highly Suitable," there may still be opportunities to enhance functional precision, expand features, or further align the system with organizational needs.

The result revealed that the system meets the expected level of functional suitability needed for effective NGO operations. The consistent ratings across all items show that

the system performs its core tasks reliably and supports the organization's objectives. This finding aligns with the ISO/IEC 25010 quality model, which states that functional suitability is achieved when a system provides functions that meet both stated and implied user needs. While the system already provides adequate functionality, continued enhancements may further elevate its performance and better serve both administrators and program beneficiaries.

4.2 Performance Efficiency

Performance efficiency evaluates how resources are utilized under specific conditions to achieve performance goals. The assessment results from elected IT experts on the performance efficiency of the Project Hera are provided in the table below.

Table 14

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Performance Efficiency.

Item Statements	Weighted Mean	Verbal Description
The system responds quickly to user	3.00	Good

Item Statements	Weighted Mean	Verbal Description
requests (loading of training schedules, dashboards, and donation history). The system uses resources efficiently (database queries, cloud/server capacity).	2.44	Good
The system parameters meet the system requirements (Capacity)	2.72	Good
Grand Mean	2.72	Good

The findings revealed that end-users rated the system's performance as generally good, as indicated by a grand mean of 2.72. Users agreed that the system responds quickly to requests such as loading training schedules, dashboards, and donation history, giving this item a weighted mean of 3.00. They also rated the system's resource efficiency—such as database queries and server usage—as good, although with a lower weighted mean of 2.44. The system's capacity in meeting required parameters received a weighted mean of 2.72, indicating acceptable performance in handling expected

loads and operations.

These results suggest that the system is functioning well in handling user requests and performing expected tasks, but there are still areas where performance can be improved. The lower rating for resource efficiency implies that the system may benefit from optimization in database operations, server handling, or cloud resource allocation. Improving these aspects can help enhance responsiveness and prevent delays, particularly during periods of peak usage. Recent studies have highlighted that efficient resource allocation in cloud-based systems plays a crucial role in maintaining performance and ensuring high-quality service, especially under high demand (Zhang et al., 2022). Systems with optimized database operations and load-balancing strategies have demonstrated significant improvements in responsiveness and scalability (Ahmed & Khan, 2023). Strengthening system capacity and resource management could also contribute to smoother operations and better user experience in the long run.

Table 14 showed that the system's performance was satisfactory, with strengths in quick response times and acceptable system capacity. However, opportunities for

improvement remain, particularly in resource efficiency. Addressing these areas can help refine the system's performance, ensuring it becomes more efficient, stable, and capable of supporting future growth and user demands.

4.3 Compatibility

Compatibility refers to the extent to which a system or component can exchange information with other systems or components and fulfill its required functions within the same software environment. The compatibility assessment results for the Project Hera, conducted by elected IT experts, are presented in the table below.

Table 15

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Compatibility.

Item Statements	Weighted Mean	Verbal Description
The system co-exists with standard browsers, devices, and internet connections.	3.08	Compatible

Item Statements	Weighted Mean	Verbal Description
The system integrates with external platforms (e.g., payment gateways like GCash).	3.08	Compatible
Grand Mean	3.08	Compatible

The results showed that the elected IT experts rated the system as compatible with commonly used technologies, as reflected in the grand mean of 3.08, which corresponds to the rating of "Compatible". The experts agreed that the system co-exists well with standard browsers, devices, and internet connections (WM = 3.08) and that it integrates properly with external platforms such as payment gateways like GCash (WM = 3.08). These ratings indicate that the system functions effectively within typical technical environments and works smoothly with essential third-party services.

The findings suggest that the system is technically reliable in terms of compatibility, meaning it can operate across different devices and browsers without major issues. This aligns with Parlakkiliç's (2022) study, which found that responsive design significantly enhances usability across varying devices. The positive rating for integration with external platforms

further shows that the system is capable of supporting secure and convenient transactional processes, such as digital payments. Tsai et al. (2022) similarly emphasize that secure transaction processes increase users' willingness to adopt digital systems. Maintaining compatibility may require periodic updates to keep pace with evolving browser standards, device types, and changes to third-party APIs. This is consistent with Balsam and Mishra's (2024) observation that continuous maintenance and testing are essential for maintaining sustained system reliability, especially given the dynamic nature of web technologies and external integrations, such as payment APIs.

Table 15 indicates that the system is technically capable of operating within the expected technological environment and interacting effectively with external services. The consistent compatibility ratings demonstrate that the system meets essential technical requirements and can be used reliably by a wide range of users. These results provide a strong foundation for future improvements, ensuring that the system remains functional and adaptable as technology continues to evolve.

4.4 Usability

Usability measures how effectively specific users can use the system to accomplish particular goals under defined conditions, focusing on accuracy, efficiency, and satisfaction. Table 20 presents the results of the Project Hera usability assessment conducted by elected IT experts.

Table 16

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Usability.

Item Statements	Weighted Mean	Verbal Description
The system interface is intuitive and user-friendly for admins and beneficiaries.	3.40	Excellent
Features are easy to learn, navigate, and operate.	3.32	Excellent
The system can be easily operated and controlled (Operability)	3.12	Good
The design is visually	2.92	Good

Item Statements	Weighted Mean	Verbal Description
pleasant, with clear dashboards. The system protects users from making errors (wrong inputs, double transactions).	2.80	Good
The system can be accessed by users under specified conditions (Accessibility)	2.92	Good
The system provides basic operational instructions and hints to its users when in operation (Supportability)	2.80	Good
The information can be easily searched, found, or located with the use of the system (Searchability)	3.00	Good
Grand Mean	3.04	Good

IT experts rated the system's usability as Good, supported by a grand mean of 3.04. The system interface being intuitive and user-friendly received the highest rating (WM = 3.40,

Excellent), followed by the ease of learning and navigation (WM = 3.32, Excellent). Other aspects, such as operability (WM = 3.12), searchability (WM = 3.00), accessibility (WM = 2.92), visual design (WM = 2.92), error protection (WM = 2.80), and supportability (WM = 2.80), were rated Good. These results indicate that the system performs well in terms of usability, but still has areas where functionality and user support can be improved.

The ratings indicate that the system effectively provides an interface that is easy to understand, navigate, and operate, benefiting both administrators and beneficiaries. High usability scores indicate that the system can be quickly learned and used effectively with minimal difficulty, aligning with findings that the System Usability Scale (SUS) is a reliable predictor of learnability and user satisfaction (Hyzy et al., 2022). Similarly, recommendations for user interface design emphasize clear presentation, simplified navigation, and consistent layouts to improve user comprehension and interaction (Diehl et al., 2022). The areas rated good point to practical opportunities for improvement, such as enhancing visual presentation, adding safety checks for errors, improving accessibility, and

offering clearer operational guidance. These suggestions are supported by recent heuristic evaluation studies, which demonstrate that issues such as error prevention, minimalist design, and accessibility are crucial to enhancing the user experience (Benaïda, 2023; Abbasi et al., 2024; Dobbala & Lingolu, 2024). Addressing these may help the system deliver an even smoother and more reliable user experience.

IT experts recognized the system as user-friendly and functional, particularly in terms of interface design and ease of use. Although several usability components were rated as good rather than excellent, these provide useful direction for future enhancements. With further refinement, the system can achieve a higher level of usability and better support the needs of its users.

4.5 Reliability

Reliability assesses the system's ability to consistently perform specific functions under defined conditions over a set period. The table below presents the reliability assessment results for the Project Hera, as evaluated by elected IT experts.

Table 17

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Reliability.

Item Statements	Weighted Mean	Verbal Description
The system meets the needs of its users under normal operations (Maturity)	3.08	Moderate Reliable
The system is accessible and operational at the time needed for use (Availability)	2.92	Moderate Reliable
The system operates as planned despite the existence of malfunctioning hardware or software (Fault Tolerance)	2.80	Moderate Reliable
The system handles errors and recovers without data loss.	2.84	Moderate Reliable
Grand Mean	2.72	Moderate Reliable

The findings showed that the elected IT

experts rated the system's reliability at a moderate level, as indicated by a grand mean of 2.72, which is verbally described as "Moderate Reliable." The system's ability to meet user needs under normal operations received a weighted mean of 3.08, while its accessibility and availability for use earned a weighted mean of 2.92. Experts also moderately agreed that the system could continue operating even when hardware or software malfunctions occurred (WM = 2.80) and that it was able to handle errors and recover without losing data (WM = 2.84). These ratings suggest that the system demonstrates reliability but still requires improvements in certain areas to reach a higher level of performance.

The moderate reliability ratings suggest that, although the system functions adequately, it still requires enhancements to ensure smoother and more dependable performance. The ratings indicate that the system can meet user needs and recover from errors, but experts see room for improvement in areas such as availability, fault tolerance, and stronger error-handling mechanisms. Strengthening these components may help reduce system interruptions, improve data security during recovery, and increase user confidence,

especially during critical operations. Improving fault tolerance and availability is vital to building resilient systems (Sharma & Prasad, 2023). In cloud and distributed architectures, redundancy, active replication, and self-healing recovery mechanisms have been shown to significantly enhance system uptime and reliability (Chandra, 2025; Acharya et al., 2025).

The assessment conducted by the elected IT experts indicates that the system is moderately reliable, performing reasonably well but not yet reaching an optimal level of stability. The results highlight the system's strengths in meeting user needs and managing errors, but they also reveal opportunities to improve availability and fault tolerance. With targeted enhancements, the system has the potential to progress from moderate to high reliability, offering users a more dependable and secure experience.

4.6 Security

Security measures how well the system protects organizational data, ensuring that access is granted based on authorized levels. The table below presents the assessment results, as evaluated by elected IT experts, on the

security of Project Hera.

Table 18

Results of the Assessment of Elected IT Experts on the Technical Quality of the Livelihood and Resource Management System in Angat, Nueva Ecija, by an NGO, in terms of Security.

Item Statements	Weighted Mean	Verbal Description
The system ensures that information resources are accessible only to those authorized to have access (Confidentiality)	3.28	Excellent
The system prevents unauthorized access to information resources or modification of data (Integrity)	3.32	Excellent
The system supports accountability through audit logs. (non-repudiation)	3.12	Good
The system authenticates and verifies users based on their role (admin, donor, beneficiary). (Accountability)	3.08	Good

Item Statements	Weighted Mean	Verbal Description
Compliance with the Data Privacy Act of 2012 and GDPR is met.	3.12	Good
The system can preserve and protect its records for future reference (Achievability)	3.04	Good
Grand Mean	3.03	

The results showed that the IT experts rated the system positively in terms of security, with a grand mean of 3.03, which falls under *the category of 'good'*. They rated the system's ability to ensure confidentiality at 3.28 and its capability to maintain data integrity at 3.32, both of which were described as *Excellent*. Meanwhile, features related to non-repudiation, accountability, compliance with data privacy regulations, and record preservation received weighted means ranging from 3.04 to 3.12, all of which were categorized as 'Good'. These results demonstrate that the system excels in core security functions, particularly in protecting and controlling access to information.

These findings suggest that the system

possesses robust security features that can effectively safeguard sensitive information. The excellent ratings for confidentiality and integrity indicate that the system effectively prevents unauthorized access and protects data from alteration. The good ratings in non-repudiation, accountability, data privacy compliance, and record preservation suggest that while the system meets essential security standards, there are areas where further enhancement could improve transparency and user trust. Strengthening audit trails, refining role-based authentication, and reinforcing long-term data protection measures can help further elevate the system's security performance.

The evaluation from IT experts shows that the system possesses a dependable level of security, performing exceptionally well in protecting data confidentiality and integrity while maintaining good performance in other security aspects. The results reflect a strong foundation in securing information resources, with room for improvement in supporting functions such as accountability and long-term data preservation. The system meets the expected technical standards for security and provides a stable framework for safeguarding user

information.

5. Candidate end-users' evaluations of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO based on the System Usability Scale (SUS) in terms of the following usability characteristics.

The design and functionality of Project Hera were assessed by candidate end-users to determine its level of usability and effectiveness. Using the System Usability Scale (SUS), the system was evaluated in terms of key usability characteristics, including ease of use, efficiency, learnability, and overall user satisfaction. This evaluation offers valuable insights into the system's user-centered performance, ensuring it meets the needs of its intended users.

5.1 Ease of Use

Ease of use describes how simple and straightforward it is for end-users to operate a system and complete their tasks.

Table 19

Results of the Candidate end-users' evaluations of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO, in terms of Ease of Use.

Item Statements	Weighted Mean	Verbal Description
The system is simple and easy to use.	3.53	Strongly Agree
I can quickly find the features I need.	3.44	Strongly Agree
The system's layout and navigation are clear and easy to understand.	3.16	Agree
I feel comfortable using the system without assistance.	2.81	Agree
Grand Mean	3.23	Agree

Table 19 revealed that users generally perceived the system as user-friendly, as reflected in the grand mean of 3.23, corresponding to an overall rating of "Agree." Users strongly agreed that the system was simple and easy to use, with a weighted mean of 3.53, and that they could quickly locate the features they needed, which received a weighted mean of 3.44. They also agreed that the system's layout

and navigation were clear and understandable, indicated by a weighted mean of 3.16. Although still positive, the lowest rating (WM = 2.81) showed that users only agreed—not strongly agreed—that they felt comfortable using the system without assistance, suggesting a slightly weaker area compared to the others.

These results imply that the system's overall design effectively supports user interaction, particularly in terms of simplicity, accessibility, and navigability. Ease of use is a key aspect of usability, focusing on minimizing cognitive load and providing intuitive interactions that facilitate task completion. (Lestari et al., 2021). The strong agreement regarding ease of use and feature discoverability suggests that the interface successfully meets users' expectations for intuitive operation. One important feature is intuitive design, which reduces frustration and promotes seamless task execution, as observed in web-based applications where visual simplicity is correlated with higher completion rates (Setiawan et al., 2024). However, despite the generally positive feedback, the lower level of comfort reported by some users when using the system without assistance signals an

opportunity for improvement. This aligns with findings from recent usability mechanism research, which suggest that providing clear feedback, undo ("abort") options, and user preferences significantly enhance user effectiveness and satisfaction (Ferreira et al., 2024). Enhancing guidance elements—such as help prompts, tutorials, or clearer instructions—may improve user confidence and promote a more independent and seamless experience with the system.

The assessment of the system's ease of use indicated that users generally find it simple and straightforward, able to quickly locate features and navigate its layout with minimal difficulty. While the system is effective in supporting intuitive interactions and task completion, some users expressed a lower level of comfort using it independently, suggesting that additional guidance or support could further enhance user confidence. The system demonstrates a satisfactory level of ease of use, promoting accessibility and efficiency for end-users.

5.2 Efficiency

Efficiency refers to the speed and ease with which users can complete tasks within a

system. This demonstrates how effectively the system enables users to complete their tasks with minimal effort and fewer errors.

Table 20

Results of the Candidate end-users' evaluations of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO, in terms of Efficiency.

Item Statements	Weighted Mean	Verbal Description
The system responds quickly to my commands.	3.47	Strongly Agree
Tasks such as registration, viewing schedules, and tracking resources can be completed easily.	3.50	Strongly Agree
The system minimizes unnecessary steps or actions to complete a task.	3.00	Agree
The system performs functions efficiently without errors or delays.	3.00	Agree
Grand Mean	3.24	Agree

The findings revealed that end-users generally perceived the system as efficient, as

indicated by a grand mean of 3.24, corresponding to the 'Agree' response. Users strongly agreed that the system responds quickly to their commands (WM = 3.47) and that essential tasks such as registration, viewing schedules, and tracking resources can be completed easily (WM = 3.50), reflecting high satisfaction in these areas. They also agreed that the system minimizes unnecessary steps (WM = 3.00) and performs its functions without errors or delays (WM = 3.00), although these ratings were slightly lower than the first two. Overall, the results show that the system performs efficiently from the users' perspective, with certain areas demonstrating stronger performance than others.

These results imply that the system is functioning effectively in supporting users' tasks, particularly in terms of responsiveness and efficiency. The strong agreement on task efficiency suggests that the system has a positive impact on productivity by reducing the time required to complete essential activities. In digital health systems, predictive human-computer interaction modeling has been found to improve both usability and safety by anticipating and preventing user errors (Paton et. al., 2021).

The evaluation of the system's efficiency indicated that users generally find it quick and effective in performing commands and completing tasks such as registration, viewing schedules, and tracking resources. While users agreed that the system minimizes unnecessary steps and performs functions efficiently without errors or delays, these aspects received slightly lower ratings. The findings suggested that the system supports task completion effectively, though there is room for improvement in streamlining processes and ensuring error-free performance.

5.3 Learnability

Learnability refers to the ease with which users can understand and begin using a system. This evaluation shows how quickly users can learn the system's features and perform tasks without confusion.

Table 21

Results of the Candidate end-users' evaluations of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO, in terms of Learnability.

Item Statements	Weighted Mean	Verbal Description
It is easy for me to learn how to use	3.31	Strongly Agree

Item Statements	Weighted Mean	Verbal Description
the system.		
The system provides clear instructions or guidance when needed.	3.28	Strongly Agree
I can remember how to use the system after using it once or twice.	2.88	Agree
I can recover easily from mistakes or errors while using the system.	3.03	Agree
Grand Mean	3.13	Agree

The results showed that end-users found the system fairly easy to learn, as reflected in the grand mean of 3.13, which corresponds to the "Agree" response. They strongly agreed that learning how to use the system was easy (WM = 3.31) and that the system provided clear instructions or guidance when needed (WM = 3.28). Users also agreed that they could remember how to use the system after trying it once or twice (WM = 2.88) and that they could recover easily from mistakes or errors while using it (WM = 3.03). These ratings suggest that users generally had a positive experience in understanding and using the system.

Learnability assesses the rapid development of proficiency and retention, a cornerstone of inclusive learning in digital educational environments (Magylaité et al., 2022). The findings suggest that the system supports new users well, making it simple for them to get started and understand its basic functions. The strong ratings for ease of learning and clear instructions show that users can quickly become familiar with the system's features. Intuitive elements, such as visual cues and frustration-free tutorials, are key, with gamified studies showing that they boost retention by 30% through memorable designs (Sison et al., 2023).

The evaluation presented in Table 21 showed that the end-users find the system easy to learn and understand, with clear instructions and guidance available when needed. Although users can quickly grasp the system's functions, remembering how to use it after initial exposure and recovering from errors received slightly lower ratings. These findings suggested that the system supports effective learning, allowing users to navigate and operate it with minimal difficulty.

5.4 User Satisfaction

User satisfaction refers to the degree of pleasure users experience with a system. It demonstrates how effectively the system meets users' needs and expectations, while delivering a positive experience.

Table 22

Results of the Candidate end-users' evaluations of the Livelihood and Resource Management System in Angat, Nueva Ecija, NGO, in terms of User Satisfaction.

Item Statements	Weighted Mean	Verbal Description
I am satisfied with the system's overall performance.	3.31	Strongly Agree
I am satisfied with the system's overall performance.	3.28	Strongly Agree
I am satisfied with the system's overall performance.	3.28	Strongly Agree
I am satisfied with the system's overall performance.	3.25	Agree
Grand Mean	3.28	Strongly Agree

The findings showed that end-users were

highly satisfied with the system, as indicated by a grand mean of 3.28 (WM = 3.28), which corresponds to "Strongly Agree." The satisfaction statements received weighted means ranging from 3.25 to 3.31, reflecting consistently positive feedback from users. These results suggest that users viewed the system's performance favorably and felt that it met their expectations.

The high level of user satisfaction implies that the system is meeting the needs of its users and functioning in a manner that effectively supports their tasks. Since all satisfaction-related items received strong ratings, the system appears to offer dependable performance and a user experience that aligns well with what users' value. This consistent positive feedback suggests that only minor adjustments may be needed moving forward, while continued maintenance and small improvements can help sustain or further strengthen user satisfaction over time.

Chapter 4

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes the study's key findings, draws conclusions based on the results, and provides practical recommendations for enhancing the system's performance and future use.

Summary

The capstone project "Project Hera: A Livelihood Training and Resource Management Platform for Angat Nueva Ecija NGO" was developed in response to the organization's operational challenges in training coordination, donation monitoring, and resource distribution. Findings showed that the NGO's manual processes caused significant delays, inefficiencies, and risks in handling records and resources. Respondents agreed that tasks such as report generation, registration, updating information, and resource tracking consumed excessive time. Data security also emerged as a major weakness of the manual system, as respondents expressed concerns over misplaced documents, a lack of confidentiality, and insufficient protection against unauthorized access. These challenges underscored the need for a secure and automated

platform.

The study revealed that the NGO's manual system significantly slowed down operations, with respondents agreeing that tasks such as report generation (WM = 3.28) and updating information required excessive time and effort. Data security was also identified as a major concern, with respondents disagreeing that records were safe from unauthorized access (WM = 1.94) and expressing low confidence in document confidentiality (WM = 1.66). These results confirmed the need for a secure, automated platform to improve accuracy, safety, and workflow efficiency.

Following the development of Project Hera, technical evaluations demonstrated strong system performance across several quality categories. IT experts rated the platform's usability as Good (GM = 3.04), noting its intuitive interface (WM = 3.40) and ease of navigation (WM = 3.32). Compatibility was also rated positively (GM = 3.08), indicating smooth operation across standard devices, browsers, and integrated services such as payment gateways. Security received one of the highest evaluations, with Excellent ratings for confidentiality (WM = 3.28) and data integrity (WM = 3.32), reflecting the system's ability to

protect sensitive information. Reliability, rated as Moderately Reliable (GM = 2.72), indicated that the system functioned adequately but required improvements in availability, fault tolerance, and error recovery. End-users expressed strong satisfaction with the system's overall performance, as reflected in the high grand mean rating of 3.28, indicating that the platform effectively met their needs and expectations.

The findings demonstrate that Project Hera successfully addressed the NGO's operational limitations by delivering an automated, secure, and user-friendly platform. The system significantly improved efficiency, enhanced transparency in training and resource management, and strengthened the organization's ability to support women beneficiaries. While certain areas require further refinement, the platform serves as a valuable digital solution that can modernize NGO operations and sustain long-term livelihood development initiatives.

Conclusion

Based on the findings, the following conclusions were drawn:

- The system was developed using the Iterative Prototyping Model, which includes the phases of ideation, prototyping, refinement, implementation, and testing. This approach allowed continuous feedback and modification throughout development, ensuring that the system remained reliable, user-centered, and flexible for real-world use. Through repeated evaluation and improvement cycles, the platform evolved into a functional solution that addressed the specific operational needs of the Angat Nueva Ecija NGO.
- The development of Project Hera successfully addressed the operational challenges faced by Angat Nueva Ecija NGO, particularly the inefficiencies caused by manual record-keeping, slow report generation, and difficulty in updating and retrieving information.
- Findings showed that time-related issues were significant, with respondents strongly agreeing that manual report

generation was highly time-consuming, confirming the need for an automated solution.

- Data security was identified as a critical weakness in the NGO's manual system, with low ratings for record safety, confidentiality, and document protection. Project Hera's security features effectively addressed these concerns by ensuring confidentiality, integrity, and compliance with data privacy standards.
- The system demonstrated strong usability, receiving a good rating, with experts highlighting its intuitive interface and ease of navigation, which improved user experience for administrators, donors, and beneficiaries.
- Compatibility results showed that Project Hera operated effectively across browsers, ensuring accessibility for diverse users.
- Reliability was rated as moderately reliable, indicating that the system currently meets user needs but still requires improvements in areas such as availability, fault tolerance, and stronger error-handling mechanisms
- End-users reported high levels of satisfaction, demonstrating that the

system's performance aligned with their expectations and provided meaningful improvements over the manual workflow.

- The study concludes that Project Hera is an effective, secure, and user-centered platform that significantly enhances the NGO's training management, donation monitoring, and resource tracking processes. With further refinement, particularly in terms of reliability and system resilience, Project Hera has the potential to evolve into a long-term, high-impact digital tool that enhances the NGO's capacity to empower women and sustain livelihood development programs.

Recommendations

Based on the results and conclusions of the study, the following recommendations are proposed to further enhance the effectiveness, reliability, and long-term sustainability of Project Hera: A Livelihood Training and Resource Management Platform for Angat Nueva Ecija NGO:

- **Enhance Server Performance and Infrastructure:** To accommodate future users and increased data volume, the system must adopt a more scalable infrastructure. Upgrading the hosting plan, optimizing

database queries, and applying caching mechanisms will greatly improve speed. Implementing a cloud-based or containerized environment can support long-term growth and maintain system responsiveness.

- **Strengthen Security Measures:** Although the system scored high in security, additional layers of protection should be added. Multi-factor authentication will help prevent unauthorized access, while role-based access control (RBAC) should be refined to limit user permissions based on responsibilities.
- **Enhance Mobile Responsiveness:** Many beneficiaries may rely primarily on smartphones for access. Enhancing the mobile interface, such as improving layout responsiveness, reducing loading times, and ensuring compatibility with different screen sizes, will make the platform more accessible and inclusive. Offline-friendly features or lightweight mobile views can further benefit users with limited internet connectivity.
- **Integrate Advanced Analytics and Reporting:** Adding analytics dashboards will allow the NGO to visualize beneficiary

progress, donation trends, and resource distribution patterns more effectively.

- **Expand the Training Management Module:** Based on respondent feedback, additional tools such as training calendars, automated reminders, trainer profiles, and attendance analytics can further improve the efficiency of training coordination. This expansion will enable more effective monitoring of training outcomes and participant engagement.
- **Improve the Resource Management Tracking Features:** Since resource distribution is a core function of the NGO, the system should include more detailed inventory tools, low-stock alerts, and automated logs for item issuance and replenishment. Enhancing this module will ensure more transparent and accurate tracking of donated materials and program resources.
- **Implement a Feedback and Evaluation Mechanism Within the System:** To allow continuous improvement, the platform should integrate a built-in feature where beneficiaries, donors, and staff can submit feedback about their system experience. This enables the organization to identify recurring issues, user

concerns, and potential enhancements in real-time.

- **Plan for Continuous System Maintenance and Updates:** Allocating a dedicated budget or technical team for system upkeep will ensure sustainability. Regular reviews should be conducted to align the platform with evolving organizational needs.

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APPENDICES

APPENDIX A

Request Letter for Adviser



Republic of the Philippines
NUEVA ECIA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Cabanatuan City, Nueva Ecija

COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

**APPLICATION FOR CAPSTONE PROJECT TOPIC
AND DESIGNATION OF ADVISER**

Part I. Capstone Project Information

Name of Proponents

	Signature
1. ALONZO, KURT	
2. DELTRAN, CRISTOPHER	
3. CARANG, JACOB MARCO	
4. ENRIQUE, JANELLA EUNICE	

Specialization:

☐ Database Systems Technology
☐ Network Systems Technology
☒ Web Systems Technology

Proposed Capstone Project Topic/Title:

EmpowerHER: Women Livelihood Training and Resource Management System

Proposed Capstone Project Adviser (Write 3 preferred advisers):

1.	<u>DR. CRIS NORMAN P. OLIPAS</u>
2.	<u>ANGELITO I. GUNANAN JR., MSIT</u>
3.	<u>ANDREW CORRAE A. VILLEGAS, MSIT</u>

Please attach a concept paper (maximum of 2 pages) to support the capstone project topic

Endorsed by:


ROFE ANNE GUTIERREZ-COCHANCO
Capstone Project Instructor
Date Signed: 01-29-2025

Part II. Approved Capstone Project Topic/Title and Adviser

Capstone Project Topic / Title:

EmpowerHER: Women Livelihood Training and Resource Management System

Capstone Project Adviser: DR. CRIS NORMAN P. OLIPAS

Remarks:

Approved by:


ROFE ANNE GUTIERREZ-COCHANCO
Program Chairperson for the Specialization
Date Signed: 01-29-25

Form to be prepared in 4 copies (program chairperson, adviser, capstone project instructor and students)

APPENDIX B

Survey Instrument for Assessing the Common Problems Encountered Using the Manual System

Dear Respondents,

Greetings of Peace!

The undersigned is conducting a study titled "*A Livelihood Training and Resource Management Platform for Women's NGOs,* " and we sincerely appreciate your participation in this important research.

The goal of this questionnaire is to assess the current manual file and record management system, establishing a solid foundation for the development of the system. If you choose to participate in this study, you will be asked to provide basic profile information and assess the existing manual file and record management system, including the challenges associated with manual record-keeping. Completing the questionnaire will take approximately 10-15 minutes of your time.

Participation in this study is entirely voluntary. You are not obligated or coerced into answering the survey. You have the right to decline or withdraw from the study at any time

for any reason, especially if you experience discomfort while participating. To withdraw or rescind consent, you may exit the form at any point before submitting your responses. However, once the data is submitted, it will be considered part of the study. Rest assured, there are no foreseeable consequences for declining or withdrawing from participation.

Although there are no direct benefits or incentives for participating in this study, your input will contribute to a deeper understanding and serve as a concrete basis for designing and developing the system.

Your participation in this study poses no physical or psychological risks. Confidentiality and anonymity of the data are highly valued.

While your name may be requested for a follow-up study, no other identifying details will be collected. The gathered data will be accessible only to the researchers and will be used solely for research purposes, as outlined in the study.

It is important that you fully understand the purpose of the study and your rights as a research participant. If you have any questions, clarifications, or concerns regarding the study or your rights as a participant, please do not

hesitate to contact the researchers via email
at **cictresearch2024@gmail.com**.

Sincerely,

The Researchers

I. Demographic Profile

Direction: Please check or provide details where applicable. <i>Panuto: I-tsek at ilagay ang tamang impormasyon kung kinakailangan.</i>	
Name: hf	
Sex: ☐ Female ☐ Male	Age: ☐ 25-34 ☐ 18-24 ☐ 35-44 ☐ 45 and above
Educational Attainment: ☐ Primary (Elementary) ☐ Secondary (High School) ☐ Tertiary (College) ☐ Post-Baccalaureate ☐ Vocational	
You are the: ☐ Leader/Coordinator ☐ Trainer ☐ Beneficiary ☐ Other: _____	Years of Involvement in Various Activities: Less than 1 year ☐ 1 - 3 years ☐ 4 - 6 years ☐ More than 6 years

II. Current Livelihood Training and Resource Management Process

Direction: Indicate your level of agreement with the following statements regarding the current system.

Panuto: Tukuyin ang antas inyong pagsang-ayon sa mga sumusunod na pahayag tungkol sa kasalukuyang sistema.

Scale	Descriptions
4 - Strongly Agree	You completely agree with the statement. (<i>Lubos kayong sumasang-ayon sa pahayag.</i>)
3 - Agree	You agree with the statement. (<i>Kayo ay sumasang-ayon sa pahayag.</i>)
2 - Disagree	You do not agree with the statement. (<i>Hindi kayo sumasang-ayon sa pahayag.</i>)
1 - Strongly Disagree	You strongly disagree with the statement. (<i>Lubos kayong hindi sumasang-ayon.</i>)

A. Documentation

In a manual system, this refers to the recorded files and procedures ensuring the protection of physical documents from loss, damage, theft, or unauthorized access.

Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1. Training records are systematically recorded and stored. (<i>Maayos at sistematikong naililista,</i>				

<i>naitatala, at naitatabi ang mga training records.)</i>				
2. Resource allocation details are documented properly. <i>(Ang impormasyon tungkol sa resource allocation ay maayos na naitatala.)</i>				
3. There is a structured process for logging participant attendance. <i>(May malinaw na proseso para sa pagtatala ng attendance ng mga kalahok.)</i>				
4. Financial transactions related to training resources are well-documented. <i>(Ang mga gastusin na may kaugnayan sa pagsasanay</i>				

<i>ay maayos na naitatala.)</i>				
5. Record-keeping is consistent and regularly updated. <i>(Ang pagtatala ng dokumento ay regular na ina-update at ginagawa nang palagian.)</i>				
B. Accessibility In a manual system, accessibility refers to documented procedures that ensure authorized personnel can efficiently retrieve and use physical documents while maintaining security.				
Items/Statements	Strongly Agree <i>(Lubos na Sumasang-Ayon)</i> 4	Agree <i>(Sumasang-Ayon)</i> 3	Disagree <i>(Hindi Sumasang-Ayon)</i> 2	Strongly Disagree <i>(Lubos na hindi Sumasang-Ayon)</i> 1
1. Important records can be retrieved when needed. <i>(Madaling makuha ang mahahalagang records)</i>				

<i>kapag kinakailangan.)</i>				
2. NGO members and trainers can easily access training materials. <i>(Ang mga miyembro ng NGO at mga tagapagsanay ay madaling nakakakuha ng training materials.)</i>				
3. There are no difficulties in locating past training records. <i>(Walang problema sa paghahanap ng mga nakaraang training records.)</i>				
4. Information is well-organized and easy to find. <i>(Ang impormasyon at dokumento ay maayos na nakaayos at</i>				

<i>madaling hanapin.)</i>				
5. Resource allocation records are accessible to authorized personnel. <i>(Ang mga resource allocation records ay madaling ma-access ng mga awtorisadong tao.)</i>				
C. Training Effectiveness In a manual system, training effectiveness refers to how well documented training programs equip personnel with the necessary skills to handle, store, and protect physical records.				
Items/Statements	Strongly Agree <i>(Lubos na Sumasang-Ayon)</i> 4	Agree <i>(Sumasang-Ayon)</i> 3	Disagree <i>(Hindi Sumasang-Ayon)</i> 2	Strongly Disagree <i>(Lubos na hindi Sumasang-Ayon)</i> 1
1. The current system ensures well-organized				

training sessions. (Tinitiyak ng kasalukuyang proseso at sistema ang maayos na pagsasagawa ng mga training sessions.)				
2. It is easy to monitor the progress of trainees. (Madaling subaybayan ang progreso ng mga miyembro sa pagsasanay.)				
3. The current method allows for interactive and engaging training. (Ang kasalukuyang pamamaraan ay nakatutulong sa interaktibong pagsasanay.)				
4. Training schedules and				

materials are efficiently distributed. <i>(Ang mga iskedyul at materyales sa training ay maayos na naipapamahagi.)</i>				
5. Trainers can easily assess skill development among participants. <i>(Madaling masubaybayan ng mga trainers ang kasanayan ng mga miyembro.)</i>				
D. Resource Allocation In a manual system, resource allocation refers to the documented distribution of personnel, time, and materials to ensure the proper handling, storage, and protection of physical records.				
Items/Statements	Strongly Agree <i>(Lubos na Sumasang-Ayon)</i> 4	Agree <i>(Sumasang-Ayon)</i> 3	Disagree <i>(Hindi Sumasang-Ayon)</i> 2	Strongly Disagree <i>(Lubos na hindi Sumasang-Ayon)</i>

				1
1. The process for requesting training materials is efficient. <i>(Maayos ang proseso para sa paghingi ng mga materyales sa pagsasanay.)</i>				
2. Financial resources for training are well-distributed. <i>(Ang mga pondo para sa pagsasanay ay maayos na naipapamahagi.)</i>				
3. Resource shortages rarely disrupt training programs. <i>(Ang mga training programs ay bihirang maantala ng kakulangan sa mapagkukunan.)</i>				

4. The allocation process is fair and transparent. (<i>Ang proseso ng pamamahagi ay patas at maayos.</i>)				
5. There are no delays in receiving necessary resources. (<i>Walang delays sa pagtanggap ng kinakailangan mga mapagkukunan.</i>)				

III. Challenges and Pain Points

Direction: Indicate your level of occurrence with the following statements regarding the challenges and pain points in the current process <i>Panuto: Tukuyin kung gaano kadalas nararanasan ang mga sumusunod na pahayag tungkol sa mga suliranin sa kasalukuyang proseso.</i>	
Scale	Descriptions
4 - Always	This issue occurs all the time. (<i>Laging nangyayari.</i>)
3 - Sometime s	The issue occurs occasionally but is not a major concern. (<i>Paminsan-minsan lang nangyayari at hindi laging suliranin.</i>)

2 - Rarely	The issue happens infrequently. (<i>Bihirang mangyari.</i>)			
1 - Never	The issue does not occur at all. (<i>Hindi kailanman nangyayari.</i>)			
A. Time Consumption In a manual system, time consumption refers to the duration and efficiency of tasks related to file and record management.				
Items/State ments	Always (Madalas) 4	Sometim es (Paminsan- minsang) 3	Rarely (Bihirang Mangyari) 2	Never (Hindi kailanman nangyayari) 1
1. The manual process takes too much time. (<i>Nasasayang ang maraming oras sa manwal na proseso.</i>)				
2. Participant s experience delays in training registratio n. (<i>Nakakaranas ng pagkaantala ang mga kalahok sa pagpaparehi stro sa</i>				

<i>pagsasanay.</i> <i>)</i>				
3. Generating reports manually is time-consuming. <i>(Ang manwal pagbuo ng mga report ay matagal gawin.)</i>				
4. Resource tracking is inefficient. <i>(Hindi epektibo ang resource tracking.)</i>				
5. Updating information requires excessive effort. <i>(Ang pag-update ng impormasyon ay nangangaila ngan ng labis na effort.)</i>				
B. Data Security In a manual system, data security refers to the measures taken to protect digital records from unauthorized access, loss, or damage.				
Items/State ments	Always (Madalas)	Sometimes	Rarely (Bihirang)	Never (Hindi kailanman)

	4	(Pamins an- minsan) 3	Mangya ri) 2	nangyay ari) 1
1. Records are safe from unauthorized access. (Ang mga record ay ligtas mula sa hindi awtorisadong pag-access.)				
2. Important documents are not at risk of being misplaced. (Ang mahahalagang dokumento ay hindi nanganganib na mawala.)				
3. Financial records are protected. (Ang mga financial record ay protektado.)				
4. There is a backup system for important				

files. (Mayroon isang backup system para sa mahahalagang file.)				
5. Data confidentiality is maintained. (Napananatili ang pagiging confidential ng mga data.)				
C. Record Loss In a manual system, record loss refers to the misplacement, accidental destruction, or unauthorized removal of physical documents.				
Items/Statements	Always (Madalas) 4	Sometimes (Paminsan-minsan) 3	Rarely (Bihirang Mangyari) 2	Never (Hindi kailanman nangyayari) 1
1. Training records have been lost or misplaced. (Nawawala o hindi na makita ang mga training records.)				
2. Important				

financial records have been damaged. <i>(Nawala o nasira ang mahahalagang financial records.)</i>				
3. There are no duplicate copies of critical files. <i>(Hindi nadodoble ang kopya ng mahahalagang mga file.)</i>				
4. Resource inventory tracking is unreliable. <i>(Hindi maayos ang pag-track sa mga gamit at kagamitan.)</i>				
5. The NGO has experienced loss of essential documents. <i>(Nawalan ng mahahalagang dokumento ang NGO.)</i>				

D. Transparency

In a manual system, transparency refers to the clarity and accountability in managing files and records. This includes documenting access logs, tracking modifications, and ensuring that information is available to authorized personnel while preventing unauthorized alterations or concealment of data.

Items/State ments	Always (<i>Madalas</i>) 4	Sometim es (<i>Paminsan- minsan</i>) 3	Rarely (<i>Bihirang Mangyari</i>) 2	Never (<i>Hindi kailanm an nangyay ari</i>) 1
1. The manual system ensures fairness in resource allocation. (<i>Ang manwal na sistema ay nasisiguro ang patas na resource allocation.</i>)				
2. Financial transactions are transparent and traceable. (<i>Ang mga financial transactions ay</i>				

<i>malinaw at madaling masundan.)</i>				
3. Members feel confident about data integrity. <i>(Kampante ang mga miyembro na tama at ligtas ang mga data.)</i>				
4. There is accountability in training processes. <i>(May pananagutan sa proseso ng pagsasanay.)</i>				
5. Decision-making processes are well-documented. <i>(Ang mga proseso ng pagpapasya ay maayos na naitatala.)</i>				

IV. Perceived Need for a Digital System

Direction: Indicate your level of agreement with the following statements regarding the current system.

Panuto: Tukuyin ang antas inyong pagsang-ayon sa mga sumusunod na pahayag tungkol sa kasalukuyang sistema.

Scale	Descriptions
4 - Strongly Agree	You completely agree with the statement. (<i>Lubos kayong sumasang-ayon sa pahayag.</i>)
3 - Agree	You agree with the statement. (<i>Kayo ay sumasang-ayon sa pahayag.</i>)
2 - Disagree	You do not agree with the statement. (<i>Hindi kayo sumasang-ayon sa pahayag.</i>)
1 - Strongly Disagree	You strongly disagree with the statement. (<i>Lubos kayong hindi sumasang-ayon.</i>)

A. Usefulness

In a manual system, usefulness refers to the effectiveness and relevance of file and record management in supporting operations.

Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1. A digital system would make livelihood				

training easier. (<i>Mas mapapadali ng digital na sistema ang livelihood training.</i>)				
2. A centralized system would improve resource tracking. (<i>Mas mapapadali at mas magiging maayos ang pagsubaybay ng mga kagamitan sa pamamagitan ng isang sentralisado ng sistema.</i>)				
3. An online platform would reduce delays. (<i>Ang online platform ay makakatulong upang maiwasan ang mga delays.</i>)				
4. Automated reporting would enhance monitoring. (<i>Ang</i>				

<i>awtomatikong report ay magpapahusay sa monitoring.)</i>				
5. A digital system would improve financial management. <i>(Mas magiging maayos ang pamamahala sa financial gamit ang digital na sistema at plataporma.)</i>				
B. Ease of Use In a manual system, ease of use refers to how simple and convenient it is for users to manage files and records. This includes the clarity of procedures, accessibility of documents, and the effort required to store, retrieve, and update records efficiently.				
Items/Statements	Strongly Agree <i>(Lubos na Sumasang-Ayon)</i> 4	Agree <i>(Sumasang-Ayon)</i> 3	Disagree <i>(Hindi Sumasang-Ayon)</i> 2	Strongly Disagree <i>(Lubos na hindi Sumasang-Ayon)</i> 1
1. A web-based system would be				

easy to navigate. (Ang web-based system ay madaling gamitin.)				
2. Training participants would adapt easily to a digital platform. (Madaling makakapag-adjust ang mga miyembro sa pagsasanay sa digital na plataporma.)				
3. Minimal training would be required to use the platform. (Kakaunting pagsasanay lamang ang kailangan upang magamit nang maayos ang plataporma.)				
4. The system should have user-friendly features. (Dapat may				

<i>madaling gamitin na mga features ang sistema.)</i>				
5. Accessibilit y of the system would improve efficiency. (<i>Ang madaling pag-access sa sistema ay mas magiging efficient.</i>)				
C. Impact This includes how the system influences productivity, accuracy, security, and user satisfaction, as well as its role in addressing challenges and improving workflow.				
Items/Statem ents	Strong ly Agree (Lubos na Sumasa ng- Ayon) 4	Agree (Sumasa ng- Ayon) 3	Disagr ee (Hindi Sumasa ng- Ayon) 2	Strong ly Disagr ee (Lubos na hindi Sumasa ng- Ayon) 1
1. A digital platform would improve the quality of training				

programs. <i>(Ang isang digital system ay makakatulong upang mapabuti ang kalidad ng mga programang pangpagsasan ay.)</i>				
2. A system would enhance transparency in resource distribution . <i>(Ang system ay makakatulong upang maging mas maayos ang resource distribution .)</i>				
3. Members would feel more confident using an automated process. <i>(Mas magiging kumpiyansa ang mga miyembro sa paggamit ng awtomatikong proseso.)</i>				

4. The efficiency of reporting would significantly increase.				
5. More women could benefit from the training due to better accessibility. (<i>Mas maraming kababaihan ang maaaring makinabang sa training dahil sa mas madaling pag-access.</i>)				
D. Willingness to Adopt Refers to the readiness and openness of users to transition to a new or improved file and record management system.				
Items/Statements	Strongly Agree (<i>Lubos na Sumasang-Ayon</i>) 4	Agree (<i>Sumasang-Ayon</i>) 3	Disagree (<i>Hindi Sumasang-Ayon</i>) 2	Strongly Disagree (<i>Lubos na hindi Sumasang-Ayon</i>) 1
1. I am open to learning how to use a				

digital system. (Willing akong pag-aralan kung paano gumamit ng digital system.)				
2. I would be willing to attend training for the new system. (Handa akong dumalo sa pagsasanay para sa bagong system.)				
3. The NGO would support the shift to digital management. (Suportado ng NGO ang paglipat sa digital management.)				
4. I would actively use the system if implemented. (Gagamitin ko nang aktibo ang sistema kapag				

<i>naipatupad na ito.)</i>				
5. I believe digital transformation is necessary. <i>(Naniniwala ako na kailangang magkaroon ng digital change.)</i>				
E. Readiness Refers to the preparedness of an organization or users to implement improvements or transition to a new file and record management system.				
Items/Statements	Strongly Agree <i>(Lubos na Sumasang-Ayon)</i> 4	Agree <i>(Sumasang-Ayon)</i> 3	Disagree <i>(Hindi Sumasang-Ayon)</i> 2	Strongly Disagree <i>(Lubos na hindi Sumasang-Ayon)</i> 1
1. The NGO has access to the basic technology needed for a digital system. <i>(May access ang NGO sa pangunahing teknolohiyang kailangan)</i>				

para sa isang digital system.)				
2. Our members have the necessary skills to transition to an online platform. (May sapat na kasanayan ang aming mga miyembro upang lumipat sa isang online na platform.)				
3. Internet connectivity is sufficient to support a digital system. (Sapat ang internet connection para suportahan ang isang digital system.)				
4. The NGO is prepared to allocate time for system training.				

(Handa ang NGO na maglaan ng oras para sa pagsasanay sa paggamit ng sistema.)				
5. We have resources available to support system implementation. (Mayroon kaming mga mapagkukunan upang suportahan ang pagpapatupad ng sistema.)				

V. Preferred Features of a Web Platform

Direction: Please check all that applies.
Panuto: Lagyan ng tsek ang lahat ng naaangkop.

Preferred Features of a Web Platform

- ☐ User Management (Member Profiles & Access Levels)
- ☐ Online Training Registration
- ☐ Resource Inventory Management
- ☐ Automated Reports (Financial, Attendance, Training Progress, etc)
- ☐ Financial Tracking (Budget, Expenses, Grants, Donations, etc.)
- ☐ E-Learning Modules & Online Workshops
- ☐ Feedback and Evaluation System
- ☐ Document and File Storage (Training Manuals, Reports, Forms, etc.)
- ☐ Communication Hub (Announcements, Messaging, Discussion Forums, etc.)

☐ Task and Project Management (Deadlines, Assignments, Goal Tracking)

☐ Mobile Accessibility (Optimized for Smartphones/Tablets)

☐ Security and Data Privacy Features

☐ Other: -

APPENDIX C

End-Users Evaluation for Project Hera: Livelihood Training and Resource Management System for Angat, Nueva Ecija NGO

Demographic Profile

Direction: Please check or provide details where applicable. <i>Panuto: I-tsek at ilagay ang tamang impormasyon kung kinakailangan.</i>	
Name: hf	
Sex: ☐ Female ☐ Male	Age: ☐ 25-34 ☐ 18-24 ☐ 35-44 ☐ 45 and above
Educational Attainment: ☐ Primary (Elementary) ☐ Secondary (High School) ☐ Tertiary (College) ☐ Post-Baccalaureate ☐ Vocational	
You are the: ☐ Leader/Coordinator ☐ Trainer ☐ Beneficiary ☐ Other: _____	Years of Involvement in Various Activities: ☐ 1 - 3 years ☐ 4 - 6 years ☐ More than 6 years

II. Assessment of the Quality of Using the System.

Direction: Indicate your level of agreement with the following statements regarding the current system. <i>Panuto: Tukuyin ang antas inyong pagsang-ayon sa mga sumusunod na pahayag tungkol sa kasalukuyang sistema.</i>	
Scale	Descriptions

4 - Strongly Agree	You completely agree with the statement. (<i>Lubos kayong sumasang-ayon sa pahayag.</i>)			
3 - Agree	You agree with the statement. (<i>Kayo ay sumasang-ayon sa pahayag.</i>)			
2 - Disagree	You do not agree with the statement. (<i>Hindi kayo sumasang-ayon sa pahayag.</i>)			
1 - Strongly Disagree	You strongly disagree with the statement. (<i>Lubos kayong hindi sumasang-ayon.</i>)			
A. Ease of Use.				
Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1. The system is simple and easy to use. (<i>Madaling gamitin ang system.</i>)				
2. I can quickly find the features I need. (<i>Nahahanap ko agad ang mga nais kong hanapin.</i>)				

3. The system's layout and navigation are clear and understandable. <i>(Malinaw at madaling intindihin ang layout ng system.)</i>				
4. I feel comfortable using the system without assistance. <i>(Komportable kong nagagamit ang system kahit walang tulong.)</i>				
B. Efficiency				
Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1. The system responds quickly to my commands.				

(Mabilis gumana ang system.)				
2. Tasks such as registration, viewing schedules, and tracking resources can be completed easily. (Mabilis makumpleto ang mga proseso.)				
3. The system minimizes unnecessary steps or actions to complete a task. (Hindi pasikut-sikot ang proseso.)				
4. The system performs functions efficiently without errors or delays. (Hindi mabagal ang system.)				
C. Learnability				

Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1. It is easy for me to learn how to use the system. (Mabilis kong matutunan ang paggamit ng system)				
2. The system provides clear instructions or guidance when needed. (May malinaw na prosesong system.)				
3. I can remember how to use the system after using it once or twice. (Natatandaan ko agad ang				

proseso ng system.)				
4. I can recover easily from mistakes or errors while using the system. (Mabilis kong naaayos kapag nagkakamali ako sa paggamit ng system.)				
D. User Satisfaction				
Items/Statements	Strongly Agree (Lubos na Sumasang-Ayon) 4	Agree (Sumasang-Ayon) 3	Disagree (Hindi Sumasang-Ayon) 2	Strongly Disagree (Lubos na hindi Sumasang-Ayon) 1
1 I am satisfied with the system's overall performance.				
2. The system meets my expectations as an end-user.				

3. I would like to use this system again in the future.				
4. I would recommend this system to others.				

APPENDIX D

IT Experts Evaluation for Project Hera: Livelihood Training and Resource Management System for Angat Nueva Ecija NGO

Demographic Profile

Direction: Please check or provide details where applicable. <i>Panuto: I-tsek at ilagay ang tamang impormasyon kung kinakailangan.</i>	
Name:	
Sex: ☐ Female ☐ Male	Age: ☐ Below 24 ☐ 25-34 ☐ 35-44 ☐ 45 and above
Highest Educational Attainment: ☐ Bachelor's Degree ☐ Master's Degree ☐ Doctorate ☐ Other: _____	
Field of Specialization: ☐ Information Technology ☐ Computer Science ☐ Software Engineering ☐ Other: _____	Years of Expertise: ☐ Less than 2 years ☐ 2 - 3 years ☐ 4 - 6 years ☐ More than 6 years
Affiliated Organization/Institution:	

II. Assessment of the Technical Aspects of the System

Direction: Indicate your level of agreement with the following statements regarding the current system. <i>Panuto: Tukuyin ang antas inyong pagsang-ayon sa mga sumusunod na pahayag tungkol sa kasalukuyang sistema.</i>
--

A. Functional Suitability				
Items/State ments	Highly Suitable 4	Suitable 3	Unsuitable 2	Highly Unsuitable 1
1. The system covers all specified NGO tasks (training management and donation tracking).				
2. The system can provide correct results with the needed degree of precision. (Functional correctness)				
3. The system facilitates the accomplishment of NGO objectives (efficiency, transparency, security).				
B. Performance Efficiency				

Items/State ments	Excell ent 4	Good 3	Fair 2	Poor 1
1. The system responds quickly to user requests (loading of training schedules, dashboards, donation history).				
2. The system uses resources efficiently (database queries, cloud/server capacity).				
3. The system parameters meet the system requirements (Capacity)				
C. Compatibility				
Items/State ments	Highly Compat ible 4	Compat ible 3	Somewh at Compat ible 2	Not Compat ible 1
1. The system co-exists with				

standard browsers, devices, and internet connections .				
2. The system integrates with external platforms (e.g., payment gateways like GCash) .				
D. Usability				
Items/State ments	Excellent 4	Good 3	Fair 2	Fair 1
1 The system interface is intuitive and user-friendly for admins and beneficiari es.				
2. Features are easy to learn, navigate, and operate.				
3. The system can be easily				

operated and controlled (Operability)				
4. The design is visually pleasant, with clear dashboards.				
5. The system protects users from making errors (wrong inputs, double transactions).				
6. The system can be accessed by users under specified conditions (Accessibility)				
7. The system provides basic operational instructions and hints to its users when in operation				

(Supportability)				
8. The information can be easily searched, found or located with the use of the system (Searchability)				
E. Reliability				
Items/State ments	Highly Reliable 4	Moderately Reliable 3	Slightly Reliable 2	Not Reliable 1
1. The system meets the needs of its users under normal operations (Maturity)				
2. The system is accessible and operational at a time needed for use (Availability)				
3. The system operates as				

planned despite of the existence of malfunction hardware or software (Fault Tolerance)				
4. The system handles errors and recovers without data loss.				
F. Security				
Items/State ments	Excellent 4	Good 3	Fair 2	Fair 1
1. The system ensures that information resources are accessible only to those authorized to have access (Confidentiality)				
2. The system prevents unauthorized access to				

information resources or modification of data (Integrity)				
3. The system supports accountability through audit logs. (non-repudiation)				
4. The system authenticates and verifies users by role (admin, donor, beneficiary). (Accountability)				
5. Compliance with the Data Privacy Act of 2012 and GDPR is met.				
6. The system can preserve and protect its records for future reference (Achievability)				

APPENDIX E

User Manual

General User

Home Page Navigation

The Home Page serves as the primary entry point for the Angat Nueva Ecija NGO website. Its primary purpose is to introduce the organization's mission, showcase its key programs, and provide direct pathways for engagement.



Home Button: Redirects the User to the home page when clicked.

Program Button: Redirects the User to the Program page.

Livelihood Button: Redirects the User to the Livelihood page.

Resources Button: Redirects the user to the Log

In page, where the Dashboard will be accessible after login.

About Us Button: Redirects the User to the About page, where the user will find information about the NGO.

Contact Us Button: Redirects the User to the Contact page, where they can contact the NGO by filling out the provided form.

Contact Us

CONTACT US

We'd love to hear from you

Have any question in mind or want to enquire? Please feel free to contact us through the form or the following details.

Send us a message

1 First Name 2 Last Name

3 Email Address

4 Subject

5 Your Message

6 Send Message

Let's talk!

WhatsApp

enquiries@cdgnews.com

Headoffice

3rd Street, Kapitan Pele
Calabarland City, Mkolodi Edo

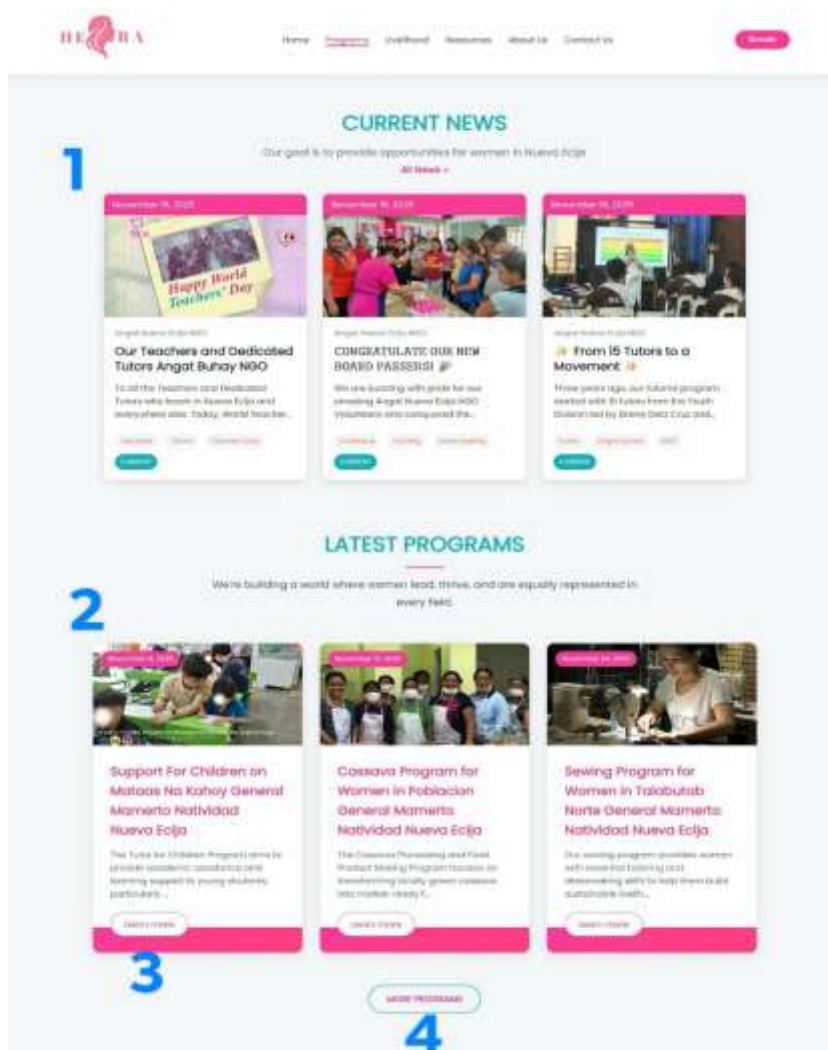
1. **First Name** - Your given name.
2. **Last Name** - Your surname or family name.
3. **Email Address** - Enter a valid email address where the NGO can reply to your message.

4. **Subject** - A brief title for your message (e.g., "Volunteer Inquiry," "Program Question").
5. **Your Message** - The main body of your inquiry or feedback. Clearly and politely detail your question or comment.
6. **Send Message Button** - The final action to submit the form.

Donate Button: Redirects the user to the donation page.

Program Page

The Programs Page serves as the central directory for all initiatives and projects undertaken by Angat Nueva Ecija NGO. This page expands on the brief summaries on the Home Page, providing detailed information on what the organization does, whom it helps, and where its work is focused.

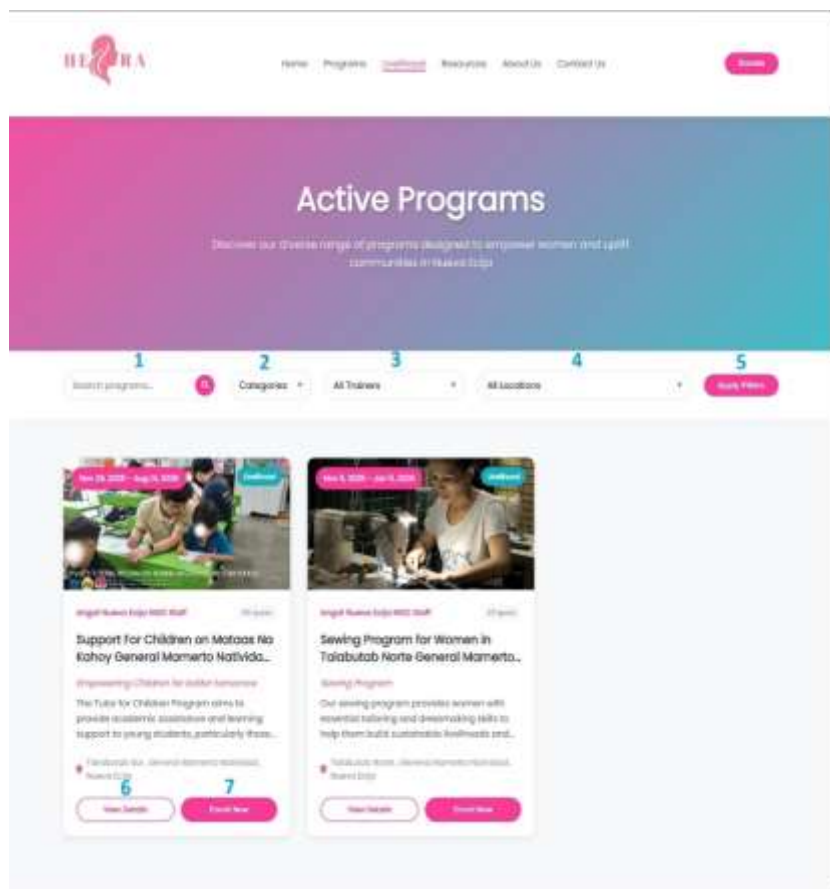


1. Shows the current programs being conducted by the Angat Nueva Ecija NGO.
2. Shows the latest programs being implemented by the NGO.
3. By clicking the "Learn more" **button**, the full details of the program will be shown.

4. By clicking the “More Programs” button, all programs will be shown.

Livelihood Page

The Livelihood Page is dedicated to showcasing the programs designed to equip women and community members with the skills, resources, and training needed for sustainable economic independence. This section is dynamic, featuring both active programs open for enrollment and the successful history of completed initiatives.



1. Search Bar - Enter keywords (e.g., "Sewing," "Cassava," "Tutor") to find programs with matching titles or descriptions.
2. Categories - Filter the list to see only programs belonging to a specific category.
3. Trainers - Filter by a specific person or team responsible for the training.
4. Location - Filter the list to see only programs available in a specific geographic area.
5. Apply Filters - After making your selections in the dropdowns (2, 3, 4), you **must** click this button to update the program list below.
6. View Details - **Click this button (6)** to read comprehensive information, including the detailed curriculum, eligibility requirements, location maps, and trainer bios before deciding to enroll.
7. Enroll Now - **Click this button (7)** when you are ready to apply for the program. It will typically take you to an application form or a contact submission page.

Program Enrollment Form

When a user clicks the "Enroll Now" button (labeled 7 on the Livelihood Page), a pop-up

window or modal titled "Program Enrollment Form" appears. This form must be completed by the beneficiary or applicant to secure a slot in the chosen training program. But first they must be Logged-in before they can continue.

The screenshot shows a mobile application interface for a "Program Enrollment Form". The form is titled "Support For Children on Mataas Na Kahoy General Mamerto Natividad Nueva Ecija". It provides details about the schedule (Nov 28, 2025 - Aug 14, 2026), venue (Takabutab Sur, General Mamerto Natividad, Nueva Ecija), and available slots (60). The form includes input fields for "Full Name", "Contact Number", "Email Address", "Address", "Emergency Contact Name", and "Emergency Contact Number". There is also a "Consent Agreements" section with a checkbox for allowing the use of photos/videos for documentation and promotion. At the bottom, there are "Cancel" and "Submit Enrollment" buttons.

Program Enrollment Form

Support For Children on Mataas Na Kahoy General Mamerto Natividad Nueva Ecija

Schedule: Nov 28, 2025 - Aug 14, 2026

Venue: Takabutab Sur, General Mamerto Natividad, Nueva Ecija

Available Slots: 60

Full Name *

Cabang Lyndon Marie Salvador

Contact Number *

983161365

Email Address *

invulnerablelyndon06@gmail.com

Address *

III Purok I, Talabutab Norte, General Mamerto Natividad, Nueva Ecija 3125

Emergency Contact Name *

Enter emergency contact full name

Emergency Contact Number *

Enter emergency contact phone number

Consent Agreements *

☐ I allow the use of my photo/video for documentation and promotion (optional).

Cancel Submit Enrollment

Full Name: Enter the applicant's complete legal name.

Contact Number: Please enter the best phone number for communication regarding program schedule and updates.

Email Address: Enter a valid email address for official correspondence and notifications.

Address: Enter the complete residential address.

Emergency Contact Name/Number: This field is mandatory. Please provide the full name and phone number of the person who should be contacted immediately in the event of an emergency during the program.

Consent Agreements (Optional): This checkbox requests permission to use your photo or video in the NGO's documentation and promotional materials.

- **Note:** Checking this box is **optional** and does not prevent enrollment if left unchecked.

Submit Enrollment: Click the large pink "Submit Enrollment" button to finalize and send your application to the program organizers.

Cancel: Click the lighter pink **"Cancel"** button to close the form without submitting the data.

Resources Page / Log-In Page

The Resources Page is a secure, members-only area designed to provide exclusive information and tools to the Angat Nueva Ecija community, including Admins, Donors, and registered beneficiaries.

If you Logged Out: It acts as the **Access Gateway**, displaying the **Login Form** (and a link to the Registration Form). You must log in here to proceed further.

HEARA

Home Programs Lifelihood **Resources** About Us Contact Us

Donate

Welcome back! Please log in to access your account.

1 Sign in with Google

2 Username

3 Password

4 I don't have an account Sign up

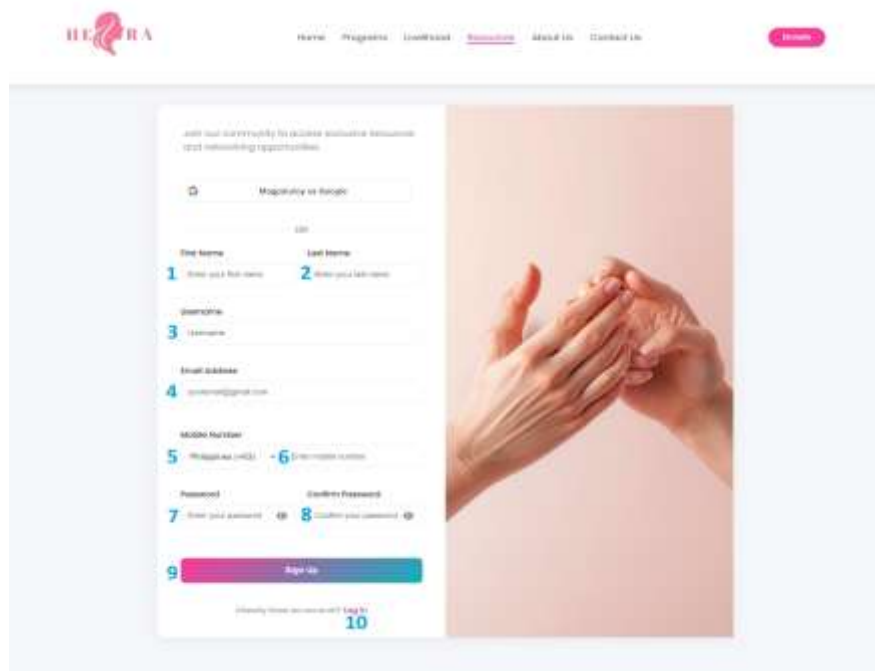
5 Log in

6

Log In

1. **Sign up with Google** - The quickest way to create an account using an existing Google profile.
2. **Username** - Your unique username created during registration.
3. **Password** - Enter your password.
4. **reCAPTCHA** - security measure to verify you are a human user.
5. **Log In Button** - Click this button after all fields are complete and verified.
6. **Sign up Link** - Click this link if you need to create a new account or don't have an account.

Register



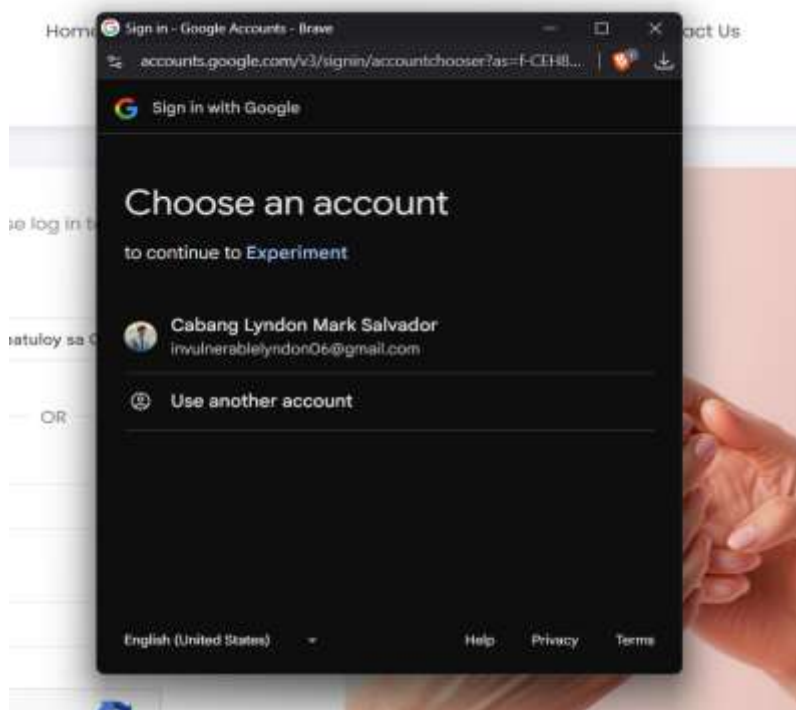
The screenshot shows the HERA registration form with the following elements:

- HERA Logo** and navigation links: Home, Programs, Download, **Registration**, About Us, Contact Us.
- Buttons:** "Sign Up" (pink) and "Log In" (blue).
- Form Fields:**
 - 1. **First Name** (text input)
 - 2. **Last Name** (text input)
 - 3. **Username** (text input)
 - 4. **Email Address** (text input)
 - 5. **Password** (text input)
 - 6. **Confirm Password** (text input)
 - 7. **ReCAPTCHA** (checkbox)
 - 8. **Log In** (button)
 - 9. **Sign Up** (button)
- Image:** A photograph of two hands clasped together, symbolizing support or care.

1. **First Name** - Enter your first name.

2. **Last Name** - Enter your last name.
3. **Username** - Create a unique username you will use to log in.
4. **Email Address** - Enter your email address.
5. **Mobile Number** - Select your country code.
6. **Mobile Number** - Enter your mobile number.
7. **Password** - Your secret security key.
8. **Confirm Password** - Re-enter your password to ensure it was typed correctly.
9. **Sign Up Button** - Finalizes the registration process.
10. **Log In Link** - For users who already have an account

Login & Register with Google



If you select the "Continue with Google" option, a secure Google pop-up window will appear:

- **Choose an Account:** Select the Google account you wish to use to log in or register.
- **Use another account:** Select this if you want to use an account that is not listed.

Donation Page

The Donation Page is the secured portal for financial contributions to Angat Nueva Ecija NGO. It provides supporters with a clear, secure, and straightforward process for contributing funds that directly support women's empowerment, community development, and livelihood programs.

3. **Donation Amount** - Pre-set buttons and a **CUSTOM AMOUNT** field (only visible if **Monetary** is selected in 2)
4. **Payment Method** - Choose your payment provider. Follow the step-by-step instructions provided to complete the transaction successfully.
5. **Full Name** - Your complete name for the donation receipt and record-keeping.
6. **Email Address** - Your contact email for receiving the receipt and confirmation.
7. **Phone Number** - Enter your contact number.
8. **Donate with GCash - Click this button (8)** to initiate the payment process through the secure gateway.

Scan to Pay with GCash

GCash QR PH

Amount to Donate

P200.00



Reference ID: DON-1763375619733-598IDFE7D

How to Donate

1 Open GCash App
Tap "Scan QR Code"

2 Scan the QR Code
Point your camera at the code

3 Enter Amount
Confirm P200.00

4 Complete Donation
Take screenshot of confirmation

Upload Donation Proof Required

Drop screenshot here or click to browse

JPG, PNG, or PDF (Max 5MB)

Submit Payment Proof



Thank You for Your Donation!

Your donation has been processed successfully.

Reference ID: DON-1763375619733-598IDFE7D

Make Another Donation

Return to Home

235

In-Kind Donation

The screenshot shows the 'In-Kind Donation' form on the HEARA website. The form is titled 'In-Kind Donation' and includes a 'Donation Type' dropdown set to 'In-Kind Donation'. Below this, there is a 'Donation Item 1' section with a red close button. The 'Item Type' dropdown is set to 'Select an item type'. The 'Item Description' field is empty, with a placeholder text 'Please describe the items you're donating'. The 'Quantity & Unit' section has a 'Number of Items' input field and a 'Select Unit' dropdown. A green '+ Add Donation Item' button is at the bottom.

Drop-off Hours:

- Weekday: 10:00 AM - 5:00 PM
- Weekend: 10:00 AM - 5:00 PM
- Evening: 6:00 PM - 8:00 PM

Opening Hours:

- Monday - Friday: 10:00 AM - 5:00 PM
- Saturday: 10:00 AM - 5:00 PM
- Sunday: Closed

Conditions:

- Items must be in good condition
- Clothing should be clean and wearable
- Food items must be non-perishable
- Electronics should be functional

Donation Type: In-Kind Donation

Donation Item 1

Item Type: Select an item type

Item Description: Please describe the items you're donating

Quantity & Unit: Number of Items: Select Unit:

+ Add Donation Item

Item Type: Educational Materials, Food & Groceries, Clothing & Shoes, Hygiene Products, Electronics, Furniture, etc.

The screenshot shows the 'Donation Item 1' dropdown menu. The dropdown is titled 'Donation Item 1' and has a red close button. The 'Item Type' dropdown is set to 'Select an item type'. The dropdown menu is open, showing a list of item types: 'Select an item type', 'Educational Materials', 'Food & Groceries', 'Clothing & Shoes', 'Hygiene Products', 'Electronics', 'Furniture', 'Tools & Equipment', 'Office Supplies', and 'Other Items'. A green '+ Add Donation Item' button is at the bottom.

Donation Item 1

Item Type: Select an item type

Select an item type

- Educational Materials
- Food & Groceries
- Clothing & Shoes
- Hygiene Products
- Electronics
- Furniture
- Tools & Equipment
- Office Supplies
- Other Items

+ Add Donation Item

Item Description: Clearly describe the item (e.g., "10 boxes of crayons," "gently used children's books," "canned sardines").

Quantity & Unit: Number of Items (input) and Select Unit (dropdown).

Preferred Drop-off Method: Select 'Drop at our center' if you will deliver the items during operating hours (listed on the left).

Select 'Schedule a pickup' if you require the NGO to collect the items from you (must be within Nueva Ecija limits).

Donation Type

In-Kind Donation

Donation Item 1

Item Type

Select an item type

Item Description

Please describe the items you're donating

Quantity & Unit

Number of items

Select Unit

Pieces (pcs)

Liters

Kilograms (kg)

Box

Others

Select Unit

+ Add Donation Item

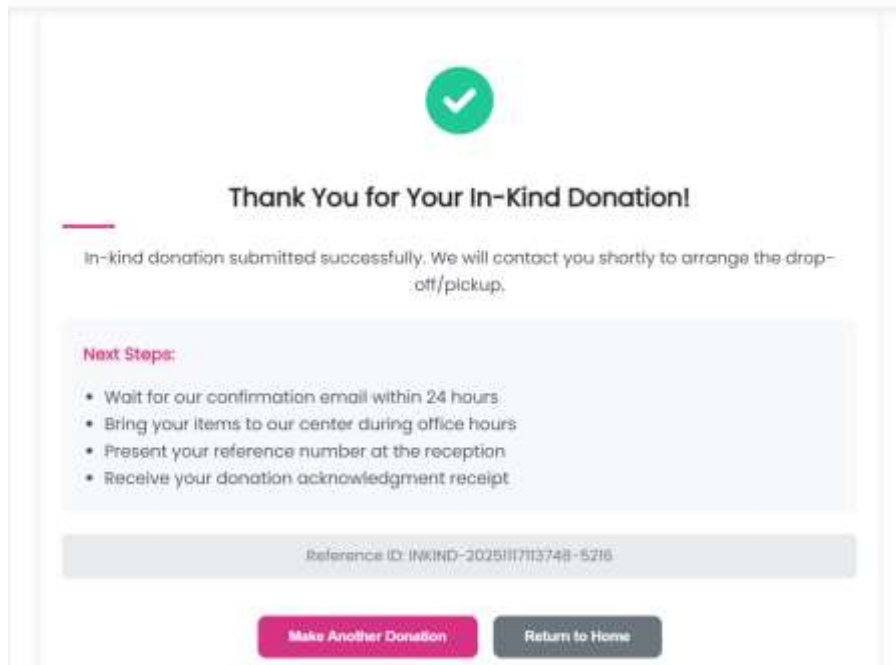
Full Name: Your complete name for the donation receipt and record-keeping.

Email Address: Your contact email for receiving the receipt and confirmation.

Phone Number: Enter your contact number.

Donate: Click this button (8) to initiate the

donation process.

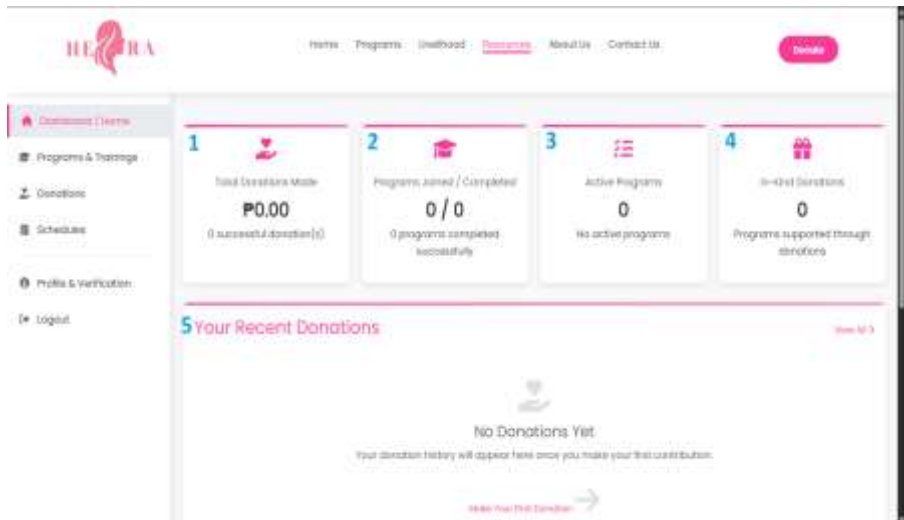


User Dashboard

This page serves as your **centralized account hub** for all interactions with Angat Nueva Ecija NGO. It provides immediate access to tools, exclusive content, and personalized status updates based on your relationship with the organization.

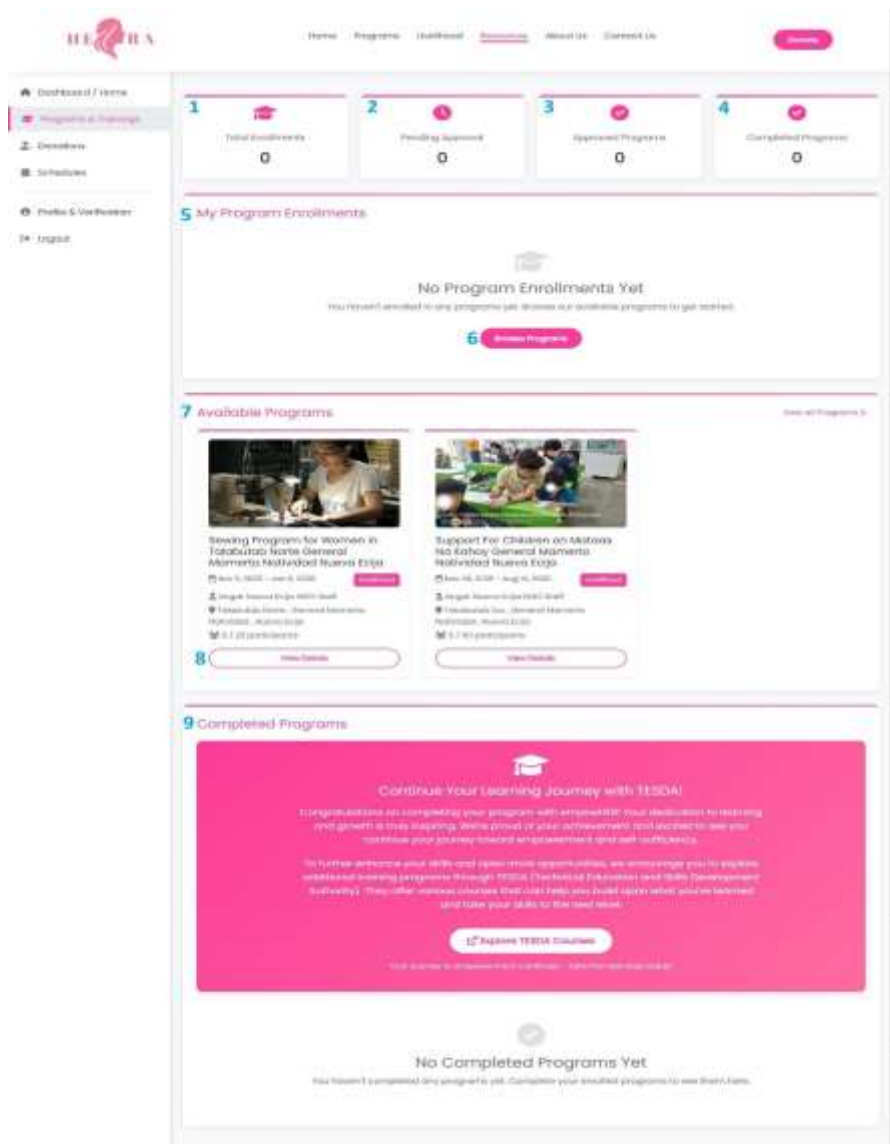
The Dashboard is designed to keep you informed, whether you are supporting the mission (Donor) or benefiting from a program (Beneficiary).

1. **Total Donations Made** - Shows the cumulative monetary value of all your successful financial contributions.



2. **Programs Joined / Completed** - Shows the number of programs you have successfully enrolled in versus the number you have successfully completed.
3. **Active Programs** - Shows how many programs or volunteer activities you are currently enrolled in or participating in.
4. **In-Kind Donations** - Shows the number of programs or initiatives you have supported through physical goods (in-kind).

5. **Your Recent Donations** - A scrollable list of your most recent monetary and in-kind contributions.



1-4 **Metrics** - (Total Enrollments, Pending, Approved, Completed) Summary boxes showing the status of all programs you have applied for or

joined.

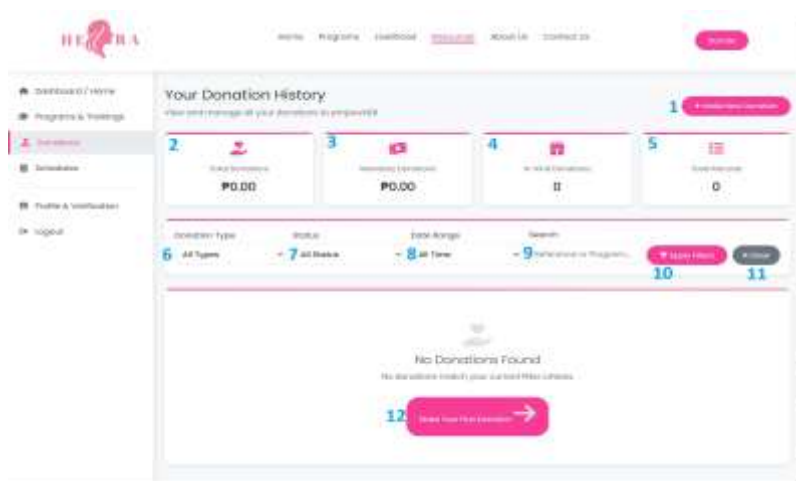
5. **My Program Enrollments** - Shows programs you have applied to or are currently active in.

6. **My Program Enrollments** - Shows programs you have applied to or are currently active in. If empty, click **"Browse Programs"** (6) to return to the Livelihood Page and enroll in a new training.

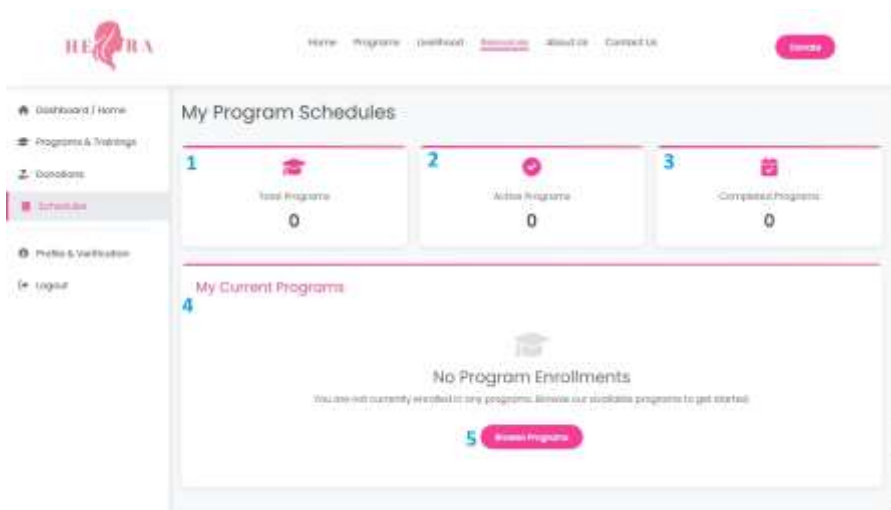
7. **Available Programs** Displays a selection of open programs from the Livelihood page.

8. **View Details Button** - click to view full program details.

9. **Completed Programs** - Records all programs you have finished successfully.
have finished successfully.



1. **+ Make New Donation Button** - Click this to be immediately redirected to the main **Donation Page** to process a new financial or in-kind gift.
- 2-5. **Metrics** - Metric card showing the total cumulative value of all your donations, cash contributions, non-monetary items or in-kind records submitted, and overall number of separate donation transactions (Monetary and In-Kind) recorded in your history.
6. **Donation Type Filter** - Dropdown menu allowing you to filter by the category of the gift.
7. **Status Filter** - Dropdown menu to filter based on the transaction status.
8. **Date Range Filter** - Dropdown menu to filter the records by time period.
9. **Search Field** - Text input field for searching by keyword, Enter a **Reference Number** or the name of the **Program** (e.g., "Cassava") that you supported.
10. **Apply Filters Button** - The button that executes your filter selections.
11. **Clear Button** - The button that resets all filter and search settings.
12. **Make Your First Donation** - A direct call-to-action link visible when no records are found. Click this to be redirected to the main Donation Page.



1. **Total Programs** - The total number of programs you have applied for or been enrolled in.
2. **Active Programs** - The count of programs that are currently running and for which you have scheduled sessions.
3. **Completed Programs** - The count of programs you have finished, showing you have fulfilled all scheduled requirements.
4. **My Current Programs** - This section displays a calendar or list of your upcoming sessions, including dates, times, and venue for any program you are enrolled in.
5. **Browse Programs** - A direct link visible when you have no active enrollments.

HELPERA

Home Programs Dashboard **Personal Information** About Us Contact Us

Logout

Dashboard / Home
Programs & Trainings
Dashboard
Schedules
Profile & Verification

Cabang Lyndon Mark Salvador
Member since November 2020
+ Working Volunteer

100%

Personal Information Demographics Document Uploads

Complete your profile to participate in programs.
(Completion req. going Demographics & Document Uploads, going individual or training programs)

Basic Information

First Name: 1 Cabang Last Name: 2 Lyndon Mark Salvador

Civil Status: 3 Single Phone Number: 4 09100000000

Date of Birth: 5 mm/dd/yyyy Gender: 6 Female

7 Save Changes

- 1-2 **First Name, Last Name** - Your legal name.
3. **Civil Status** - Marital status (e.g., Single, Married).
4. **Phone Number** - Your registered mobile number.
5. **Date of Birth** - Your birth date.
6. **Gender** - Your gender identification.
7. **Save Changes Button** - Saves all inputs on the current tab.

Address & Location

Important for determining program eligibility and accessibility

House No. / Street / Purok: Enter your house details

Barangay: Enter your barangay

Municipality / City: Enter your municipality/city

Province: Enter your province

ZIP Code: Enter ZIP code

Living Situation: Select living situation

1. **House No. / Street / Purok** - Your specific house number and the name of the street or Purok (sub-village).
 2. **Barangay** - The name of your Barangay
 3. **Municipality / City** - The name of your town or city, Province, The name of your Province (e.g., Nueva Ecija)..
 4. **ZIP Code** - Your postal code.
 5. **Living Situation** - Select from the dropdown (e.g., Owner, Renter, Living with Family).
1. **Household Size** - Enter the total number of people currently living together in your household.

Household Information

Help us assess economic status and prioritize aid

Household Size:

Number of Dependents:

Head of the Family:

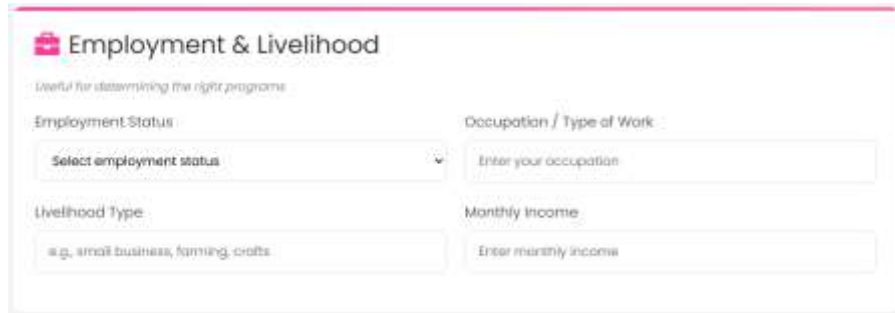
Monthly Household Income (P):

Source of Income:

2. **Number of Dependents** - Enter the number of children or elders who are financially dependent on your income.
3. **Head of the Family** - Select the appropriate option from the dropdown menu (e.g., Self, Spouse, Parent).
4. **Monthly Household Income** - Select the income

range that best represents your household's total monthly earnings from the dropdown menu.

5. Source of Income - Select the primary source of income for your household (e.g., Farming, Salaried Work, Small Business) from the dropdown.



The screenshot shows a form titled "Employment & Livelihood" with a subtitle "Useful for determining the right programs". The form is divided into four sections: "Employment Status" with a dropdown menu labeled "Select employment status"; "Occupation / Type of Work" with a text input field labeled "Enter your occupation"; "Livelihood Type" with a text input field labeled "e.g., small business, farming, crafts"; and "Monthly Income" with a text input field labeled "Enter monthly income".

- 1. Employment Status** - Select your current employment status from the dropdown (e.g., employed, unemployed, self-employed).
- 2. Occupation / Type of Work** - Enter the name of your current or most recent job/profession.
- 3. Livelihood Type** - Enter the specific nature of your livelihood, such as "small business, farming, or crafts".
- 4. Monthly Income** - Enter your estimated personal monthly income.

♥ Special Circumstances

Person with Disability (PWD) Solo Parent

Select option Select option

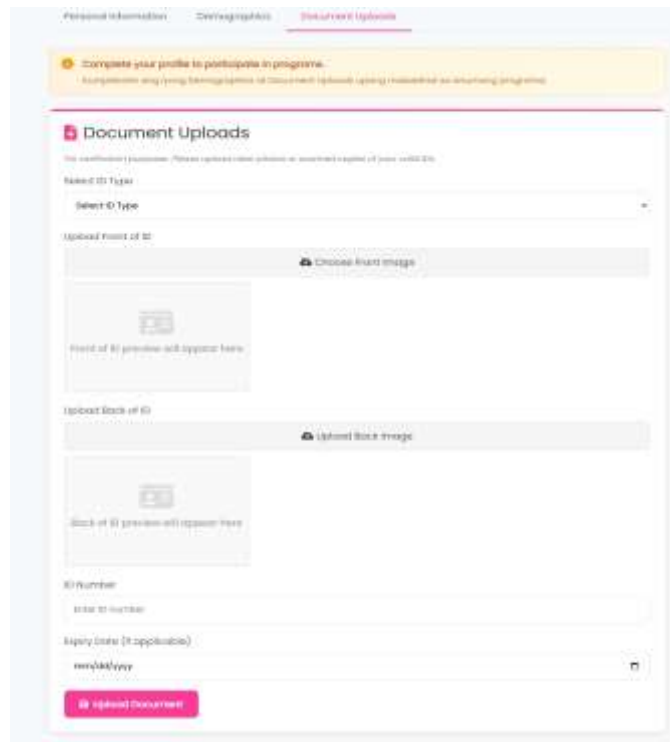
Any Existing Health Conditions?

Please specify any health conditions (optional)

Save Demographics Information

1. **Person with Disability (PWD)** - Select the appropriate option from the dropdown menu.
2. **Solo Parent** - Select the appropriate option from the dropdown menu.
3. **Any Existing Health Conditions?** - Briefly specify any existing health conditions you have (this field is optional).

4. **Save Demographics Information** Click this **button** - after filling out all the fields in the Demographics tab to save your data.

The screenshot shows a web form titled "Document Uploads" with a yellow banner at the top that says "Complete your profile to participate in programs." Below the banner, there's a section for "Document Uploads" with a sub-header "This section is for participants. Please upload your photos as instructed in order of your selected ID." The form includes a "Select ID Type" dropdown menu, followed by two image upload sections: "Upload front of ID" with a "Choose front image" button and a placeholder for the front of the ID, and "Upload back of ID" with an "Upload back image" button and a placeholder for the back of the ID. Below these are text input fields for "ID Number" and "Expiry Date (if applicable)". At the bottom is a pink "Upload Document" button.

1. **Select ID Type** - Select the type of official ID you are uploading from the dropdown menu (e.g., Driver's License, Postal ID).
2. **Upload Front of ID** - Click **Choose Front Image** to upload a file of the front side of your selected ID.
3. **Upload Back of ID** - Click **Upload Back Image** to upload a file of the back side of your selected ID.
4. **ID Number** - Enter the unique identification

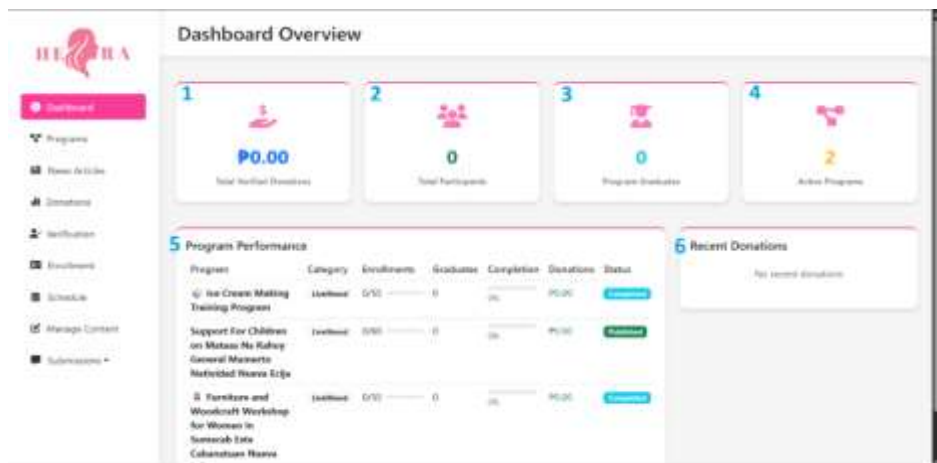
number found on the uploaded document.

5. **Expiry Date (If applicable)** - Enter the expiration date of your ID in \$\text{mm/dd/yyyy}\$ format.

6. **Upload Document** - Click this button after filling out all the fields and uploading both images.

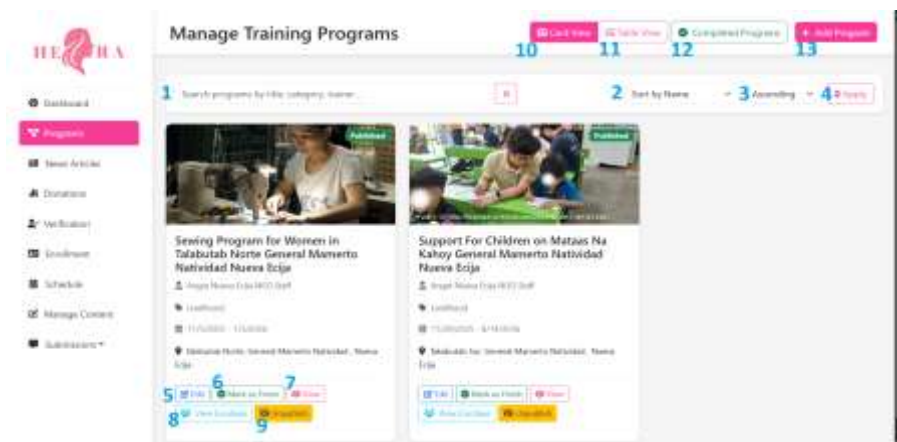
Admin Dashboard

The Admin Dashboard provides NGO staff and authorized personnel with a comprehensive overview and management tools for the entire Angat Nueva Ecija organization. This is the control center for tracking performance, managing users, reviewing program status, and verifying submitted documents and donations.



1. **Total Verified Donations** - Shows the cumulative monetary value of all processed and verified donations.

2. **Total Participants** - Shows the total number of individuals currently participating in all active programs.
 3. **Program Graduates** - Shows the total number of individuals who have successfully completed programs.
 4. **Active Programs** - Shows the current count of ongoing programs and initiatives.
 5. **Program Performance** - A detailed table listing current and recently completed programs.
 6. **Recent Donations** - A sidebar dedicated to displaying the latest donations received.
1. **Search Field** - Allows searching by program title, category, or trainer.



2. **Sort by Name** - A dropdown menu used to choose the criteria for ordering the program cards (e.g., by Name, Date Created, etc.).

3. **Ascending/Descending** - A dropdown menu to select the direction of the sort (A-Z or Z-A).
4. **Apply** - The button that executes the search and sorting options
5. **Edit** - Opens the program editing form.
6. **Mark as Finish** - A button to officially close the program.
7. **View** - Navigates to the public-facing details page of the program.
8. **View Enrollees** - Opens a list of all applicants and enrolled participants for that specific program.



9. **Unpublish** - Hides the program from the public Livelihood page.
10. **Card View** - Displays programs as large cards with images (the current view).
11. **Table View** - Changes the display to a list/spreadsheet format.

12. **Completed Programs** - Filters the list to show only programs that have been officially marked as finished.

13. **+ Add Program** - A button to create a new program.

The screenshot shows a 'Add New Program' form with a pink header and a close button. Below the header are four steps: 1. Basic Info, 2. Location, 3. Content, and 4. Publish. The form fields are as follows:

- 1. Program Title* (text input)
- 2. Category* (dropdown menu with 'Select Category' text)
- 3. Trainer* (text input)
- 4. Max Participants* (text input)
- 5. Start Date* (date picker with 'mm/dd/yyyy' format)
- 6. End Date* (date picker with 'mm/dd/yyyy' format)
- 7. Cover Image (file upload area with 'Choose File' button and 'No file chosen' text)
- 8. Next (blue button)

1. **Program Title** - The full name of the training or initiative (e.g., "Sewing Program for Women").

2. **Category** - Select the program's type from the dropdown (e.g., Livelihood, Education).

3. **Trainer** - Enter the name of the lead staff or instructor for the program.

4. **Max Participants** - Enter the maximum number of people allowed to enroll.

5. **Start Date** - The day the program begins.
6. **End Date** - The day the program concludes.
7. **Cover Image** - Upload a file to serve as the main visual for the program card.
8. **Next** - Proceeds to the **Location** step.

The screenshot shows a web form titled "Add New Program" with a pink header. A light blue banner at the top right says "1 Enter the barangay, municipality, and p". Below the header, there are four circular icons representing steps: 1 Basic Info (pink), 2 Location (pink), 3 Content (grey), and 4 Publish (grey). The "Location" step is active. The form section is titled "Program Location" and contains three input fields: "Barangay*", "Municipality*", and "Province*". The fields are numbered 1, 2, and 3 respectively. At the bottom right, there are two buttons: "Back" (grey) and "Next" (blue). The "Next" button is numbered 5.

1. **Barangay** - The barangay where the program will be held.
2. **Municipality** - The municipality or city where the program will be held.
3. **Province** - The province where the program will be held (e.g., Nueva Ecija).
4. **Back** - Returns to the **Basic Info** step.
5. **Next** - Proceeds to the **Content** step.

Add New Program

1 Basic Info 2 Location 3 Content 4 Publish

Program Content

Mini Heading

1 Enter mini heading

Additional Information

2 Enter details

Image (optional)

3 Choose File No file chosen

4 + Add Content Block

5 Back 6 Next

1. **Mini Heading** - A brief subtitle for the content block.
2. **Additional Information** - The main body of text describing the program curriculum, eligibility, and benefits.
3. **Image (optional)** - Allows uploading an image specific to this content block.
4. **+ Add Content Block** - Adds another set of **Mini Heading**, **Additional Information**, and **Image** fields.
5. **Next** - Proceeds to the **Publish** step.

1. **Status** - Select either **Draft** or **Publish** from the dropdown.
2. **Back** - Return to the content step.
3. **Save Program** - Click this button to finalize the creation and save the program under the selected status.

- 1-4. **Metric Cards** - Summarize the total number of articles, published articles, current (active) articles, and scheduled articles.
5. **Filter by Status** - Dropdown to filter articles by their publication status (e.g., All

Status, Published, Draft).

6-7. **Filter by Date** - Dropdowns to filter articles by **Month** and **Year**.

8. **Search News** - Search field to find articles by title or keyword.

9. **Archived News** - Button to view articles that have been manually or automatically archived.

10. **+ Add News** - Button to open the article creation form.

1. **News Title** - The primary headline for the news article.

2. **Status** - Select **Draft** or **Publish** (default is Draft).

3. **Source** - The source of the news or article (e.g., NGO Staff, Partner Org).

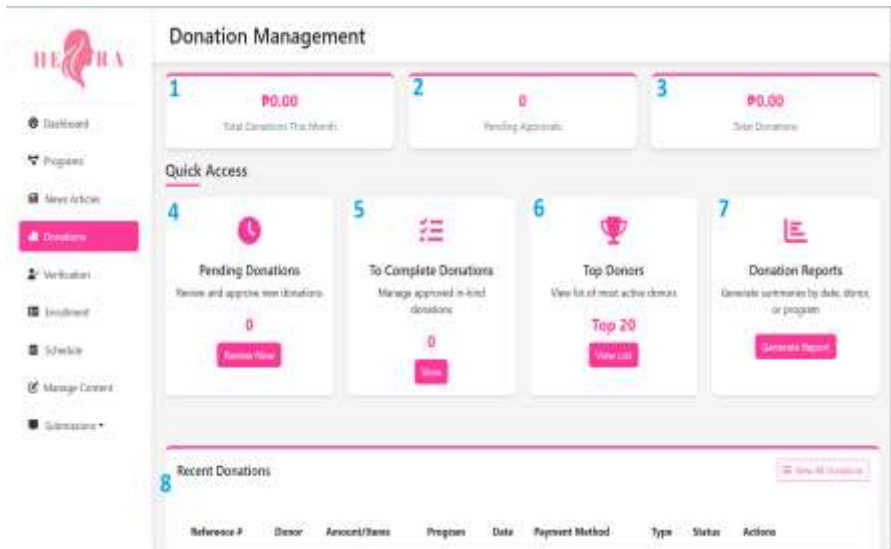
4. **Duration Date** - The date after which the article will **automatically archive**.

- 5. **Scheduled Publish Date** - The date and time the article will automatically go live.
- 6. **Tags** -Keywords separated by commas.
- 7. **News Media (Image)** - Upload a main image file.
- 8. **Lifetime Article** - A checkbox to prevent the article from automatically archiving.
- 9. **Excerpt/Summary** - A brief summary of the news article.

The screenshot shows a web form for creating a news article. It features two text input fields: 'Excerpt/Summary*' and 'Full Content *'. The 'Full Content' field is marked with a blue '1'. At the bottom right, there are two buttons: a grey 'Cancel' button marked with a blue '2' and a pink 'Add News Article' button marked with a blue '3'. The background shows a sidebar with navigation links like 'All Status', 'November', 'Angela', 'Publish', 'Our Tea', and 'Buhay'.

- 1. **Full Content** - The entire detailed body of the news article.
- 2. **Cancel** - Discards all changes and closes the modal.

3. **Add News Article** - Click this button to save and submit the new article.



1. **Total Donations This Month** - Tracks the monetary value of all donations received in the current month.
2. **Pending Approvals** - Counts the number of donations (typically in-kind or pending verification) awaiting staff approval.
3. **Total Donations** - Displays the cumulative total monetary value of all donations to date.
4. **Pending Donations (Review Now)** - Direct access to review and approve new donations.
5. **To Complete Donations (View)** - Direct access to manage approved in-kind donations (e.g., tracking pickup/drop-off).

6. **Top Donors (View List)** - Provides a list of the most active donors
7. **Donation Reports (Generate Report)** - Direct access to the report generation tool.
8. **Recent Donations** - A table listing the latest donation transactions.

Generate Donation Report

Report Type: Comprehensive Report Export Format: PDF

Start Date: 10/10/2025 End Date: 12/31/2025

Select Program: All Programs

Select Donor: All Donors

Report Summary

Total Donations: 0	Total Amount: P0.00	Unique Donors: 0	Programs: 0
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Donation Details

Reference #	Donor	Amount	Program	Type	Date	Method
No donations found for the selected criteria.						

Close Generate Report Export Report

1. **Report Type** - Select the type of report needed (e.g., Comprehensive Report, By Date Range).
2. **Export Format** - Select the file format for the exported report (e.g., CSV (Excel), PDF).
3. **Start Date / End Date** - Define the time period

for the report using the date fields.

4. **Select Program** - Filter the data to include only donations made to a specific program, or select **All Programs**.

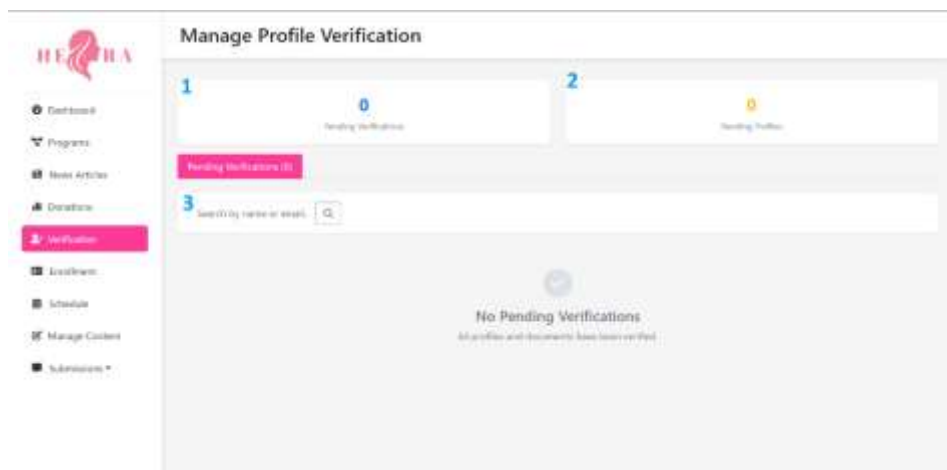
5. **Select Donor** - Filter the data to include only donations made by a specific donor, or select **All Donors**.

6. **Report Summary** - Displays live metric counts based on the filters applied above (e.g., Total Donations, Total Amount, Unique Donors).

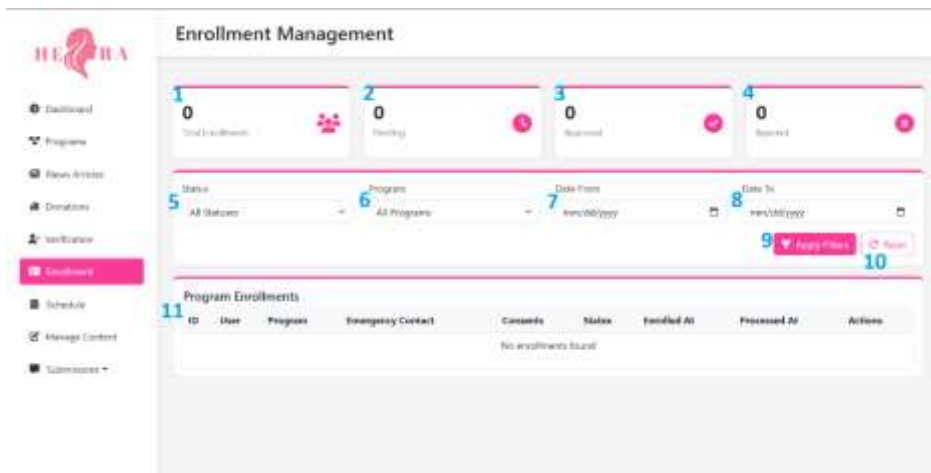
7. **Donation Details Table** - Previews the raw data that will be included in the report, showing columns like **Reference #**, **Donor**, **Amount**, **Program**, **Type**, **Date**, and **Method**.

8. **Generate Report** - Click this button to refresh the on-screen preview with the latest filter settings.

9. **Export Report** - Click this button to download the report in the selected **Export Format**.



1. **Pending Verifications** - Displays the count of user profiles and documents that are waiting for staff review and approval.
2. **Pending Profiles** - Displays the count of user profiles that have not been fully completed or verified yet.
3. **Search Field** - Allows searching the verification list by the user's name or email.



1-4. Metric Cards - Summarize the counts of **Total Enrollments**, **Pending** applications, **Approved** enrollments, and **Rejected** applications.

5. Status Filter - Dropdown to filter enrollments by their current status (e.g., All Statuses, Pending, Approved).

6. Program Filter - Dropdown to filter enrollments by a specific program, or select **All Programs**.

7-8. Date Range Filters - **Date From** and **Date To**

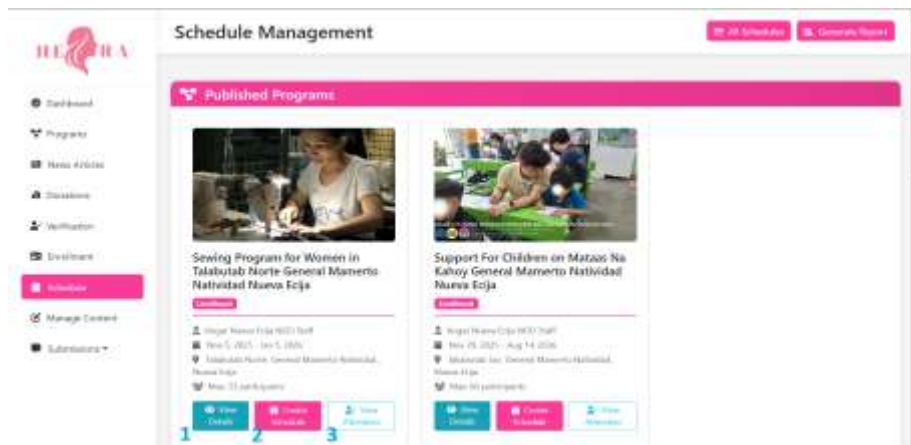
fields used to filter applications based on the date they were submitted.

9. **Apply Filters** - Executes the selected search and filter options.

10. **Reset** - Clears all applied filters and restores the default view.

11. **Enrollments Table** - The main table displaying all enrollment records based on the applied filters. Columns include **ID**, **User** (the applicant), the specific **Program** they applied for, **Emergency Contact**, **Consents**, **Status**, **Enrolled At** (date), **Processed At** (date).

1. **View Details** - Opens the public-facing details page for the program.



2. **Create Schedule** - Opens the **Create Schedule** modal.

3. **View Attendees** - Opens the **Attendees for [Program Name]** modal.

Create Schedule

Program

Support For Children on Mataas Na Kahoy General Mamerto N

Schedule Title *

Support For Children on Mataas Na Kahoy General Mamerto N

Start Date & Time *

11/17/2025 11:09 AM

End Date & Time*

mm/dd/yyyy --:-- --

Location *

Enter location details

Description / Notes

Add any additional notes or instructions...

Cancel

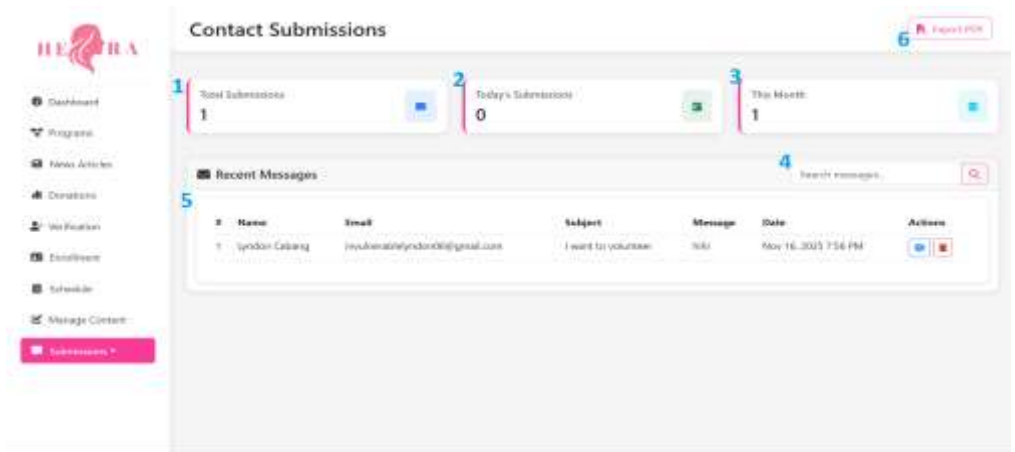
Create Schedule

1. **Program** - The program the schedule is being created for.

2. **Schedule Title** - A descriptive name for the session (e.g., "Orientation Day," "Week 3 Training").
3. **Start Date & Time** - The date and time the session begins.
4. **End Date & Time** - The date and time the session ends.
5. **Location** - The physical location details for this specific session.
6. **Description / Notes** - Any additional instructions or reminders for attendees.
7. **Create Schedule** - Click this button to save and publish the new schedule session.

1. **Program Information** - Displays read-only details about the program (Category, Trainer, Duration, Location).
2. **Attendance Summary** - Displays high-level metrics: **Total Enrollees**, **Attended** count, and **Attendance Rate**.

3. **Select Schedule Session** - A dropdown menu to select a specific session that was previously created.



1. **Total Submissions** - Displays the cumulative count of all messages received.
2. **Today's Submissions** - Displays the count of messages received on the current day.
3. **This Month** - Displays the count of messages received in the current month.
4. **Search messages** - Search field allowing staff to filter the message list by keywords, sender name, or subject.
5. **Export PDF** - Button to download all messages currently displayed in the table into a PDF file.

APPENDIX F

Certificate of Compliance

Angat Nueva Ecija NGO
3rd Street corner 6th Street Kapitan Pepe Subd.
Cabanatuan City, Nueva Ecija, 3100

This is to certify that the capstone project entitled **"PROJECT HERA: A Livelihood Training and Resource Management Platform for Angat Nueva Ecija NGO"**, developed by students of the **Nueva Ecija University of Science and Technology**, namely **Kurt M. Alfonso, Crister John F. Beltran, Lyndon Mark S. Cabang, Janella Eunice C. Enrile**, has been formally reviewed, evaluated, and tested by **Angat Nueva Ecija NGO** as the end-user.

After thorough assessment and system demonstration, we confirm that the platform meets the required functionalities of the organization and aligns with its operational needs, livelihood programs, and documentation workflows. The system performs according to expectations and is considered **acceptable, functional, and effective** for its intended purpose of improving training coordination, resource tracking, and donor engagement.

This certification is issued to acknowledge that **Project HERA** has successfully complied with the business standards and objectives of Angat Nueva Ecija NGO and satisfactorily fulfills the requirements set during its development process.

Certified by:


Rose Eirene Dela Cruz
ANE NGO Youth Division - External Vice President

APPENDIX G

Curriculum Vitae

LYNDON MARK S. CABANG

09613161365

lyndoncabang@gmail.com

Talabutab Norte General

Mamerto Natividad

Nueva Ecija



Personal Information

Age: 21

Sex: Male

Birthday: March 6, 2003

Place of Birth: Talabutab Norte General

Mamerto Natividad Nueva Ecija

Civil Status: Single

Father's Name: Marlino D. Cabang

Mother's Name: Marilyn S. Cabang

Religion: Born Again

Educational Background

Elementary: Talabutab Norte Methodist Learning Center (TNMLC)

S.Y 2009-2015

Talabutab Norte General Mamerto Natividad

Nueva Ecija

Junior Highschool: Our Lady of Fatima Academy (OLFA)

S.Y 2015-2019

Poblacion General Mamerto Natividad Nueva Ecija

Senior High School: Good Samaritan Colleges (GSC)

S.Y 2019-2021

Burgos Ave Cabanatuan City Nueva Ecija

Tertiary Education: Nueva Ecija University of Science and Technology (NEUST)

Bachelor of Science in Information Technology (BSIT)

S.Y 2022-Present

Sumacab Este Cabanatuan City Nueva Ecija

JANELLA EUNICE C. ENRILE

09701357328

janellaeunice1313@gmail
.comBrgy. Polilio, Cabanatuan
City, Nueva Ecija, 3100**PERSONAL INFORMATION**

Age:	21
Sex:	Female
Birthday:	December 14, 2003
Place of Birth:	Sumacab Sur, Cabanatuan City, Nueva Ecija
	Single
Civil Status:	Jonathan T. Enrile
Father's Name:	Edinel C. Enrile
Mother's Name:	Roman Catholic
Religion:	

EDUCATIONAL BACKGROUND

Elementary:	Polillo Elementary School S.Y. 2010-2016 Polilio Cabanatuan City, Nueva Ecija
Junior High School:	College of Immaculate Conception S.Y. 2016-2019 Cabanatuan City, Nueva Ecija Santa Rosa National High School S.Y. 2019-2020 Sta. Rosa, Nueva Ecija
Senior High School:	Top Link Global College Inc. Accountancy, Business, and Management (ABM) S.Y. 2020-2022 Daang Sarile, Cabanatuan City, Nueva Ecija
Tertiary Education:	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology (BSIT) Major in Web Systems Technology (WST) A.Y. 2022-Present Sumacab Este, Cabanatuan City, Nueva Ecija

Kurt M. Alfonso

09164574696

kurtalfonso15@gmail.com

Purok 2, Balante,

Gapan City

Nueva Ecija

**Personal Information**

Age: 21
Sex: Male
Birth day: August 15, 2004
Place of birth: San Leonardo, Nueva Ecija
Civil status: Single
Fathers Name: Josefino C. Alfonso
Mothers Name: Ma Cecilia M. Alfonso
Religion: Roman Catholic

Educational Background

Elementary: **Balante Elementary School**
S, Y 2010-2016
Balante, Gapan City, Nueva Ecija

High School: **The Great Shepherdess Montessori Center Inc.**
S.Y. 2016-2020
San Isidro, Gapan City, Nueva Ecija

Senior High: **Our Lady of Fatima University School (OLFU)**
Science, Technology, Engineering and Mathematics (STEM)
S, Y 2020-2022
Km 311 Hilltop Dr. Pan-Philippine Highway, Cabanatuan City, Nueva Ecija

Tertiary Education: **Nueva Ecija University of Science and Technology**
Bachelor of Science in Information Technology
S, Y 2022-Present
Sumacab Este, Cabanatuan City, Nueva Ecija

CRISTER JOHN F. BELTRAN

0912-316-1559

beltrancj18.neust@gmail.com

Purok 7, Bantug Norte,

Cabanatuan City, Nueva Ecija



PERSONAL INFORMATION

Age: 22

Sex: Male

Birthdate: August 18, 2003

Place of Birth: San Antonio, Nueva Ecija

Civil Status: Single

Father's Name: Crisanto Beltran

Mother's Name: Emily Beltran

Religion: Born Again

Educational Background

Primary: **Faustino P. De Leon Elementary School**

S.Y. 2010-2016

High School: **Nueva Ecija High School**

S.Y. 2016-2020

Senior High School: **Nueva Ecija High School -
Senior High School**

S.Y. 2020-2022

TVL - Computer System Servicing

Tertiary: **Nueva Ecija University of Science
and Technology**

A.Y. 2022 - present

Bachelor of Science in Information
Technology

Major in Web Systems Technology